

The Origin of the State: Land Productivity or Appropriability?*

Joram Mayshar[†]

Omer Moav[‡]

Luigi Pascali[§]

November 6, 2020

Abstract

The conventional theory about the origin of the state is that the adoption of farming increased land productivity, which led to the formation of food surplus. This surplus was a prerequisite for the emergence of tax-levying elites, and eventually states. We challenge this theory and propose that hierarchy arose due to the shift to dependence on appropriable cereal grains. Our empirical investigation, utilizing multiple data sets spanning several millennia, demonstrates a causal effect of the cultivation of cereals on hierarchy, without finding a similar effect for land productivity. We further support our claims with several case studies.

Geography, Hierarchy, Institutions, State Capacity

JEL CLASSIFICATION NUMBERS: *D02, D82, H10, O43*

*We would like to thank Daron Acemoglu, Alberto Alesina, Yakov Amihud, Josh Angrist, Quamrul Ashraf, David Atkin, Simcha Barkai, Sascha Becker, Davide Cantoni, Ernesto Dal Bo, Carl-Johan Dalgaard, Diana Egerton-Warburton, James Fenske, Raquel Fernandez, Oded Galor, Eric Ginder, Paola Giuliano, Stephen Haber, Naomi Hausman, Victor Lavy, Andrea Matranga, Stelios Michalopoulos, Joel Mokir, Suresh Naidu, Nathan Nunn, Omer Ozak, Joel Slemrod, Fabio Schiantarelli, James Snyder, Uwe Sunde, Jim Robinson, Yona Rubinstein, Ana Tur Prats, Felipe Valencia, Tom Vogl, Joachim Voth, John Wallis and David Weil for useful comments. We are also grateful for comments from participants in many seminars and workshops.

[†]Department of Economics, Hebrew University of Jerusalem. Email: msjoram@huji.ac.il

[‡]Department of Economics University of Warwick, School of Economics Interdisciplinary Center (IDC) Herzliya, CAGE and CEPR. Email: omer.moav100@gmail.com; Moav's research is supported by the Israel Science Foundation (Grant No. 73/11).

[§]Pompeu Fabra University. Email: luigi.pascali@upf.edu

1 Introduction

Following the Neolithic Revolution – our ancestors’ transition from hunting-gathering to sedentary farming – complex hierarchical societies emerged, leading eventually to the rise of tax-levying states. The Neolithic transition raises intriguing, fundamental questions: What are the mechanisms by which the adoption of agriculture led to complex hierarchies and states? Why did states fail to emerge in some regions, mostly in the tropics, long into recent centuries, despite the adoption of agriculture? And why have some states failed to develop significant state capacity?

The prevailing theory traces back to Adam Smith and earlier scholars. It holds that following the adoption of farming, complex social hierarchies emerged as a result of increased food output. In particular, high output generated a surplus (food in excess of farmers’ subsistence needs), which was a prerequisite for the rise of an elite that taxed farmers and supported bureaucrats, troops, and other specialists who did not engage in food production. According to this conventional productivity theory, regional differences in land productivity explain regional disparities in the development of hierarchies and states.

However, there are regions in the world where complex hierarchies did not emerge, even after the adoption of productive agriculture that increased food output. The common feature of many of these regions is that the staple crops that were cultivated were not cereals, but mainly roots and tubers. As Scott (2017) puts it: “It is surely striking that virtually all classical states were based on grain ... History records no cassava states, no sago, yam, taro, plantain, breadfruit, or sweet potato states.”¹

Cereal grains can be stored, and because they are harvested seasonally, have to be stored so that they are available for year-round subsistence. The relative ease of confiscating stored cereals, their high energy density, and their durability enhance their appropriability, thereby facilitating the emergence of tax-levying elites. Roots and tubers, in contrast, are typically perennial and do not have to be reaped in a particular period, but once harvested are rather perishable.²

¹In distinguishing between cereals and other crops that are not storable, without questioning the conventional productivity-and-surplus theory, as we do, Scott (2009, 2017) follows Taylor (1973) and Testart (1982a, 1982b, 1982c). Taylor, in fact, proposed that the Neolithic Revolution ought to be called the “storage revolution.”

²Moreover, roots and tubers’ high water content hampers the efficient transportation of their nutritional content. In Appendix A we describe the characteristics of roots and tubers. We show that their portability is hindered both by their vulnerability to spoilage and their bulkiness (due to ca. 70% moisture content). We also support the

We challenge the conventional productivity theory, contending that it was not an increase in food production that led to complex hierarchies and states, but rather the transition to reliance on appropriable cereal grains that facilitate taxation by the emerging elite. It follows from our critique that it is primarily differences in land suitability for different crops that account for regional disparities in the developmental trajectory of hierarchies and states, rather than differences in land productivity per se.

In challenging the claim that the emergence of hierarchy was a result of high farming productivity, we also contest that surplus is the mechanism that links productivity to taxes. We illustrate our argument with the following scenario: consider a farming society that subsists on a cereal grain that has to be harvested within a short period and then stored for year-round consumption. A tax collector could confiscate part of the stored grain and transport it for consumption by distant elite and other non-food producers, even without a food surplus. Ongoing confiscation can be expected to impact adversely the size of the farming population. However, due to diminishing average product of labor, the smaller population would lead to higher output per farmer. This would result in an equilibrium in which total output exceeds the farming population's subsistence needs, with the surplus confiscated by the non-farming elite.

We concur with the conventional productivity theory to the extent that in hierarchical societies farmers produce a surplus. Our contention is that surplus isn't a prerequisite for the emergence of a tax-levying elite, it is rather the elite that generates the food surplus on which it can flourish, once the opportunity to appropriate rises. Thus, this simple hypothetical scenario demonstrates that the availability of surplus need not be a precondition for taxation and hierarchy, as argued by the conventional productivity theory.

To further illustrate our critique, consider now an early society that cultivates cassava. Cassava is a perennial root that can be harvested year-round but rots shortly after harvest. This makes it difficult to confiscate, and practically impossible to transport for use by a distant elite. It is thus unlikely that a complex hierarchy could emerge in this society, despite the availability of food surplus. This scenario suggests that had the Neolithic Revolution amounted solely to a transition to the cultivation of perishable food sources, the increase in productivity would not have led to the

claims: (i) that reliance on roots and tubers is a major phenomenon in many regions; (ii) that roots and tubers are highly productive in these regions; (iii) that their harvest is in general non-seasonal; (iv) that after harvest they are significantly more perishable than cereals.

emergence of complex hierarchy.

Moreover, we propose that increased output is unlikely to generate a surplus, as conventional theory asserts.³ This is simply inconsistent with the Malthusian mechanism.⁴ Improved hunting techniques and the accumulation of knowledge led to increased productivity over time among hunter-gatherers too. That increase was apparently converted to larger population size rather than to sustainable surplus, leaving hunter-gatherers in small egalitarian societies, without complex hierarchies. Since the Neolithic Revolution spanned several millennia (Purugganan and Fuller, 2010), one could expect that this gradual increase in productivity would also have been absorbed by increased population.

The conventional theory fails to explain why the adoption of agriculture is different from productivity growth among hunter-gatherers, but this disparity is easily explained by the appropriability theory. Hunter-gatherers relied on hand-to-mouth food sources which are not easily appropriable and therefore did not develop hierarchies even when their productivity increased.

We propose that when it became possible to appropriate crops, a taxing elite emerged, and this led to the state. We note that stored cereals are appropriable not just by a would-be elite, but also by bandits – therefore their cultivation generated a demand for protection and at the same time facilitated taxation to finance the supply of such protection by the elite.⁵ Accordingly, we propose that the protection of food stockpiles and hierarchy emerged in parallel to the gradual transition to reliance on appropriable food sources. We do not suggest, however, that elites were benign; rather, as Olson (1993) observed, deterring bandits served both the farmers and the elite.⁶

In appendix D, we present a formal model that focuses on the possibility of a state emerging and deterring bandits (which provides one specific interpretation of our general thesis). In our model

³Other scholars have already pointed out that an increase in productivity may be dissipated in various ways without creating surplus. Pearson (1957) contends that cultural needs would evolve to eliminate any surplus. Sahlin (1972) argues that hunter-gatherers, too, could have procured food beyond their subsistence needs, but deliberately refrained from doing so by preferring leisure. He infers that the first farmers could have similarly responded to increased productivity by working less hard. But he does not explain why they failed to do so, or what accounts for the rise of hierarchy following the adoption of agriculture.

⁴Ashraf and Galor (2011) provide empirical support for the applicability of Malthus’s theory in the pre-industrial era.

⁵If the elite/state deters bandits, the population size with a state is likely to be larger than without one (see the model in Appendix B), unlike our simple scenario described above, in which we abstract from bandits.

⁶The role of banditry as posing a significant threat to farmers and as provoking a basic need for protection is often raised (e.g., McNeill 1992: 85, and Mann 1986: 48). We do not provide any evidence in our empirical analysis for the role of bandits, but we note that for our main thesis the question of the existence of bandits is irrelevant.

farmers allocate their land between tubers, that cannot be looted or taxed, and cereals, that can be taxed by the elite at a cost of employing tax collectors, or looted by unorganized bandits. The productivity of the two crops differs across geographic locations. Farmers choose to cultivate cereals only in locations where their productivity advantage over tubers is sufficiently high to compensate for the loss due to taxation or looting. We distinguish between two regimes. In ‘anarchy,’ there is no protection against looting. The number of bandits is determined endogenously so that their average revenue from theft is equal to the alternative productivity in foraging. Under ‘hierarchy,’ there is a net-revenue maximizing elite that commits to its selected rate of taxation of cereals and provides farmers with full protection from bandits. A state can only exist if its tax revenue covers the fixed cost of deterring bandits.

The main prediction of this model is that a state can only exist if cereals are sufficiently more productive than tubers. This result illustrates our main claim that it is the magnitude of the productivity advantage of cereals over tubers, rather than high agricultural productivity per se, that determines whether hierarchy emerges. The model also suggests that even though the elite is self-serving, whenever hierarchy exists it dominates anarchy in terms of farmers’ welfare. Anarchy is more distortionary than hierarchy for two reasons.⁷ First, the state’s ability to commit to a lower tax rate encourages the cultivation of cereals when these are sufficiently more productive. Second, the net revenue-maximizing state employs tax collectors only up to the point where their marginal tax revenue equals their wage (which we assume to equal the alternative income from foraging), and thus employs (weakly) fewer tax collectors than the equilibrium number of bandits under anarchy.

In our empirical section we test for the root cause of the emergence of hierarchy: is it land productivity or appropriability?⁸ Our research question doesn’t allow for one perfect randomized controlled trial that could prove or disprove our thesis. We therefore perform multiple imperfect tests based on different data sets. We present our empirical analysis in four subsections, each using a different dataset that measures hierarchical complexity.

The first subsection is based on Murdock’s (1967) *Ethnographic Atlas*, which covers cultural,

⁷We ignore here the possibility that the non-benevolent state may contribute further to farmers’ welfare, if it contributes directly to agricultural productivity, for example through publicly provided irrigation infrastructure.

⁸We cannot test empirically for the direction of a causal effect between hierarchy and surplus, which are correlated according to both the productivity and the appropriability theories. But we do provide some evidence when data permits, that it is appropriability, and not productivity, that causes surplus.

institutional, and economic features of more than 1,200 pre-colonial societies from around the world. Our main outcome variable is the level of hierarchical complexity, which we regress on a dummy that identifies societies that rely on cereal grains for their subsistence. Since the choice of the main crop might depend on hierarchy, we instrument for the cultivation of cereals by the potential productivity advantage of cereals over roots and tubers in a rain-fed subsistence economy, calculated from the land suitability data provided by the Food and Agriculture Organization (FAO). Consistent with the appropriability theory, the 2SLS estimates show that cultivating cereals has a considerable positive effect on hierarchical complexity. Moreover, and consistent with our criticism of the conventional theory, our analysis fails to show any positive effect of land productivity on hierarchy, while it shows that societies based on roots or tubers display similar levels of hierarchy to non-farming societies. Finally, for a subset of the sample for which data are available, we also find that societies growing cereals are characterized by more burdensome taxation. These results hold when controlling for land productivity.

The results based on the Ethnographic Atlas are not conclusive. First, due to the cross-sectional nature of the data, we cannot exclude potential omitted geographic factors that might be driving the 2SLS results (though we do control for a large set of potential confounders). Second, the Ethnographic Atlas is not a random sample of societies and it is biased towards relatively isolated societies with relatively low levels of hierarchy. Third, the data pertains mainly to the 18th and 19th century, which is a long time after the early transition to farming.

To overcome these limitations, in the second subsection we employ a dataset on hierarchy compiled by Borcan, Olsson, and Putterman (2017). This dataset is based on the present-day boundaries of 159 countries, with institutional information every half-century. We can only run reduced form regressions with this dataset, due to the lack of information about the prevalent crop over the same time-span in these regions. We first look at the classical age – the earliest period in human history for which detailed data on the location of large states are available. For the year 450 AD (just before the collapse of the Roman Empire), qualitative results are consistent with those obtained using the data from the Ethnographic Atlas described above: we document a significant positive correlation between the productivity advantage of cereals (proxied with the FAO data) and state presence. Moreover, once controlling for the productivity advantage of cereals, the correlation between land productivity and hierarchy disappears.

FAO data is based on modern yield predictions. To overcome any concerns about relevance to earlier periods, in collaboration with the Global Crop Diversity Trust we have developed a dataset on the distribution of wild relatives (WRs) of domesticated crops (i.e. wild plants that are genetically related to cultivated crops). The number of WRs of a certain domesticated crop in a region proxies for the potential for the domestication of that crop in that region. Our cross-sectional regressions using the WRs data suggest that the key for hierarchy is the combination of availability of cereal grains and no availability of an alternative root or tuber. Thus, in contrast to the conventional productivity theory, we show that the availability of less appropriable productive crops prevents the emergence of hierarchy. The analysis accounts for a large set of possible confounding factors, but we cannot rule out that unobservable characteristics that are systematically correlated with the availability of different crops might be driving our results.

To alleviate this concern, we focus on the last millennium and exploit the “Columbian Exchange” of crops between the New World and the Old World as a natural experiment. The exchange introduced new crops which, as predicted by FAO data, changed both land productivity and the productivity advantage of cereals over roots and tubers in most countries in the sample. Consistent with the appropriability theory and with our critique of the productivity theory, the panel regressions confirm that the productivity advantage of cereals over roots and tubers had a positive impact on hierarchical complexity, while land productivity did not.

The third subsection gets us closer to the Neolithic transition and is based on cross-section data from various sources on the location of ancient cities and archeological sites (e.g. pyramids, ancient temples, palaces, and mines), which presumably indicate social hierarchy. With these data, we employ three different cross-sectional approaches to test the appropriability and productivity theories. First, using the WRs data and following the same approach described above for classical states, we obtain the same qualitative results: the availability of WRs of domesticated cereal grains and lack of availability of WRs of domesticated roots or tubers explain a significant part of the variation in the different indications of hierarchy in all the archeological data sets, for data spanning various periods of antiquity.

As an alternative to the WRs proxy for crop availability, we utilize data on the location of centers of origin of agriculture. We show that distance from these centers only has a negative impact on the development of early civilization if the center domesticated cereals. Finally, using

the FAO-based data, we find that the cultivation of cereals, unlike land productivity, can explain the distribution of ancient cities and other indications of hierarchy. Although results are robust to a large number of confounders a limit of this data set is that it is cross-sectional.

We overcome this limitation in our fourth empirical subsection, in which we use data from the *Archaeological Atlas of the World* (Whitehouse and Whitehouse, 1975). Although this source was published more than 40 years ago, it has the advantage of providing radiocarbon estimates dating various archaeological sites, enabling us to count the number of pre-Neolithic and post-Neolithic sites in each area. The difference-in-difference estimates support the appropriability theory. Specifically, we find that the Neolithic transition only led to more traces of indications for complex hierarchical societies in areas where agriculture was more likely to start with cereals, based on our three proxies explained above (WRs of domesticated crops, proximity to areas of domestication, and FAO productivity data). We find no evidence for the conventional productivity theory using these data.

In summary, our empirical analysis provides repeated evidence that the cultivation of cereals had a significant causal effect on the development of complex hierarchies and states, consistent with the appropriability theory. It also illustrates that the correlation between land productivity and hierarchy disappears when controlling for the cultivation of cereals, consistent with our critique of the conventional productivity theory. Moreover, the finding that it is unlikely that complex hierarchies would emerge when productive roots and tubers are available, supports both the appropriability theory and the critique of the productivity theory. It is consistent with the prediction that farmers would prefer, when practically possible, to cultivate less appropriable crops that protect from bandits and tax collectors. Complex hierarchies, the data suggests, emerged when farmers were constrained to cultivating cereals.

In the empirical section, we discuss other potential concerns regarding the data and the identification. As mentioned above, we acknowledge that none of our empirical investigations provides full proof for our thesis. However, we contend that the robustness of our results is sufficient to cast doubt on the prevailing productivity-and-surplus explanation for the emergence of hierarchy, and, pending further studies, to favor the appropriability explanation.

Our empirical analysis, presented in section 2, does not identify the mechanism that links cereal cultivation and complex hierarchy. In section 3 we present five case studies that are consistent with

the appropriability theory. Nonetheless, the properties of cereals that render them appropriable could give rise to other mechanisms that are also consistent with the evidence. For example, increasing returns to scale in the construction of storage facilities, which are required where cereals are the staple food source, may have contributed to early communal storage under leaders. Moreover, communal storage and redistribution across both time and individuals provide better food security and might require leadership to organize. In addition, the durability and transportability of cereals may have had a role in promoting trade, which increases the return to public goods such as law and order and requires leadership for their provision.

Finally, our model in which cereals attract bandits is related to various conflict theories, which are also consistent with our main finding: Storage of appropriable cereals increases the return to attacks and makes it easier for the winners of a conflict to extract ongoing surplus from the losers. The review of the extensive literature on the rise of hierarchy and states is presented in section 4, so that we can discuss it in light of our empirical findings and case studies. We offer a few concluding remarks in Section 5.

2 Evidence I - Empirical Investigation

In this section we report the results of our empirical tests. We describe the data in subsection 2.1 and present our various empirical exercises and findings, using four different dataset on hierarchy and statehood, in subsections 2.2 to 2.5.

2.1 Data

2.1.1 Ethnographic data

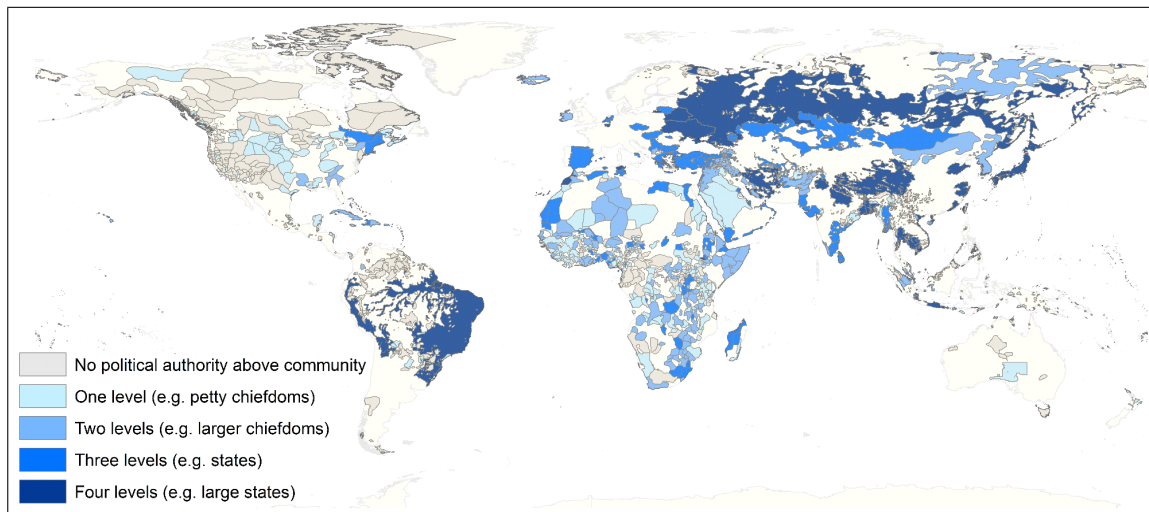
Murdock’s (1967) Ethnographic Atlas provides a database of 1,267 societies from around the world. The database contains information on cultural, institutional, and economic features of these societies, before experiencing any significant industrialization.⁹ The sample is global, with an emphasis on North American and African societies. We removed from the sample 2 duplicate observations,

⁹The focal year that data refer to is pre-1800 for 3% of the societies, the 19th century for 25%, the first half of the 20th century for 69%, and the second half of the 20th century for just 2.5%. The remaining societies are missing a focal year. Figure F.1 in the online Appendix F reports the density plot of the year that the data refers to for the different societies.

7 societies observed before AD 1500, and 10 societies for which the year of observation is missing, leaving 1,248 societies. These are matched to ethnic maps using either the geo-coordinates of each ethnicity provided by the Ethnoatlas or the maps on the spatial location of ethnicities constructed by Fenske (2013).¹⁰

We use the variable “Jurisdictional Hierarchy beyond the Local Community” as a measure of hierarchical complexity.¹¹ This is an ordered variable with five possible levels: (i) no political authority beyond community, (ii) petty chiefdoms, (iii) larger chiefdoms, (iv) states, and (v) large states. We plot this measure of hierarchy in Figure 1 and present the summary statistics in the first row of Table F.1 in the online appendix. Most of our sample is composed of societies that lack any political integration above the local community, or of small districts ruled by chiefs. These societies prevail in North America, Australia and in Central Africa, but are rare in Northern Africa and in Asia, where large chiefdoms and states are more common.

Figure 1: Jurisdictional hierarchy beyond the local community in pre-industrial societies



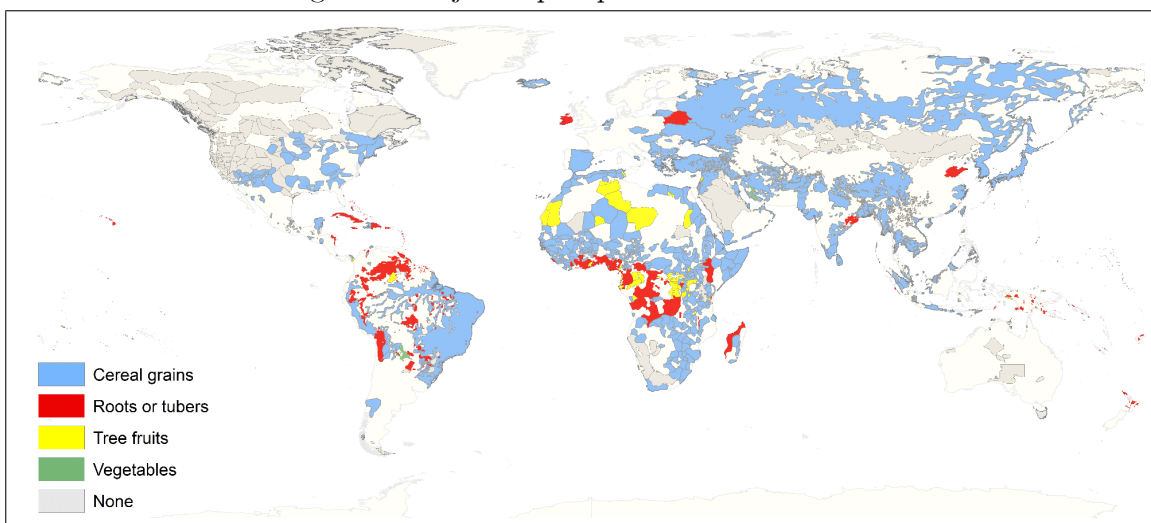
The Ethnoatlas also provides information on the reliance of these societies on agriculture for their diet, and on the major crop type of societies that practice agriculture. These two variables are plotted in Figure 2, with summary data in rows 2 and 3 of Table F.1 in the online appendix. As can

¹⁰The ethnic maps in Fenske (2013) are constructed by combining Murdock’s (1959) ethno-linguistic map for Africa with three other sources for the rest of the world (Heizer and Sturtevant, 1978; Global Mapping International, and Weidmann et al., 2010).

¹¹Gennaioli and Reiner (2007) and Michaelopoulos and Papaioannou (2013) make a similar use of this variable.

be seen from Figure 5, approximately one fifth of the societies in the sample do not practice any form of agriculture. These societies are concentrated in North-West America, Central Asia, Australia and South-West Africa. The median society relies on agriculture for approximately 50 percent of its caloric needs. The great majority of the societies that practice some form of agriculture rely on either cereal grains (65.4 percent) or on roots and tubers (26.1 percent). Using this information, we define a dummy that identifies societies whose primary crop is cereals and present summary statistics in the second row of Table F.1. The Ethnoatlas also provide data on the use of domestic animals. Based on this information, we create three dummy variables (rows 5-7 in Table F.1). The first dummy identifies societies that rely on some form on animal husbandry; the second dummy identifies societies in which domesticated ruminants are the most relevant animal husbandry; the third dummy identifies societies that use animals for cultivation.

Figure 2: Major crop in pre-industrial societies



The second source of data we use is a derivative of the Ethnographic Atlas, provided by the Standard Cross-Cultural Sample (SCCS). The data are based on a representative sample, defined by Murdock and White (1969), of 186 societies from the Ethnoatlas. A large number of publications by diverse authors coded the SCCS societies for many different characteristics. Cumulative ethnographic codes and codebooks are published in the World Cultures electronic journal. We use three variables from the SCCS (rows 15-17 in Table F.1). The first variable, coded by Tuden and Marshall (1972), lists

the local elite’s sources of political power. We create a dummy for “the existence of a farming surplus” that is zero if the most prestigious members of the society derive their livelihood from their own subsistence activities and one otherwise. This dummy is plotted in figure F.2 in the online appendix. The second variable refers to the “Extent of burden caused by tribute payments or taxation,” coded by Lang (1998). Based on his work, we create a variable that captures the tax burden on members of these societies. We code this variable as 0 if there is no evidence of tribute or taxation, 1 if there is evidence of sporadic taxation or the taxes are reported not to be burdensome, and 2 if either there is evidence of regular taxation or the exactions are reported to be burdensome. The third variable is a measure of population density coded by Pryor (1985). Societies are categorized into 6 bins (the first bin contains societies with 0-1 persons per square mile, and the last contains societies with 500+ persons per square miles).

Table F.5 in the online appendix reports pairwise correlations between the variables of the societies in the Ethnographic Atlas. As expected, societies characterized by more complex hierarchies do generally display a higher reliance on agriculture (and in particular on cereals), a higher probability of producing a farming surplus, higher tax burden and more dense populations.

2.1.2 Country-level data

We constructed a hierarchy index using data from Borcan, Olsson and Putterman (2018). The data cover the area of 159 modern-day countries for every half century from AD 50 to 2000. A score of 1 is assigned if there is a government above the tribal level in that area; 0.75 if it is a chiefdom, and 0 if there is no authority above the tribe. We merge these data with information on: the legal origin of the country (from La Porta et al., 1998); population density in 1500 (Acemoglu, Johnson and Robinson, 2002); mortality of early settlers (Acemoglu, Johnson and Robinson, 2001); the number of exported slaves (Nunn, 2008); climate and latitude (Nunn and Puga, 2012); genetic diversity (Ashraf and Galor, 2013); the density of locally available wild animals suitable for domestication (Hibbs and Olsson 2004 and Ashraf and Galor 2011);¹² the potential for irrigation (Bentzen et al.,

¹²This variable is computed dividing the number of wild terrestrial mammals, which are believed to have been domesticated prehistorically for herding, by the area of the country. These are the ancient ancestors of sheep, goats, cattle, horses, pigs, Bactrian camels, Arabian camels, llamas, yaks, Bali cattle, reindeers, water buffaloes, donkeys and mithuns. Both Hibbs and Olsson (2004) and Ashraf and Galor (2011) exclude neo-Europes from the dataset. We have therefore complemented their dataset with new data we collected on Australia, Canada, New Zealand, and United States using Nowak (2011) as primary source.

2017), and the colonial history of each country (Pascali, 2017). Figure F.3 in the online appendix shows the colonial history of each country and Table F.2 in the online appendix provides summary statistics for these variables.

2.1.3 Location of ancient cities and archaeological sites

To capture differences in hierarchical complexity further back in time, we collected data on the location of ancient cities. We use two different sources of information. The first is Reba, Reitsma and Seto (2016), which provides data on the location of urban settlements from 3700 BC to AD 2000. The dataset is based on historical, archaeological, and census-based urban population data previously published in tabular form by Chandler and Modelski. Figure F.4 shows ancient settlements (founded before 500BC), while Figure F.5 shows classical settlements (founded before AD 450). The second source is the website developed by Daniel DeGroff,¹³ which provides the list of cities and towns that were founded before AD 400. We also use archaeological evidence indicating ancient complex civilizations, collected from miscellaneous sources, presented in the online appendix.¹⁴

We aggregate data on the location of cities and archaeological ruins at the 1x1 decimal degree raster area. The first 10 rows of Panel A in Table F.3, in the online appendix, present descriptive statistics on the number of cities and relevant archaeological ruins in each terrestrial raster point.

2.1.4 Radiocarbon-dated prehistoric archaeological sites

David and Ruth Whitehouse’s (1975) *Archaeological Atlas of the World* provides a database of the most relevant global prehistoric and proto-historic archaeological sites, which were known at that

¹³<https://sites.google.com/site/ancientcitiesdb>

¹⁴Data on archaeological sites from the ancient world come mainly from Ancientlocations.net. Sites are included if they existed prior to AD 476 (end of the West Roman Empire) in the Old World and prior to 1492 in the New World. The data are complemented with archaeological data from the Megalith Portal, a web community with input from thousands of photographers and archaeologists. Ruins are classified according to 57 categories, which allows us to distinguish the effect cereals and land productivity on different archaeological evidence of complex societies, such as pyramids, mines, temples and palaces, from the effect on other types of evidence (e.g. standing stones) that are perhaps less indicative of complex hierarchies. The portal initially categorized archaeological ruins in Great Britain and only recently extended to cover the entire world. As a result it oversamples Europe. We therefore exclude types of ruins that are only found in Europe and its surroundings, and always show the robustness of our regressions when excluding Europe.

time. The atlas includes 4,215 sites that are radiocarbon dated.¹⁵

We geo-reference these sites and, using the information in the map titles and accompanying text, classify them depending on whether they pre-date the Neolithic transition in the relevant location or not. The result is a list of 825 sites that belong to pre-transition years and 3,309 sites that belong to the post-transition years. (We exclude 8 sites for which either geo-referencing was not possible or dates were uncertain and 73 sites for which we were uncertain about whether they belong to the pre- or post- transition years).

We compute the number of pre-Neolithic sites and post-Neolithic sites at the 1x1 decimal degree raster.¹⁶ In the empirical analysis we either use all sites, or alternatively, only sites of prehistoric settlements. Panel B of Table F.3, in the online appendix, presents the descriptive statistics for these variables.

2.1.5 Soil suitability data

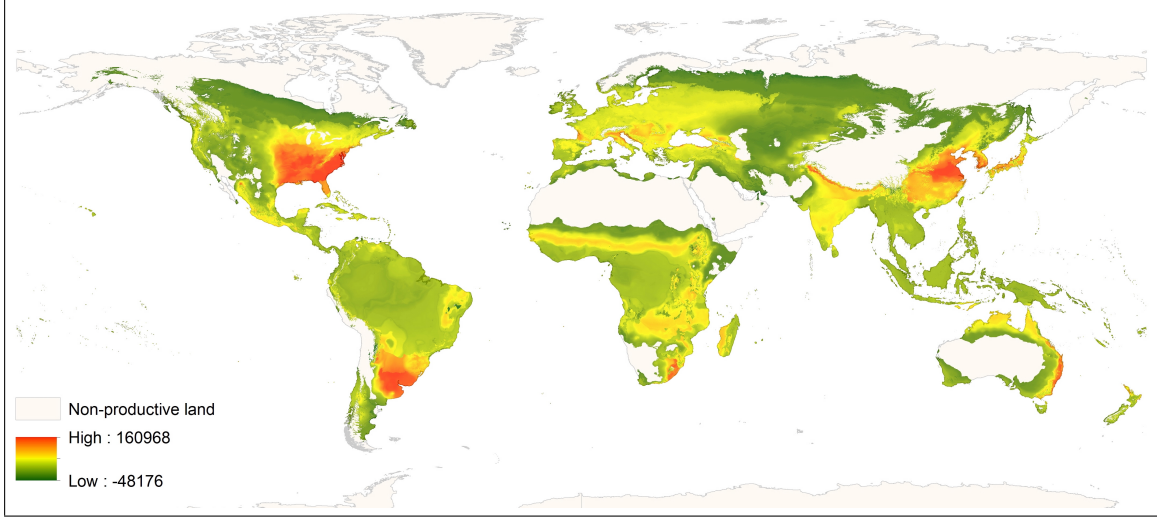
For data on land productivity and the farming of cereals, we use detailed spatial data on the suitability of soil for different crops from the Global Agro-Ecological Zones (GAEZ) project of the Food and Agriculture Organization (FAO). The data provides global estimates of potential crop yields for different crops with cell size of 5'x5' (i.e. approximately 100 km²) based on two possible categories of water supply (rain-fed and irrigation) and three different levels of inputs (high, medium and low). In addition, it supplies two alternative projections of potential crop-yields: one is based on agro-ecological constraints, which could potentially reflect human intervention, and one based on agro-climatic conditions, which are arguably unaffected by human intervention. To prevent concerns of reverse causality, we consider potential yields based on agro-climatic conditions under rain-fed low-input agriculture.

GAEZ provides data on potential yields, in terms of tons per hectare per year, for 11 cereal grains and 4 roots and tubers. Following the same procedure as in Galor and Ozak (2016), we

¹⁵ Although this database is approximately 40 years old, Maurer, Pischke and Rauch (2016) conclude that: "While there has been much additional excavation in the intervening period, there is little reason to believe that it is unrepresentative for the coverage of sites and locations."

¹⁶ The Atlas classifies these sites according to eight different categories: 1. Undifferentiated sites and find-spots; 2. Settlements; 3. Funerary monuments; 4. Religious monuments; 5. Caves and rock shelters; 5. Cave art and rock reliefs; 6. Hoards and votive deposits; 7. Mineral sources; 8. Mineral workings; 9. Sites which combine several of the above categories.

Figure 3: Difference in potential yields (calories per hectare) of cereals versus roots and tubers.



transformed tons to calories using data provided by the USDA National Nutrient Database for Standard Reference,¹⁷ and find the crop with the highest potential caloric yields for each raster point. (We report the results in figure F.9 in the online appendix). Cereal grains are the highest yielding crops in approximately 99 percent of the raster points in the sample, while roots and tubers are optimal in a few very small areas in Siberia, Eastern Brazil and Central-East Africa.¹⁸ From these data we construct two measures: the productivity of land, measured as the maximum potential caloric yield per hectare, and the productivity advantage of cereals over roots and tubers, measured as the difference between the maximum caloric yield of cereals and the maximum caloric yield of roots or tubers. The latter measure is captured in Figure 3.

As robustness checks, we exploit two alternative measures of the productivity of the land, which have been widely used in the literature. The first is an index developed by Ramankutty et al. (2002), which measures the fraction of land that is suitable for agriculture. The second is a caloric suitability index developed by Galor and Ozak (2016), which captures the highest attainable

¹⁷See Table F.4 in the appendix for the complete list of cereal grains, roots and tubers used in the empirical section and the corresponding caloric content.

¹⁸Calculating the “net” potential caloric yield of each crop would require additional data on the caloric cost of cultivating it and procuring eventual complementary inputs. To the best of our knowledge these data are not available. Although, ideally, we would have preferred to work with net yields, we will show in the next subsection that gross yields are still a good predictor of the crop choice.

potential caloric yields from 48 crops (which includes sugar crops, pulses, oil crops, vegetables, fruits, fiber crops and stimulant crops, in addition to cereals, roots and tubers). Table F.5 in the online appendix illustrates that our measure of the productivity advantage of cereals is positively correlated with our benchmark measure of land productivity (the correlation is slightly below 0.8), with the Ramankutty et al. index of suitable land (0.4) and with the Galor and Ozak caloric suitability index (0.8). We also construct a measure of the productivity advantage that comes from using the plow in agriculture. This equals the difference between the maximum caloric yield among crops that Alesina, Giuliano and Nunn (2013) identify as “plow-positive” (wheat, barley and rye) and those that they identify as “plow-negative” (sorghum, foxtail millet and pearl millet).

These productivity variables are attributed to the different societies in the Ethnoatlas by taking an average of their values within a 20-miles radius around the geo-coordinates reported in the Ethnoatlas.¹⁹ They are attributed to countries and the 1x1 decimal degrees raster squares by averaging them within these boundaries.

2.1.6 Wild relatives of domesticated crops

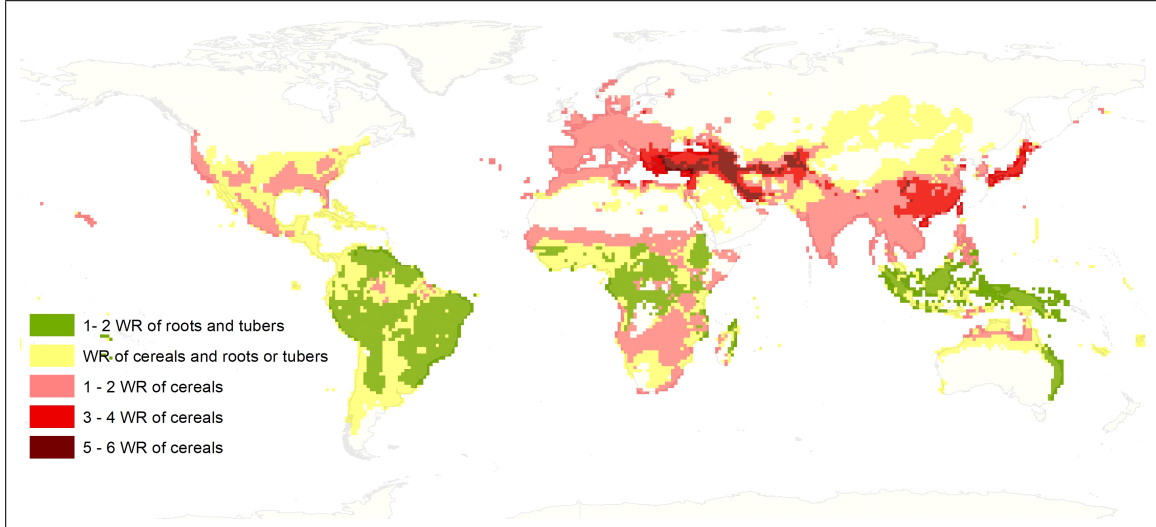
We use resources from the Global Crop Diversity Trust for the potential distribution of wild relatives of domesticated crops.²⁰ Crop wild relatives (WRs) are the wild plants that are genetically related to cultivated crops (i.e., in the same way that the wolf is related to the dog). Consider for example the *Oryza rufipogon*, a wild species that grows in South-East Asia. Rice originated from this plant, which was probably domesticated in China and India around 8,000 to 9,000 years ago (Callaway, 2014). We concentrate on wild plants in the primary gene pool of a domesticated crop: the wild plants that can be directly mated with the relevant domestic crop. We assembled a dataset of their potential geographical distributions.²¹ For our empirical analysis, we assume that the number of primary WRs of a domestic crop in a certain region proxies for the potential for domestication

¹⁹In the appendix we report the result of an alternative method, where we attribute these productivity measures to the different societies by using the maps on their spatial location constructed by Fenske (2013).

²⁰We thank Nora Castaneda-Alvarez for sharing the data with us and her invaluable help.

²¹The maps of the potential distribution of wild relatives are provided by the Global Crop Diversity Trust and are available to download at <https://www.cwrdiversity.org/distribution-map/>. They are constructed with Maxent – a machine-learning algorithm for modeling the distribution of species – using a database of georeferenced occurrence records (Castaneda-Alvarez, 2016) and a grid database of 27 edaphic, geophysical and climatic variables (Vincent et al., 2019). The MaxEnt algorithm is used to compute a probability distribution, determining for each grid cell a predicted suitability for each of the WR species. Under particular assumptions about the biological sampling efforts that led to the occurrence records, a predicted probability of presence is obtained for each cell.

Figure 4: Distribution of wild relatives of domesticated cereals, roots and tubers



of that crop in that region. Based on the data on the potential geographic distribution of these primary WRs, we computed the number of primary WRs of cereals and the number of WRs of roots and tubers in each raster point in our data. Online Appendix E lists all the domesticated crops available in the dataset and their respective WRs. As detailed in the appendix, data on the potential distribution of a limited number of WRs are missing. We constructed a map of the world dividing areas in which only WRs of cereals are available, areas in which only WRs of roots and tubers are available and areas in which WRs of both cereals and roots and tubers are available. The results are presented in Figure 4. As can be seen, the number of WRs can easily predict patterns of early domestication. For instance, the areas with the largest number of WRs of cereals correspond to the Fertile Crescent, the first region in the world that adopted agriculture, and the cradle of the first civilizations.

2.1.7 Other historical, demographic and geographic data

Larson et al. (2014) provide data on the 20 centers in which domestication of at least one plant or animal most likely took place and the list of domesticates in each of these areas (see Figure F.13). We use these data to compute the distance of each raster point in the archeological data set from the closest region of independent adoption of agriculture and from the closest region of independent

domestication of cereal grains. Descriptive statistics on these two variables are reported in columns 11 and 12 in Table F.3.

In addition, data on population density in 1995, precipitation and temperature is provided by GAEZ; the Global Digital Elevation Map (GDEM) provides raster data on elevation and ruggedness; the History Database of the Global Environment (HYDE) provides raster data on global estimates of population density between 1500 and 2000. These data are averaged within societies in the Ethnoatlas, countries and 1x1 decimal degree raster points.

2.2 Pre-industrial societies: 2SLS estimates

According to the appropriability theory, the cultivation of cereals had a causal effect on the emergence of hierarchies. In this subsection we test this prediction with the pre-industrial societies data, surveyed in Murdock’s Ethnographic Atlas. In the online appendix (subsection F.2), we report the determinants of the crop choice in these societies. As expected, cereal grains are cultivated in areas in which they are naturally more productive compared to roots and tubers. Below we report the findings from 2SLS regressions that study the impact of cultivating cereals and land productivity on hierarchical complexity, using the productivity advantage of cereals over roots and tubers as an instrument. We find that the cultivation of cereals leads to more complex hierarchies, while land productivity does not. We further document that cereal cultivation leads to a more burdensome tax system and to the formation of an elite that does not derive its livelihood from subsistence activities.

2.2.1 Cereals and hierarchy

According to the appropriability theory, cereal-based agriculture led to more complex hierarchies. To test this prediction with the Ethnographic Atlas data, we estimate a regression of the form:

$$Y_i = \alpha_1 CerMain_i + \alpha_2 LandProd_i + X_i' \beta + u_i, \quad (1)$$

where Y_i is a measure of hierarchy in society i ; $CerMain_i$ is a dummy variable that identifies societies that rely mainly on cereals for their subsistence; $LandProd_i$ is a measure of land productivity,

and X_i' is a vector of control variables.²²

This specification, however, raises two concerns. First, the choice of crop might be influenced by social institutions. To overcome this reverse causality concern, we exploit variations in potential, rather than actual, crop yields, which are derived from agro-climatic conditions that are presumably orthogonal to human intervention. Specifically, we run 2SLS regressions, where we instrument for $CerMain_i$ by using the productivity advantage of cereals, $CerAdv_i$ (i.e. the difference between the maximum potential caloric yield of cereals and of roots or tubers under a rain-fed subsistence agriculture). The first-stage is then:

$$CerMain_i = \beta_1 CerAdv_i + \beta_2 LandProd_i + X_i' \beta + \varepsilon_i. \quad (2)$$

The caloric advantage of cereals is constructed using the procedure detailed in section 2.1.5, under the assumption that the Columbian Exchange of crops between the Old World and the New World was completed and all continents had access to all major crops in our data.²³ The variables $LandProd_i$ and $CerAdv_i$ are based on productivity estimates for different crops, which are constructed from agro-climatic conditions that are unaffected by human intervention. These estimates capture potential productivity under subsistence rain-fed agriculture and not actual productivity. This is an important distinction as actual productivity is likely to depend on factors that are endogenous to local hierarchy.

Second, there are several potential omitted variables that could be correlated with the main regressor and the measure of hierarchy. The disease environment, for instance, is correlated with both the cultivation of tubers (which is concentrated in the tropics) and is likely to be correlated with the quality of institutions (Acemoglu, Johnson and Robinson, 2001). A battery of robustness checks mitigates this concern. In two of the following subsections, we utilize an alternative dataset to conduct panel regressions that alleviate concerns regarding potential time-invariant omitted variables.

²²To correct for spatial autocorrelation of the error term we estimate the standard errors in three different ways: we allow either for clustering at the country level or for spatial autocorrelation with a distance cutoff of 5 or 10 decimal degrees. Results are consistent across the three choices. In the online appendix, we only report the most restrictive choice (autocorrelation with 10 decimal degree cutoff).

²³This is a reasonable assumption as the great majority of the societies in the Ethnoatlas are captured between the end of the 19th century and the beginning of the 20th century.

In the online appendix, we report the reduced form relationship. First, Figure F.10 reports the box plot of the productivity advantage of cereals corresponding to each level of hierarchy: the productivity advantage of cereals is generally larger in societies organized as states compared to societies organized as either chiefdoms or tribes. Table F.9 reports the reduced form estimates. Column 1 shows a significant correlation between (potential) land productivity and the level of jurisdictional hierarchy in the societies in the Ethnoatlas, as predicted by the conventional productivity-and-surplus theory. Column 2 illustrates a significant correlation between the productivity advantage of cereals and hierarchy as predicted by the appropriability theory. Once both regressors are included (column 3) the effect of the productivity advantage of cereals remains positive and significant and the effect of land productivity disappears. Specifically, an increase of one standard deviation in the productivity advantage of cereals increases the hierarchy index by 0.27. The qualitative nature of this result is unchanged when using a logit or a generalized logit rather than OLS to estimate the reduced form.²⁴

Table 1 reports the OLS and 2SLS estimates of equation 1, when the dependent variable is hierarchy. The OLS estimates in column 1 show that cultivating cereals is associated with an increase of 0.70 in the hierarchy measure. The 2SLS estimates are presented in the next five columns. Panel A reports the second-stage estimates, while Panel B reports the first-stage. In the latter, an increase in the productivity advantage of cereals over roots and tubers by one standard deviation is associated with an increase in the probability of growing cereals as the main crop by about 20 percent. Reassuringly the F-statistics on the excluded instrument is between 49.3 and 75.9 (depending on the methodology to estimate the standard errors) and the instrument alone is able to explain 13 percent of the variation in $CerMain_i$. Online Appendix F2 provides a detailed discussion of the first-stage relationship with a long-list of robustness checks and additional results

²⁴Table F.10 in the online appendix reports the estimates from an ordered logit (column 1) and a generalized ordered logit model (columns 2-5) to account for the ordinal nature of the dependent variable. The estimates reported in column 1 shows that a one standard deviation increase in the productivity advantage of cereals increases the log odds of being in a higher level of hierarchy by approximately 50 percent. This estimate is based on the assumption of proportional odds (i.e. each independent variable has an identical effect at each cumulative split of the ordinal dependent variable). Columns 2-5 present the estimates of a generalized logit model, which relaxes the assumption of proportional odds. As can be seen, the greatest impact of cereal advantage is to push societies from tribes and chiefdoms to states. More specifically, while an increase in one standard deviation in the productivity advantage of cereals increases the log odds of being in a level of hierarchy higher than a tribe by 32 percent, it increases the log odds of being in a level higher than a chiefdom by 65 percent and higher than a small state by 84 percent. In all cases, the impact of land productivity is either very small and not statistically significant, or negative.

Table 1: Cereals and Hierarchy - OLS and 2SLS

PANEL A: 2ND STAGE		Dep variable: Jurisdictional Hierarchy Beyond Local Community				
	(1) OLS	(2) 2SLS	(3) 2SLS	(4) 2SLS	(5) 2SLS	(6) 2SLS PDS
<i>CerMain</i>	0.707 {0.114}*** [0.097]*** (0.131)***	1.170 {0.352}*** [0.292]*** (0.359)***	0.892 {0.447}** [0.352]** (0.420)**	1.064 {0.556}* [0.459]** (0.538)**	0.830 {0.474}** [0.426]* (0.511)	0.797 {0.378}** - -
<i>LandProd</i>				-0.037 {0.086} [0.067] (0.071)		-
Dependence on Agriculture					0.259 {0.402} [0.398] (0.478)	-
CONTINENT FE	NO	NO	YES	YES	YES	-
N	952	952	952	952	952	877
F excl instrum.		52.15 [74.90] (49.34)	33.13 [52.50] (34.76)	13.06 [29.20] (19.70)	11.43 [37.83] (23.18)	16.11
R-squared	0.793					
PANEL B: 1ST STAGE						
<i>CerAdv</i>		0.209 {0.029}*** (0.024)*** (0.029)***	0.155 {0.027}*** (0.021)*** (0.026)***	0.258 {0.071}*** (0.047)*** (0.059)***	0.291 {0.068}*** (0.021)*** (0.027)***	0.299 {0.067}***

The table reports cross-sectional OLS, 2SLS, and 2SLS PDS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Panel A reports the main estimates, while panel B reports the first stage estimates (only the estimated coefficient of the excluded instrument). Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors in curly brackets are clustered at the country levels; standard errors in square brackets and round brackets are adjusted for spatial correlation using the methodology in Conley (1999) with a distance cut-off of respectively 5 and 10 decimal degrees. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

on the effect of geography on the decision to farm.

Turning to the second stage, the baseline 2SLS estimates predict that cultivating cereals as the main crop increases the hierarchy measure by more than one (column 2), which is equivalent, for instance, to a move from a tribe to a small chiefdom or from a large chiefdom to a state. We note that the 2SLS coefficient on cereals is *larger* than the OLS one.²⁵ In column 3, we show that results

²⁵There could be two explanations. First, the measurement error in the cereal variable is likely to downward bias the OLS estimates. Second, and more important, while OLS estimates describe the average difference in hierarchy between societies cultivating cereals and the rest, the IV estimates measure the effect of growing cereals only for

are qualitatively unchanged when adding continent fixed effects. Column 4 adds land productivity as a control variable. As consistent with the reduced form regressions, land productivity does not have any significant effect on hierarchical complexity. Column 5 includes a control for the dependence of the society on agriculture. Results indicate that societies that practice agriculture are not characterized by more complex hierarchies unless they cultivate cereals.

The 2SLS results in Table F.11 survive a battery of robustness checks, which we discuss in details in the online appendix. We provide here a brief summary. In Table F.11, we control sequentially for precipitation, temperature, elevation, ruggedness and latitude/longitude— the main geographical factors affecting crop productivity. In Table F.12, we control sequentially for geographical isolation, historical population density, potential for intensive irrigation, and the productivity advantage of the plow – the main factors that political scientists and economists have associated with the rise of complex hierarchies. Finally, Table F.16 adds sequentially controls for animal husbandry and animal use in agriculture to assure that our results are not just capturing the role of animal domestication in the development of the state.

Column 6 in Table 1 reports the estimation results when adding all the controls mentioned above, including the continent fixed effects, at the same time in a post-double-selection (PDS) 2SLS methodology.²⁶ The estimated coefficient for $CerMain_i$ is still positive and statistically significant, although somehow smaller compared to the coefficient estimated without this long list of controls.²⁷

Finally, the qualitative results are maintained (the coefficient varies between 0.750 and 1.471)

societies whose choice of crop is affected by the instrument -the potential caloric advantage of cereals over roots and tubers (this is the local average treatment effect, or LATE). This is relevant, as there are several societies, especially in islands and in the most remote areas of the world, for which the choice of the crop is dictated by their availability and not by their potential caloric advantage over other crops. To understand whether this might be responsible for the lower OLS coefficients compared to the IV coefficients, we re-estimate the first two columns of Table 3 only for societies living in areas in which wild relatives of both cereals and roots/tubers are present. The result is that the OLS estimates are now larger than the IV estimates. Specifically, in the regression without controls, the OLS coefficient on cereal cultivation is 1.23 while the IV coefficient is 1.11.

²⁶This methodology is used for estimating structural parameters in linear models with many controls. A square-root-lasso is used to identify the relevant controls in predicting the dependent variable, the main regressor and the instrument. The union of the variables selected from these reduced forms is then included in the 2SLS estimates. See Belloni, Chernozhukov and Hansen (2014) for details. We use for estimation the Stata package `pdslaso` (Ahrens, Hansen and Schaffer, 2019). As far as we know, there is no previous study in which the PDS methodology has been used in conjunction with Conley’s standard errors (and no available routine in any statistical software to do so). We therefore only provide standard errors clustered at the country-level when using PDS.

²⁷The PDS procedure only produces a consistent estimate for the main variable of interest and not for the controls. Therefore, we do not report in column 6 the coefficient on land productivity, though it is included in the controls.

when using ethnic boundaries as defined by Fenske (2013) to extract data on crops productivity (Table F.14), when the sample includes societies living in desertic soils (Table F.13), or when using either the Ramankutty et al. index of fertile land or the Galor and Ozak index of caloric suitability as alternative measures of land productivity (Table F.15).

2.2.2 The appropriability mechanism: evidence on taxation and farming surplus

In this subsection, we show that the cultivation of cereals is correlated with the existence of a farming surplus and with a tax burden (both variables are described above), as consistent with the appropriation mechanism. Reduced-form estimates are reported in Table F.17, in the online appendix. In columns 1-4, the correlation between the productivity advantage of cereals over roots and tubers (indicating that cereals are the staple crop) and the presence of a farming surplus is positive and statistically significant. This is true both in OLS and logistic regressions and the result is robust to the inclusion of continent dummies. Land productivity is not statistically correlated with the presence of a farming surplus (columns 2-4). In columns 5-8, we show that the productivity advantage of cereals is also positively correlated with the burden of local taxes. This correlation is generally statistically significant at a conventional level. A notable exception is in column 7, in which we control for continent fixed effects and the estimated coefficient on the cereal advantage loses statistical significance. This should not come as a surprise as tax regressions are based on only 56 observations and the number of observations from each continent is limited. Finally, land productivity in all specifications isn't significantly correlated with the tax burden.

In Table F.18 in the online appendix we report the OLS and 2SLS estimates. In columns 1-4, we look at the impact of cultivating cereals on the existence of a farming surplus. The OLS estimates show that cultivating cereals is associated with an increase of 0.36 in the probability of producing a surplus. The coefficient more than doubles when turning to the 2SLS estimates; in this case too, land productivity (and reliance on agriculture) does not affect the dependent variable. Finally, results are robust to adding continent fixed effects.

In columns 5-8, we turn to the impact of cultivating cereals on the tax burden. In the 2SLS estimates cultivating cereals is associated with an increase in the tax burden variable in the order of 1. This is a large number compared to the mean of the dependent variable (1.09) and suggests that taxation was practically not possible in non-agricultural societies nor in societies cultivating roots

or tubers. Again, land productivity does not seem to have any effect on the dependent variable. This result is robust to adding continent fixed effects and it survives the same robustness checks that we run for Table 1 (see Tables F.19-F.21 in the online appendix).

The cross-sectional nature of the regressions and the limited number of observations, however, imply that these results should be taken with a grain of salt and might be compatible with other mechanisms, discussed in section 4, through which cereal cultivation might be fostering the development of complex hierarchies.

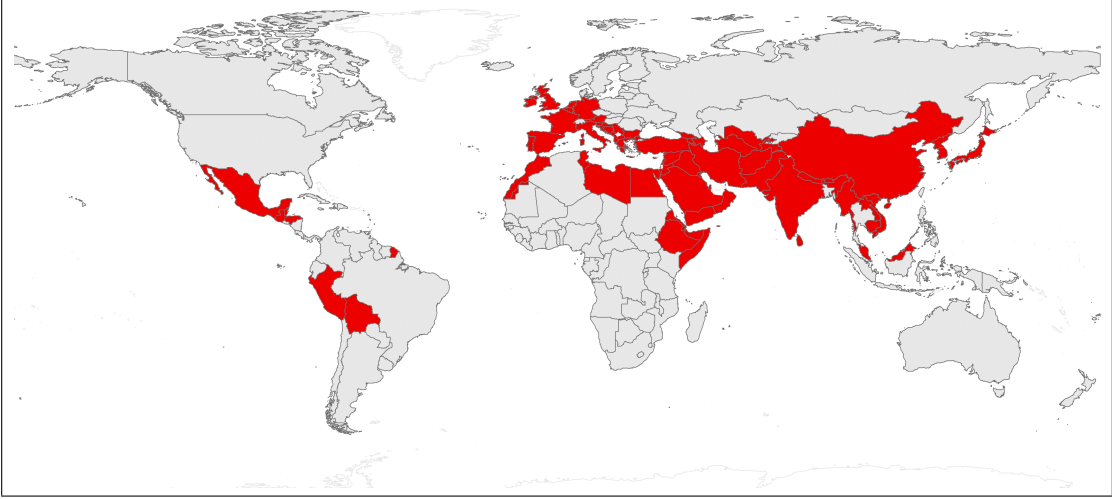
2.3 Country-level data: cross-section and panel estimates

In this subsection we use data on hierarchical complexity from Borcan et al. (2018). The unit of observation is the territory delimited by modern-day country borders for 159 countries every 50 years. The data allows for panel estimates and thereby it isn't subject to the limitation of the cross-sectional nature of the Ethnographic Atlas data employed in the previous subsection.

We first look at the classical age – the earliest period in human history for which detailed and complete data on the location of large states are available. We show, using cross-sectional variation, that regions that were organized as states in AD 450 are characterized by the presence of several wild relatives (WRs) of cereals, while WRs of roots and tubers are absent. Alternatively, using FAO data, we show that regions in which cereals are substantially more productive than roots and tubers are the regions that are organized as states. Both the presence of WRs of roots and tubers and land productivity are uncorrelated with the presence of states. These results persist when controlling for a large number of potential confounders and suggest that adopting cereals is fundamental for the emergence of states. However, the cross-sectional nature of the analysis does not allow ruling out that omitted variables might be driving the results.

We therefore turn to a natural experiment of history – the Columbian Exchange – and exploit the panel nature of the dataset. The Columbian Exchange of crops between the Old World and the New World permanently changed the productivity advantage of cereals over roots and tubers and land productivity in virtually every region of the world. We show that only the former change can explain subsequent changes in hierarchical complexity across different regions as consistent with both the appropriability theory and our critique of the conventional productivity theory. The analysis is robust to controlling for a large number of potential confounders, and the main results

Figure 5: Modern countries areas that were mainly organized as states in 450 AD



are not explained by colonization patterns or pre-trends.

2.3.1 Explaining differences in hierarchy during the Classical Age

Figure 5 identifies the areas corresponding to modern-day countries that were organized (for the majority of their current territory) as states at the peak of the classical age, just before the collapse of the Roman Empire (AD 450). We focus on this period because the exact location of state borders is less certain for the pre-classical period. We use a measure of the potential availability of each domesticated crop, based on the information on the distribution of its wild relatives (WRs) in the world, which is presumably unaffected by the development of hierarchy and, therefore, alleviates reverse causality concerns.

According to our thesis, states develop where cereals are farmed, and cereals are farmed where they are available and roots and tubers are not. This prediction is consistent with the data. By comparing Figure 5 to Figure 4, which shows the distribution of WRs of different domesticated crops, it is apparent that civilization started exactly in those places characterized by a large number of WRs of cereals, but with no WRs of roots and tubers. The Fertile Crescent is the most prominent example. Table 2 reports the estimated coefficients from the following regression:

$$Hierarchy_i^{450} = \alpha_1 I(WR_Cer)_i + \alpha_2 I(WR_RT)_i + \alpha_3 I(WR_Cer + WR_RT)_i + X_i' \beta + u_i. \quad (3)$$

The hierarchy index in AD 450 ($Hierarchy_i^{450}$) is regressed on a dummy that identifies areas with only WRs of cereals available ($I(WR_Cer)_i$), a dummy that identifies areas with only WRs of roots and tubers ($I(WR_RT)_i$), a dummy that identifies areas in which both WRs of cereals and of roots and tubers are available ($I(WR_Cer + WR_RT)_i$), and a vector of control variables (X_i'). The control group is composed of areas in which no WR of domesticated crops is available. The presence of only WRs of roots and tubers and the presence of both WRs of roots and tubers and of cereals are not correlated with the hierarchy index. In contrast, regions in which only WRs of cereals are available are characterized by an increase in the hierarchy index of 0.50. This is a large effect: the average of the dependent variable is 0.38. Moreover, the R^2 of this regression is 0.3, showing that the availability of WRs of cereals and the lack of WRs of roots or tubers, could explain almost a third of cross-regional differences in hierarchy in AD 450. Results are robust to adding continent dummies (column 3). Figures 5 and 4 suggest that latitude is correlated with both $Hierarchy_i^{450}$ and $I(WR_Cer)_i$. However, controlling for differences in latitude, leaves the estimated coefficient on $I(WR_Cer)_i$ practically unaffected (column 4). Results are practically unaffected when adding sequentially a long-list of controls. In columns 4-8 in Table 2, we control for precipitation, temperature, elevation and ruggedness. In Tables F.23 and F.24, in the online appendix, we consider a host of additional factors that might have impacted hierarchical complexity. Our choice of controls is driven by the determinants of long-term economic development that have been emphasized in the literature: legal origin of the country, population density in 1500, settlers' mortality, the number of exported slaves, genetic diversity, distance to rivers and coast, endemicity of malaria, the percentage of tropical land, the density of wild animals and ruminants suitable to domestication, and the potential for irrigation. Column 9 in Table 2 reports the results of adding this long list of controls at the same time in a post-double-selection methodology (PDS). As can be seen, the estimated coefficient for $I(WR_Cer)_i$ is smaller but still positive and statistically significant. We do not report in the table the coefficients on WR_RT and WR_Cer and RT , as the PDS procedure only produces a consistent estimate for the main variable of interest and not

for the controls. Finally, in Table F.25 in the online appendix we show that results are robust to excluding each continent one-by-one, confirming that it is not a specific region of the world that is driving our key results.

Table 2: Cereals and Hierarchy in Classical Antiquity - Cross-sectional regressions

	Dep. Variable: Hierarchy Index in AD 450								
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS	(7) OLS	(8) OLS	(9) PDS
<i>WR Cer</i>	0.535*** (0.0655)	0.526*** (0.0989)	0.465*** (0.124)	0.505*** (0.118)	0.433*** (0.129)	0.462*** (0.121)	0.423*** (0.136)	0.487*** (0.117)	0.305* (0.173)
<i>WR RT</i>		0.125 (0.174)	0.182 (0.173)	0.196 (0.170)	0.168 (0.172)	0.204 (0.174)	0.123 (0.170)	0.222 (0.172)	-
<i>WR Cer&RT</i>		-0.0319 (0.0918)	0.0623 (0.114)	0.0568 (0.114)	0.0709 (0.112)	0.0304 (0.112)	0.0447 (0.126)	0.123 (0.114)	-
Controls:									
Abs Latitude	NO	NO	NO	YES	NO	NO	NO	NO	-
Precipitation	NO	NO	NO	NO	YES	NO	NO	NO	-
Temperature	NO	NO	NO	NO	NO	YES	NO	NO	-
Elevation	NO	NO	NO	NO	NO	NO	YES	NO	-
Ruggedness	NO	NO	NO	YES	NO	NO	NO	YES	-
CONT FE	NO	NO	YES	YES	YES	YES	YES	YES	-
r ²	0.305	0.310	0.408	0.435	0.418	0.408	0.428	0.402	
N	151	151	151	151	151	150	148	145	73

The table reports cross-sectional OLS and PDS estimates and the unit of observation is the territory delimited by modern-country borders. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

We further illustrate how cereals can explain the distribution of states in classical antiquity in a box plot (Plot F.11 in the appendix). The plot shows that states emerged in areas in which the difference between the number of WRs of cereals and the number of WRs of roots and tubers is large: the median is 3 and 75% of states are in the range of 2-4. The median difference in areas defined as tribes is zero with 75% of tribes in the range of 0-1.

In Table F.26, in the online appendix, we turn to the potential productivity of crops (GAEZ data set) and estimate an equivalent of equation (2) at the country-level for the year AD 450. Here we confirm the results of Table 2: there is a significant positive correlation between the productivity advantage of cereals and hierarchy, as predicted by the appropriability theory, while land productivity is negatively correlated with hierarchy, in contrast to the conventional productivity theory.²⁸

²⁸In this table, the productivity advantage of cereals and land productivity are calculated using only the subset of

2.3.2 Panel data based on the Columbian Exchange

The analysis based on the cross section of states in AD 450 accounts for a large set of possible confounding factors, but we cannot rule out that unobservable characteristics that are systematically correlated with the productivity or availability of different crops might be driving our results. To alleviate this concern, we exploit the exogenous change in the set of available crops in different locations of the world that was induced by the Columbian Exchange.

Among the main four roots and tubers that we consider, three were available in the New World before 1500: cassava, white potatoes and sweet potatoes. Among the eleven main cereals, only maize was available in the New World. In the Old World, yam was the only available crop from the four main roots and tubers, while all cereals but maize were available. Accordingly, we compute for each location the productivity advantage of cereals over roots and tubers and land productivity before the Columbian Exchange (prior to 1500), based on the relevant subset of crops, and after the exchange (after 1550), based on the full set of crops.²⁹

The benchmark sample used in this subsection comprises 151 countries, for which the hierarchy data constructed by Borcan et al. (2018) and the crop productivity data are available. We use the years 1000-1950, with observations available every half century, but we exclude the half century 1500-1550. This leaves us with a total of 2,869 observations. We regress the hierarchy index on the productivity advantage of cereals and on land productivity:

$$Hierarchy_{it} = \alpha_1 CerAdv_{it} + \alpha_2 LandProd_{it} + X'_{it}\beta + \eta_i + \eta_t + u_{it}. \quad (4)$$

The dependent variable is the hierarchy index of country i in year t , while $CerAdv_{it}$ = $CerAdv_{i,BeforeExchange}$ (the caloric advantage of cereals over roots and tubers before the Columbian Exchange) if $t \leq 1500$ and $CerAdv_{it} = CerAdv_{i,AfterExchange}$ (the caloric advantage after the Columbian Exchange) if $t > 1550$. Similarly, potential land productivity ($LandProd_{it}$), is calculated based on the pertinent crops available before and after the Columbian Exchange. X_{it} is a set

cereals and roots/tubers that were available in the Old World and in the New World before the Columbian Exchange.

²⁹The historical evidence points out that the New World's crops were adopted in Europe and Africa only in the seventeenth century. For instance, potato cultivation in the Old World began in the late seventeenth century by Irish peasants (Nunn and Qian, 2011), while the first accounts on the adoption of maize in Africa date back to the very end of the sixteenth century (Miracle, 1966). In the benchmark analysis, we exclude the years from 1500 to 1550. In appendix F we show that our results are robust when excluding the years between 1500 and 1750 (Table F.32).

of control variables. Country fixed effects control for all time invariant factors that differ between countries, and time period fixed effects control for any time pattern of hierarchical complexity that affects all countries simultaneously. The critical identification assumption is that there were no unobserved events in the sixteenth century that are systematically correlated with the spatial variation in the change in the potential productivity advantage of cereals, and that had an independent effect on hierarchy.

We are aware that the change in crop availability induced by the Columbian Exchange coincided with colonization. However, colonization does not seem to be driving our results: excluding colonies from the estimation sample doesn't have a quantitative effect on the estimates (discussed below). Moreover, the concern that changes in hierarchy were a result of colonization rather than changes in the availability of crops, cannot explain the different impact of the changes in cereal advantage and in changes in land productivity that we observe in Table 3.

Table 3: Cereals and Hierarchy - Panel Regressions

	Dep. Variable: Hierarchy Index								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	PDS
<i>CerAdv</i>		0.189*** (0.0683)	0.272*** (0.0834)	0.282*** (0.0760)	0.240*** (0.0857)	0.255*** (0.0889)	0.261*** (0.0839)	0.197** (0.0795)	0.198** (0.0844)
<i>LandProd</i>	0.141 (0.0971)		-0.163 (0.141)	-0.193 (0.131)	-0.152 (0.139)	-0.115 (0.142)	-0.148 (0.138)	-0.165 (0.123)	-
Controls									
x Year FE:									
Precipitation		NO	NO	YES	NO	NO	NO	NO	-
Temperature		NO	NO	NO	YES	NO	NO	NO	-
Elevation		NO	NO	NO	NO	YES	NO	NO	-
Ruggedness		NO	NO	NO	NO	NO	YES	NO	-
Abs Latitude		NO	NO	NO	NO	NO	NO	YES	-
Country. FE	YES	YES	YES	YES	YES	YES	YES	YES	-
TIME FE	YES	YES	YES	YES	YES	YES	YES	YES	-
r ²	0.669	0.680	0.682	0.716	0.684	0.681	0.686	0.705	
N	2869	2869	2869	2850	2812	2755	2869	2869	1387

The table reports panel OLS and PDS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Column 1 in table 3 shows a positive but insignificant effect of land productivity on hierarchy, when cereal advantage is not controlled for. The sign of this coefficient turns negative but not

significant once the cereal advantage is included in the regression. Column 2 confirms that the higher is cereal advantage, the higher is the country’s hierarchy index. A one standard deviation increase in the productivity advantage of cereals increases the hierarchy index by 0.19. In the next six columns, we show that the results are robust when controlling for land productivity, and in addition also for precipitation, temperature, elevation, ruggedness and absolute latitude (all interacted with the time-period fixed effects). In Tables F.27 and F.28, in the online appendix, we consider a host of additional factors (interacted with time-period fixed effects) that might have impacted hierarchical complexity. In particular, as in the previous subsection, we control sequentially for legal origin, population density in 1500, settlers’ mortality, exported slaves, genetic diversity, distance to rivers and coast, endemicity of malaria, the percentage of tropical land, the density of wild animals and ruminants suitable for domestication and the potential for irrigation. The key results are essentially unaffected. Column 9 in Table 3 reports the results of adding all these controls at the same time in a post-double-selection methodology: once again, results are practically unaffected compared to the specification without any control. (In this column we report only the coefficient on our main regressor. Land productivity enter as one of the controls in the PDS).

In Table F.29 in the online appendix we exclude the cells in which the countries in our analysis were either colonies or protectorates. The estimated coefficient on the caloric advantage of cereals over roots and tubers become smaller by approximately a third, but remains positive and statistically significant, while the impact of land productivity on hierarchy is still not significant. Table F.30 and F.31 in the appendix report further robustness checks. Specifically, in Table F.30, hierarchical complexity is proxied by a dummy that identifies societies with a government above tribal level. In Table F.31, land productivity is proxied by the caloric suitability index developed by Galor and Ozak (2016), which also varies depending on whether it is measured before or after the Columbian Exchange. Finally, in Table F.32, we exclude the years between 1500 and 1750, when the Columbian Exchange of crops was not complete. In all three cases, our main results are unaffected.

Testing for pre-trends The identification assumption of equation (4) requires that, until 1500, $Hierarchy_{it}$ did not follow systematically different trends across the set of countries that expe-

experienced different shocks in the productivity advantage of cereals after 1500. Here we show the existence of parallel linear and non-linear pre-1500 trends.

We first use data on the years before the Columbian exchange and regress $Hierarchy_{it}$ on the change in the caloric advantage generated by the Columbian Exchange ($Change\ CerAdv_j$) interacted with a linear trend and a set of country and year dummies (Table F.33 in the online appendix). The coefficient on the interaction term indicates whether hierarchy in countries that experienced a larger cereal advantage shock were on a different linear trend before the Columbian exchange. The estimated coefficient is always small and not statistically significant. Similar results are obtained if we control for the interaction between the potential change in land productivity due to the Columbian Exchange with a linear trend and if we control for the usual geographic characteristics interacted with year fixed effects.

Second, we regress $Hierarchy_{it}$ on year fixed effects interacted with the change in the caloric advantage generated by the Columbian Exchange, year fixed effects and country fixed effects:

$$Hier_{it} = \sum_{j=1050}^{1850} \alpha_j * (Change\ CerAdv)_i * j + X'_{it}\beta + \eta_i + \eta_t + u_{it}. \quad (5)$$

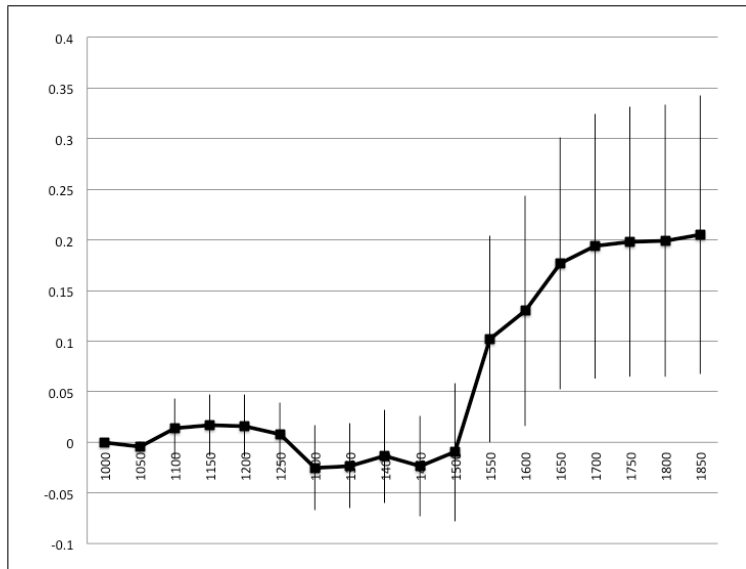
This specification does not require any assumption about the timing of the Columbian Exchange and considers the year 1000 as the baseline year. Results are presented in Table F.34. In the first column, there are no further control variables. The estimated α_j s and their 10 percent confidence intervals are reported in Figure 6.³⁰

The impact of the change in the productivity advantage of cereals over roots and tubers enabled by the Columbian Exchange is constant over time between 1000 and 1500; it increases steadily during the sixteenth century and continues to increase, but at a lower rate until 1700, after which it stabilizes. Results are practically unchanged when controlling for the interaction between the potential change in land productivity with year fixed effects and the usual geographic characteristics interacted with year fixed effects.

This analysis confirms that the Columbian Exchange produced a differential increase in hierarchy in the countries for which it caused a larger increase in the productivity advantage of cereals

³⁰The 17 coefficients reported in Figure 6, can also be described as the estimated coefficients in 17 independent cross-country regressions, in which we regress the change in the hierarchy index between each of the 17 years in the sample (1050, 1100, ..., 1850) and the year 1000 on the change in the caloric advantage of cereals over roots and tubers caused by the Columbian exchange.

Figure 6: Flexible estimates of the relationship between the change in the caloric advantage of cereals over roots and tubers due to the Columbian exchange and hierarchy.



over roots and tubers, and that most of the impact was in the sixteenth century. It also rules out the possibility that non-linear pre-trends might be driving our results.

2.4 Early traces of civilization: cross section of archaeological sites

The results presented in the two previous subsections support our thesis based on data from AD 450 and more recent centuries. We now turn to data that covers the era from pre-Neolithic sites to classical cities. With these data, we connect indications of early civilization with the domestication of cereals and roots or tubers. In this subsection, we present cross-section results based on the location of classical and pre-classical large settlements or archaeological ruins, with the presumption that the existence of a city or a large settlement is an indicator of hierarchy. We show that these archeological findings are concentrated in areas in which agriculture was likely to start only with cereal crops. In particular, these are areas that are characterized by a large number of wild relatives (WRs) of domesticated cereal crops, but where WRs of roots and tubers are not available. Alternatively, these are areas close to centers of independent domestication of cereals.

Table 4 employs two different datasets. The first is the dataset provided by Daniel DeGroff on

the location of cities founded before AD 400 in the Old World (columns 1-4). The second dataset comes from Reitsma and Seto (2016) and refers to the location of large settlements present in classical antiquity (AD 450) and pre-classical antiquity up to 500 BC (columns 5-8). We use a grid of the world land surface, in which the unit of observation is the 1x1 decimal degree raster, to test our thesis. We run regressions of the form:

$$Settlement_i = \alpha_1 I(WR_Cer)_i + \alpha_2 I(WR_RT)_i + \alpha_3 I(WR_Cer + WR_RT)_i + X_i' \beta + u_i. \quad (6)$$

Table 4: Wild Relatives of Domesticated Crops and the Location of Ancient Cities.

	Dependent variable is the presence of cities/large settlements founded by:								
	AD 400					AD 450		500 BC	
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) PDS	(6) OLS	(7) PDS	(8) OLS	(9) PDS
<i>WR Cer</i>	0.197*** (0.0325)	0.195*** (0.0326)	0.195*** (0.0377)	0.0965*** (0.0280)	0.145*** (0.0268)	0.0232*** (0.00500)	0.0136*** (0.00434)	0.00941*** (0.00264)	0.00358 (0.00310)
<i>WR RT</i>		-0.00809 (0.00700)	0.00478 (0.0145)	-0.0277 (0.0232)	-	-0.00243 (0.00157)	-	-0.00179 (0.00133)	-
<i>WR Cer&RT</i>		-0.00901 (0.00694)	0.0245 (0.0201)	-0.00191 (0.0221)	-	-0.00307** (0.00142)	-	-0.00244** (0.00115)	-
CONT. FE	NO	NO	YES	NO	-	NO	-	NO	-
Country FE	NO	NO	NO	YES	-	NO	-	NO	-
r ²	0.124	0.124	0.144	0.407		0.0125		0.0398	
N	17076	17076	17076	17076	8568	17076	8568	17076	8568

The table reports cross-sectional OLS and PDS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Column 1 shows that the availability of only WRs of domesticated cereals is associated with an absolute increase in the probability of having a city in a cell in classical antiquity of 19.7 percentage points. The magnitude of this coefficient is exceptional compared to the mean of the dependent variable (0.049) and is explained by the fact that almost all cities in the sample are located in areas in which WRs of cereals are present and WRs of roots and tubers are absent.³¹ In column 2, the presence of an ancient city is regressed on all the three main regressors in equation (6). The

³¹The probability of observing an ancient city in the cells located in areas in which either WR of cereals are not available or in which both WR of cereals and WR of roots and tubers are available is 0.013. If instead we look at cells in which only WR of cereals are available, the probability of observing an ancient city is 0.210.

estimated coefficient on $I(WR_Cer)_i$ does not change. Moreover, consistent with our critique, we find that the presence of ancient cities is not different between areas in which WRs of roots and tubers are present (regardless of the existence of WRs of cereals) and areas with no WRs of any domesticated crops. This result is practically unchanged when controlling for either continent or country fixed effects (columns 3 and 4). In Table (F.37) in the appendix we report results when controlling sequentially for precipitation, temperature, elevation, ruggedness, absolute latitude, irrigation potential and plow advantage. Our estimates for the coefficient on $I(WR_Cer)_i$ are substantially unchanged although in some specifications the coefficient on $I(WR_Cer+WR_RT)_i$ becomes positive and statistically significant. Column 5 of Table 4 reports the PDS regression adding all these controls at the same time: the estimates on $I(WR_Cer)_i$ is similar compared to the specification without controls. In the second part of the table we use data on the presence of large ancient settlements coming from Reitsma and Seto (2016). In columns 6 and 7, we look at settlements that were established before AD 450. The qualitative results are the same as in columns 1-5. In the last two columns, we move back in time to pre-classical antiquity and find the same qualitative results. The estimated coefficients on $I(WR_Cer)_i$ are approximately half compared to the previous columns consistent with the substantially smaller mean of the dependent variable. Our results are robust to using a logit model to account for the binary nature of the dependent variable (Table F.36 in the appendix).

We turn to data on centers of domestication during the Neolithic as an alternative proxy for the cultivation of cereals or roots/tubers. The underlying assumption is that the probability that a crop would reach a certain area would be negatively associated with the geographic distance to the nearest center of domestication of that crop. Therefore, raster points that are geographically close to centers where cereals were first domesticated, relative to the distance to areas of tuber domestication, would be more likely to adopt cereal farming, and thus, as the appropriability theory predicts, are more likely to develop hierarchies.

Global data on the diffusion of crops during the Neolithic transition are not available, but archaeologists and botanists have identified some 20 centers in which independent domestication took place and from which domesticated crops spread to the rest of the world (see map E.5 and Larson et al., 2014). We use these data to compute the distance to the nearest center of independent domestication of cereals and roots/tubers. A box plot (figure F.12 in the appendix) shows that the

vast majority of the raster points that have a large settlement are within 2,000 km of a center of cereal domestication, with the median less than 1000 km. Their distance from a center of root/tuber domestication is much larger and comparable to the distance of a raster point without settlements from areas of any domestication (cereals or roots/tubers).

Table (5) report OLS estimates from the following regression equation:

$$Settlement_i = \alpha_1 DistanceCer_i + \alpha_2 DistanceAgr_i + X'_{it}\beta + u_{it}, \quad (7)$$

where $DistanceAgr_i$ is the distance to the nearest center of independent domestication of agriculture and $DistanceCer_i$ is the distance to the nearest center of independent domestication of cereal grains.

Table 5: The Origin of the Neolithic Transition and the Location of Ancient Cities.

	Dependent variable is the presence of cities/large settlements founded by:								
	AD 400					AD 450		500 BC	
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) PDS	(6) OLS	(7) PDS	(8) OLS	(9) PDS
<i>DistCer</i>	-0.160*** (0.0342)	-0.214*** (0.0597)	-0.143** (0.0604)	-0.0269 (0.0198)	-0.335*** (0.0673)	-0.0303*** (0.00943)	-0.0456*** (0.0120)	-0.0168*** (0.00585)	-0.026*** (0.00808)
<i>DistAgr</i>		0.0909 (0.0676)	0.0253 (0.0566)	0.000138 (0.0102)	-	0.000595 (0.00743)	-	-0.00319 (0.00423)	-
CONT. FE	NO	NO	YES	NO	-	NO	-	NO	-
Country FE	NO	NO	NO	YES	-	NO	-	NO	-
r ²	0.0284	0.0307	0.0495	0.446		0.00674		0.00460	
N	15927	15927	15927	15927	8169	15927	8169	15927	8169

The table reports cross-sectional OLS and PDS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

The estimates illustrated in Table (5) show that distance from the nearest area of independent domestication of a cereal grain, $DistanceCer_i$, is negatively correlated with urbanization. Moreover, when controlling for $DistanceCer_i$, the distance from the nearest area of independent adoption of agriculture is not correlated with urbanization, thus suggesting that farming crops different than cereals might be irrelevant to the development of complex hierarchies. This result is robust to a long-list of checks and alternative archaeological data to measure hierarchical complexity (e.g., pyramids, temples and palaces), which we discuss in detail in the online appendix (subsection F.9).

2.5 Difference-in-difference using radiocarbon-dated prehistoric archaeological sites

The results in the previous section are based on a cross-section analysis, and even though we control for a large set of confounders, we cannot exclude that our proxies for cereals and other crops, could be correlated with unobservable geographic characteristics affecting the location of ancient cities and archaeological sites. To alleviate this concern, we use the radiocarbon-dated prehistoric and proto-historic data on archaeological sites, listed in David and Ruth Whitehouse's (1975) *Archaeological Atlas of the World*. We assign each of these sites to a 1x1 decimal degree raster point of the world land surface and count the number of pre-Neolithic sites and post-Neolithic sites in each of these points. We then run the following difference-in-difference regressions:

$$\begin{aligned} Settlement_{i,t} = & \alpha_1 I(WR_Cer)_i * P_t + \alpha_2 I(WR_RT)_i * P_t + \alpha_3 I(WR_Cer + WR_RT)_i * P_t \\ & + X'_{it}\beta + \eta_i + \eta_t + u_{it}, \end{aligned} \quad (8)$$

and:

$$Settlement_{i,t} = \alpha_1 DistanceCer_i * P_t + \alpha_2 DistanceAgr_i * P_t + X'_{it}\beta + \eta_i + \eta_t + u_{it}, \quad (9)$$

where the subscript i indicates the raster point of the world; the subscript t indicates whether the site pre-dates or not the Neolithic transition; η_i and η_t are cell and period fixed effects, and P_t is a dummy variable that identifies archaeological sites dating after the Neolithic transition.

Column 1 of Table (6) shows that the presence of WRs of domesticated cereals in areas in which WRs of domesticated roots and tubers are not available is associated with a relative increase in the probability of finding a post-Neolithic site rather than a pre-Neolithic site, confirming that the Neolithic transition led to more visible traces of human societies only in areas where agriculture started with cereals. In addition, it illustrates that in areas in which WRs of roots and tubers exist, with or without WRs of cereals, there is no increase in post-Neolithic sites relative to pre-Neolithic sites. Column 2 shows that distance from the nearest area of cereal domestication is associated with a decrease in the probability of finding a post-Neolithic site rather than a pre-Neolithic site. Again,

once distance from cereal domestication is included in the regression, distance from the nearest area of independent domestication does not produce any significant effects. The results reported in the rest of Table (6) confirm the results in columns 1 and 2, with different dependent variables: the number of archaeological sites (columns 3 and 4), the presence of a prehistoric settlement (columns 5 and 6), or the number of prehistorical settlements in the area (columns 7 and 8).

Table 6: Potential Crop Yields and the Location of Ancient Cities.

	Dependent variable is:							
	archaeol. site (dummy)	Log(1+ # archaeol. site)	anc. settlem. (dummy)	Log(1+ # anc. settlem.)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>P X WR Cer</i>	0.142*** (0.0245)		0.189*** (0.0366)		0.149*** (0.0235)		0.179*** (0.0324)	
<i>P X WR RT</i>	-0.00775 (0.0183)		-0.0102 (0.0154)		0.00543 (0.0159)		0.00217 (0.0137)	
<i>P X Cer&RT</i>	-0.00554 (0.0178)		-0.00347 (0.0179)		0.00693 (0.0147)		0.00740 (0.0157)	
<i>P X DistCer</i>		-0.0015** (0.00060)		-0.0019*** (0.00071)		-0.0014** (0.00055)		-0.0016*** (0.00063)
<i>P X DistAgr</i>		0.00038 (0.00064)		0.00090 (0.00085)		0.00001 (0.00057)		0.00043 (0.00071)
CELL FE	YES	YES	YES	YES	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES	YES	YES	YES	YES
r ²	0.0903	0.0253	0.0983	0.0265	0.0958	0.0316	0.0976	0.0294
N	34152	31854	34152	31854	34152	31854	34152	31854

The table reports difference-in-difference OLS regressions. The unit of observation is the 1x1 decimal degree square either before or after the Neolithic transition. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

The same qualitative results are obtained when we use the GAEZ data for our regressors. Higher productivity advantage of cereals over roots and tubers is associated with a relative increase in the probability of finding a post-Neolithic site/ancient settlement rather than a pre-Neolithic site/ancient settlement, confirming that the Neolithic transition led to more visible traces of human societies but only in areas where agriculture started with cereals. In addition, it illustrates that if we control for cereal advantage, land productivity does not produce any significant positive effect. Results are reported in Table (F.44) in the appendix.

3 Evidence II – Case studies

The most direct evidence that cereals played a crucial role in state formation is the observation that in farming societies that rely on roots and tubers, hierarchical complexity never exceeded the level that anthropologists define as ‘chiefdoms,’ while all agriculture-based large states that we know of relied on cereals.³² In this section, we examine several societies in different stages of the emergence of hierarchies and states, which we interpret as supporting our appropriability critique of the prevailing productivity theory.³³

3.1 Complex hunter-gatherers in North America

Summarizing extensive anthropological evidence on hunting-gathering societies, Testart (1982a, 1982b) identified a positive association between social inequality and the prevalence of storage of seasonal food sources. When considering agricultural societies, Testart may have been the first to distinguish between those based on cereals and those based on tubers and to attribute inegalitarian, complex social structures to reliance on sedentism and storage (1982a, pp. 195-204).

Tushingham and Bettinger (2013) study the transition of some hunting-foraging aboriginal Californians to intensified reliance on salt-drying and storing seasonal salmon, a transition that coincided with the concentration of population in permanent villages and increased social complexity.³⁴ They theorize that reliance on salmon was avoided for many centuries, despite its many advantages, because it is a “front-loaded” food source that takes much effort to procure and store,

³²In Appendix C we defend the statement above by examining three purported counter-examples in Murdock’s (1967) *Ethnographic Atlas*, where societies that depended on the cultivation of roots or tubers are coded as large states. We are aware that our binary measure of appropriability is imperfect, that classifying only cereals as ‘appropriable’ is too simplistic, and that we ignore altogether important food sources such as pulses, fruit, vegetables, fish, and animals. To the extent that these are not easily appropriable, we would classify them with roots and tubers. Some of these, though, can be stored, but did not give rise to states, possibly because they are not seasonal, and thus do not require lengthy storage. It is known, for example, that potatoes have been freeze-dried in the mountains of ancient Peru by the Incas (who also grew a cereal, maize), and are altogether somewhat storable. And, as Mokyr (1985) argues, potato-eating pigs, which can be appropriated (but whose maintenance is costly), helped the Irish hierarchy to survive. Livestock is also appropriable but requires costly maintenance. In the empirical analysis, we control for animal husbandry when possible, and find that this leaves our main results on the effect of cereals and land productivity on hierarchy unchanged. By focusing on agriculture-based societies, we exclude from our discussion states, like the Nabateans, the Venetians, and several African Kingdoms, which relied primarily on taxing trade.

³³De la Sierra (2020) and Buonanno et al. (2015) present additional evidence on the effect of appropriability on the emergence of elites.

³⁴Cook’s account of his voyages to the eastern shores of the Pacific Ocean (1784, volume II, book IV) provides a vivid eye-witness depiction of these villages.

but relatively little effort to prepare for consumption. They state that such a food source increases “the possibility that others will rob caches, which mobile foragers are not positioned to protect,” and also increases the vulnerability of loss to “freeloaders” from the inside (pp. 533-534). Tushingham and Bettinger’s propose that the transition to reliance on such a front-loaded food source had to coincide with increased social complexity. Their analysis of the consequences of reliance on food storage in a pre-agricultural society is highly consistent with the idea that increased vulnerability to appropriation contributes to expanded hierarchy.³⁵

3.2 The Neolithic in the Ancient Near East

Archaeological findings show that the earliest phases of the transition to cereal farming correlated with communal storage and with the emergence of inequality and hierarchy. Semi-sedentary forms of living, dwellings, sickles, mortars, and pestles, grinding stones, and storage facilities appear in the ancient Near East as early as the pre-Neolithic, Natufian period, when cereals were collected but not yet sown or domesticated. Active cereal cultivation emerged only later, during the Pre-Pottery Neolithic A (PPNA, ca. 9500-8500 BCE), when farmer-foragers collected wild grains on a large scale and sowed grain, yet still before domestication.³⁶ Differentiated dwelling sizes and funerary assemblages suggest that systematic inheritable inequality was present already at that stage. Kuijt and Finlayson (2009) report the discovery of an elaborate circular communal storage pit in the Jordan Valley from about 9000 BCE. This finding reveals that sizeable communal storage was an integral part of the earliest phase of the transition to cereal farming. Communal storage probably reflected a need to protect stored grains from the elements (moisture, insects, and rodents), as well as the existence of volume-related increasing returns to scale in storage. Such communal storage also attests to the emergence of leadership alongside the gradual intensification of cereal farming

³⁵Chiwona-Karltun et al. (2002) illustrate how reliance on a back-loaded food source provides protection. They report that women in modern Malawi, particularly single women, prefer to grow bitter and toxic cassava variants, even though these require significantly more post-harvest processing. This pattern is explained as due to the protection that this extra post-harvest drudgery provides these women against thievery, as thieves prefer the non-bitter variant.

³⁶For surveys, see Barker (2006) and Simmons (2007). Increased climatic seasonality at the end of the ice age that generated evolutionary modifications in grasses that developed larger seeds to adapt to summer drought is commonly presumed to explain the timing of the transition to agriculture in Eurasia (Diamond, 1997; Matranga, 2017). Richerson, Boyd, and Bettinger (2001, pp. 388-389) debunk the theory that this transition was caused by food shortage due to population growth. Bowls and Choi (2019) argue that the strategic complementarity between farming and private property impeded the transition to agriculture since it required the parallel adoption of the social institution of private property.

and sedentism. Constructing and overseeing the storage pit (which protected the prime source of the community’s non-human wealth) and distributing the stored foodstuffs required leadership, even if initially banditry may not have been a major issue.³⁷ The need to protect stockpiles is manifest also by the subsequent agglomeration of people in early walled villages and urban centers, long before the formation of city-states.

3.3 The post-Columbian introduction of sweet potatoes to New Guinea

New Guinea adopted agriculture at about the same time as Egypt (ca. 5000-4500 BCE), cultivating bananas, taro, and yam. But unlike Egypt, the increase in food production didn’t lead to the development of a complex hierarchy.³⁸ In the 17th century, the sweet potato, which originated in America, reached New Guinea and rapidly displaced older crops to become the staple. Wiessner and Tumu (1998) record how the new crop resulted in a substantial increase in productivity, population, and the production of prestige goods, such as the aggrandizing slaughter of pigs in communal festivals. But this considerable increase in land productivity left the highland population of New Guinea fragmented, subject to endemic tribal warfare, and without any consolidation of power or a significant increase in social complexity.

Thus, increased productivity of a less appropriable crop that generates only a temporary surplus (until the crop rots) didn’t prompt significant hierarchy. This observation stands in contrast to the conventional productivity theory yet is consistent with the appropriability theory.

3.4 State avoidance in South East Asia

With the idea of explaining how some societies managed to avoid subjugation to state authority, Scott (2009) posits that in South East Asia, states only emerged in the river valleys, where they

³⁷Large round pits from that period were found elsewhere in the Jordan Valley and in several sites near the Euphrates (Mithen et al. 2011; Willcox and Stordeur 2012). These pits are identified as having served as communal storage and also as communal meeting places, possibly for ritual ceremonies. Some archaeologists identify storage as an indication of surplus, but cereal-based farming requires intra-annual storage even if living at subsistence, with no long-term excess. The salience of communal storage in some PPNA sites is explained mostly due to increasing returns to scale. But this advantage is rather limited. In the dense Neolithic village of Çatalhöyük (PPNB, ca. 7500-6000 BCE), the storage of grains was already entirely private within the homes.

³⁸One could argue, as Diamond (1997) seems to, that farmers in these areas did not produce any surplus due to low land productivity. However, the claim of low productivity is inconsistent with the evidence (see appendix A). Amazonia provides another example of a region with productive farming and no complex hierarchies (Neves and Heckenberger, 2019).

relied on intensively cultivated appropriable rice. He argues that tribal hill people resisted the valley states and retained freedom by adopting foraging and shifting agriculture, based on the cultivation of less appropriable roots and tubers. While he refers to the distinction between crop types, Scott’s key distinctions relate to geographic differences between valley and hill people, and between sedentary and shifting farming.³⁹

3.5 The maize growing Bushong in Congo

We conclude this brief survey with an example that Acemoglu and Robinson (2012) discuss to support their claim for the precedence of political innovations to the adoption of agriculture. They describe how, in the seventeenth century and after Mize reached the West African coast, a leader of the Bushong people in the Congo heartland advanced the adoption of maize farming, and turned his tribal people into a kingdom. The Bushong’s neighbors across the river, the Lele people, share the same environment but avoided the cultivation of cereals and resisted subjugation to hierarchy. Once again, this example demonstrates the critical role of the crop type for the emergence of complex hierarchy and for its avoidance.

4 Alternative theories for the emergence of hierarchy and states

Our extensive empirical analysis reveals that cereal cultivation had a causal effect on various indicators of hierarchy in farming-based societies.⁴⁰ It fails to find any evidence that land productivity had such an impact, once cereal cultivation is controlled for. In this section, we review the literature on the emergence of hierarchy and states in light of these findings.

Adam Smith invoked the appropriability argument when arguing that governments and property protection first emerged with the transition to pastoralism and the need to protect livestock from theft (Smith 1978, p. 16). But when he turned to agriculture, Smith reverted to emphasize the role of productivity as generating a surplus, division of labor, and trade, and, as a result, also a demand for an extended role of government (1978, p. 409).⁴¹ For Smith and his intellectual heirs,

³⁹Yet, the role of elevation is reversed in South America, where the Incas had a powerful state in the mountains, and no major state emerged in the Amazon valley.

⁴⁰The empirical approach adopted here is similar to that of Alesina, Giuliano, and Nunn (2013), who offer a geographical explanation for a facet of hierarchy we ignore - hierarchy between men and women.

⁴¹The idea that agriculture generated surplus and that surplus led to the government was expressed already earlier

the surplus had to be available *before* the landlord, the capitalist, or the ruler could seize it.

Friedrich Engels argued similarly that the surplus generated by the adoption of agriculture was a prerequisite for the transition to class society (1978, p. 65). Childe (1936), too, posits that the transition to agriculture resulted in food surplus and prosperity that enabled farmers to demand specialty items – leading artisans to specialize in non-farming activities and to trade. This division of labor led over time to political integration, and eventually to urban centers and the formation of city-states. Lenski (1966) emphasizes the surplus generated by farming, which intensified with the transition from horticulture to intensive agriculture, as the source of social power.⁴²

Many scholars have sought to explain what lies behind the relative underdevelopment of tropical countries.⁴³ Diamond (1997) employs the conventional productivity theory when he summarizes the source of the advantages of temperate regions (p. 92): “In short, plant and animal domestication meant much more food The resulting food surpluses . . . were a prerequisite for the development of settled, politically centralized, socially stratified, economically complex, technologically innovative societies.” Price and Bar-Yosef (2010; p. 160) reach a similar conclusion: “Cultivation also supported a stable economy with surplus that resulted in the formation of elite groups as predicted by Lenski (1966).” Indeed, the productivity-and-surplus theory is the default explanation for state formation in both popular and scientific writing; Diamond (1997) and Price and Bar-Yosef (2010) are but two salient examples.

Our introductory argumentation, our case studies, and our extensive empirical evidence cast doubt on this conventional and still predominant productivity theory. Indeed, our empirical evidence and case studies suggest that increased land productivity in itself (without increased appropriability) had a limited causal effect on hierarchy, if any.

in the seventeenth-century – see Meek (1976) and Aspromourgos (1996).

⁴²Lenski is preoccupied with a presumed technological shift from horticulture to agriculture that increased land productivity further and enabled the transition from chiefdoms to states. However, the horticulture witnessed by ethnographers is almost invariably based on roots and tubers. This observation and our analysis make it apparent that horticulture is not but a less intensive form of agriculture, and that horticulture and chiefdoms may represent a long-term geographic-conditioned equilibrium, rather than a stage in the transition to intensive (cereal-based) agriculture and statehood.

⁴³Sachs, Mellinger, and Gallup (2001), Olsson and Hibbs (2005), Spolaore and Wacziarg (2013), and Ashraf and Galor (2013) provide empirical attempts to link income per capita across countries with geographic variables. Nowadays, two main features of the tropics are typically argued to have impeded its development: low agricultural productivity and a high burden of disease. Weil (2007, 2010) finds that the effect of health on growth is relatively small and cannot explain the extent of the gap between tropical and non-tropical countries, but his findings are controversial. Here, we question the productivity explanation and provide an alternative geographical/institutional explanation.

We now turn to alternative theories that still invoke increased productivity, but suggest factors other than the availability of surplus as the mechanism leading to the emergence of hierarchy. Some of these theories are consistent with our findings. Johnson and Earle (2000) claim that increased land productivity led to population growth and increased density, fostering conflict, and necessitating increasingly complex social forms to contain violence. North, Wallis, and Weingast (2009) adopt this functionalist theory, explaining the evolution of human history from the Neolithic age to modern times in terms of the institutions formed to contain humans' natural proclivity to violence. When permitted by the data, we control for population size in our regressions. Consistent with this theory, the coefficient of hierarchy on population is positive and significant but leaves the coefficient on our proxy for cultivating cereals practically unchanged.

Motivated by the contrasting political structures in the valleys of Peru and in Amazonia, Carneiro's (1970) 'circumscription theory' offers another variant of a conflict theory. He postulates that an elite can extract ongoing surplus only when those subjected to taxation are geographically entrapped, and contends that states could not emerge in the Amazon Basin because "the vanquished could flee to a new locale, subsisting there about as well as they had subsisted before, and retaining their independence" Whereas "in Peru . . . this alternative was no longer open to the inhabitants of defeated villages. The mountains, the desert, and the sea . . . blocked escape in every direction" (p. 735).

Carneiro's puzzlement over limited social complexity in Amazonia is reminiscent of Diamond's concern about the underdevelopment of New Guinea. But the environmental theory of one contrasts with that of the other. Diamond's theory about the advantage of an east-west orientation of landmass cannot resolve Carneiro's comparison between Peru and Amazonia. And Carneiro's circumscription theory fails to resolve Diamond's puzzle of limited social complexity in the Pacific tropical islands. Our appropriability theory offers an explanation: whereas agriculture in the tropical Amazon and the Pacific Islands was based on tuber crops, farming in the western valleys of the Andes relied mostly on maize.⁴⁴

Other scholars provide variants of conflict theory. Dow and Reed (2013) suggest that warfare between different groups left the victors as the land-owners and forced the vanquished into employed

⁴⁴ Allen (1997) applies Carneiro's theory to explain the emergence of the Ancient Egyptian state in the circumscribed Nile Valley. He mentions how the appropriability of cereals contributed to hierarchy, maintaining, however, that surplus was a precondition for the emergence of the Egyptian state.

as workers. Boix (2015) argues that the introduction of agriculture caused bandits to raid farmers. This conflict ended either in dictatorships by the bandits who turned stationary (as in Olson, 1993), or in republics managed by the farmers. And Dal Bó, Hernández and Mazzuca (2015) theorize that farmers' increased insecurity due to pillage discouraged investment. State defense capacity evolved to resolve this inefficiency.

Our finding of a positive effect of cereal cultivation on hierarchy is consistent with these various conflict theories since cereals' storage attracts predation by outsiders and generates a need for organized protection. Thus, our thesis is arguably a variant of a conflict theory with two significant additions: the distinction between appropriable cereals and alternative less appropriable, food sources; and the observation that reliance on appropriable crops increases the tax capacity of the elite, independently of predation by outsiders and the functional demand for security.

Another functional theory focuses on the demand for law and order to facilitate trade. Based on African evidence, Bates (1983) argues that ecologically diverse environments increase the returns from commercial trade and generate demand for hierarchy. Fenske (2014) and Litina (2014) provide empirical support for this theory.⁴⁵ Trade also increases the return to the construction and maintenance of roads, ports, and marketplaces. Thus, similar to our claim that the cultivation of appropriable cereals generates demand for protection and facilitates the taxation to provide such protection, we note that trade, too, creates demand for a state and simultaneously enhances the state's opportunity to tax economic activity.

The cultivation of storable and transportable cereals (unlike other perishable crops) in itself facilitates trade. When possible, we control for proxies for trade (geographical isolation, measured by distance from a major river and distance from the coast) in our empirical analysis. We find some evidence that supports the trade channel, with the effect of cereals on hierarchy practically unaffected. This suggests that cereal cultivation affects hierarchy beyond its impact on trade.

Long-term storage plays an important role in another functional theory for the emergence of complex societies (Halstead 1989; Johnson and Earle 2000, pp. 51-256, 301-302). Under Polanyi's (1944) influence, it is argued that early agricultural societies were "redistributive," where surplus output was voluntarily transferred to a central authority, then redistributed, and in part also

⁴⁵ Algaze (2008) offers a similar theory regarding ancient Mesopotamia. We note that these scholars typically have in mind long-distance trade in luxury items rather than in staple food.

stored on a long-term basis as a buffer against future shortfalls. This theory thus views the central authority serving as an insurance agency. Our emphasis here is on intra-annual storage and its potential for appropriation by the elite, rather than on central storage as a source for redistribution for the benefit of cultivators. We conjecture that overcoming idiosyncratic shortfalls to individuals did not require proto-state centralized institutions and was managed by sharing within kin groups (as among hunter-gatherers). However, since this theory, too, emphasizes the role of storage, we find it consistent with our empirical findings.

Wittfogel (1957) proposed another functionalist theory. He contends that strong despotic hierarchies were required to realize the agricultural potential of riverine environments, through the public construction and management of large irrigation projects. Indeed, Bentzen, Kaarsen, and Wingendr (2017) show that environments with the potential for irrigation systems have had greater inequality in the past and more authoritarian states in the present. In our empirical analysis, we control for evidence of agriculture based on intensive irrigation. Our estimates confirm that societies that practice intensive irrigation are characterized by relatively more complex hierarchies. Our results on the impact of cereals are unchanged when controlling for irrigation.⁴⁶

Several scholars reverse the standard causal direction from farming to hierarchy and maintain that the rise of religion and institutional innovations enabled a political elite to gain power, and may have been the driving force behind the transition to agriculture (Cauvin, 2000; Acemoglu and Robinson, 2012). These theories are consistent with our empirical findings to the extent that these social and institutional changes were correlated with land suitability for cereals.

Nunn and Qian (2011) show how the adoption of the potato in Europe in the mid-sixteenth century led to population growth and to substantial social changes. They argue that these changes were due to the higher caloric yield of the potato in regions that are particularly suitable for its cultivation. We suggest a complementary mechanism whereby European farmers adopted the potato because it provided them with greater immunity from taxation/theft, thus enabling the

⁴⁶However, Wittfogel’s critics point out that irrigation projects in early civilizations were constructed by local communities, before the emergence of a powerful central state, and were also typically managed locally rather than centrally. Mayshar, Moav, and Neeman (2017) contend that, in contrast to Wittfogel’s causal theory, it is not a need for irrigation that led to a despotic state, but rather (local) irrigation systems enabled control and expropriation by the central state – in analogy to the interpretation here that food storage facilitated confiscation. The findings by Bentzen et al. are consistent with the appropriability approach since they do not address the direction of causality on whether hierarchy preceded or followed irrigation.

growth of the farming population. Consistent with this mechanism, McNeill (1999, pp. 71-72) reports that European farmers initially resisted adopting the potato and that only during the Dutch Wars in 1557-1609, “villagers along the route [of the Spanish army] swiftly discovered that by leaving the tubers in the ground and digging them only as needed for their own consumption, they could safely survive even the most ruthless military requisitioning. Foraging parties were unwilling to dig for their food when stores of grain were available in barns.”

5 Concluding remarks

We conclude by commenting on the pertinence of our research for understanding current issues. Besley and Persson (2009, 2014) argue that underdevelopment is closely related to low state capacity. While we address agriculturally-based economies in earlier periods, we contend that the appropriability mechanism and our critique of the conventional productivity theory shed important light on the sources of current underdevelopment. Since the modern transition away from agriculture to manufacturing and services is protracted and, in many countries, stretched well into the twentieth century, and since social institutions exhibit inertia, we contend that our theory applies to the modern age.⁴⁷

In particular, the appropriability theory provides new insights about the root cause for the underdevelopment in the tropics. We contend that the high productivity of less appropriable food sources provided the population with substantial immunity against taxation, that inhibited the emergence of stable state institutions and the development of high state capacity.⁴⁸ To the extent that hierarchy and state capacity are crucial for economic development, the environment in tropical regions might be a curse of plenty.

⁴⁷Bockstette, Chanda, and Putterman (2002), Gennaioli and Rainer (2007), Spolaore and Wacziarg (2013), and Michalopoulos and Papaioannou (2013, 2014) demonstrate that deep-rooted, pre-colonial institutions affect current institutions and economic outcomes. Dincecco and Prado (2012) and Dincecco and Katz (2014) show that state capacity is persistent and positively affects economic performance.

⁴⁸Besley and Persson (2009, 2014) propose that low state capacity can be overcome by investment in fiscal administration. But our theory leads us to be less sanguine about tropical states’ ability to raise revenue from the countryside, where the bulk of the population still resides. Gennaioli and Voth (2015) emphasize how investment in state capacity since the Middle Ages responded to conflict in the spirit of Tilly’s theory (1975). Becker et al. (2018) provide further empirical support for that theory.

References

- [1] Acemoglu, Daron, Simon Johnson, and James A. Robinson (2001) “The Colonial Origins of Comparative Development: An Empirical Investigation,” *American Economic Review*, 91, 1369-1401.
- [2] Acemoglu, Daron, Simon Johnson, and James A. Robinson (2002) “Reversal of Fortune: Geography and Institutions in the Making of the Modern World Income Distribution,” *Quarterly Journal of Economics* 117: 1231-1294.
- [3] Acemoglu, Daron and James A. Robinson (2012), *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*, Random House.
- [4] Ahrens, Achim, Hansen, Christian and Mark E. Schaffer (2019) *Lassopack: Model selection and prediction with regularized regression in Stata*. *Stata Journal* 20(1), 176-235.
- [5] Alesina, Alberto, Giuliano, Paola and Nunn Nathan (2013) “On the Origins of Gender Roles: Women and the plow,” *Quarterly Journal of Economics*, 128, 155–194.
- [6] Algaze, Guillermo (2008), *Ancient Mesopotamia at the Dawn of Civilization: The Evolution of an Urban Landscape*, University of Chicago Press.
- [7] Allen, Robert C. (1997). “Agriculture and the Origins of the State in Ancient Egypt.” *Explorations in Economic History* 34, 135-154.
- [8] Ashraf, Quamrul H. and Oded Galor (2011) “Dynamics and Stagnation in the Malthusian Epoch,” *American Economic Review*, 101, 2003-2041.
- [9] Ashraf, Quamrul H. and Oded Galor (2013) “The Out of Africa Hypothesis, Human Genetic Diversity, and Comparative Economic Development,” *American Economic Review* 103(1), 1-46.
- [10] Aspromourgos, Tony (1996), *On the Origins of Classical Economics: Distribution and Value from William Petty to Adam Smith*, Routledge.
- [11] Barker, Graeme (2006) *The Agricultural Revolution in Prehistory: Why did Foragers become Farmers*, Oxford University Press.
- [12] Bar-Yosef, Ofer and Richard H. Meadow, (1995), “The Origins of Agriculture in the Near East,” in *Last Hunter–First Farmers: New Perspectives on the Prehistoric Transition to Agriculture*, edited by T. Douglas Price and Anne Brigitte Gebauer, School of American Research Press, 39-94.

- [13] Bates, Robert H. (1983) *Essays on the Political Economy of Rural Africa*, University of California Press.
- [14] Becker, Sascha O., Andreas Ferrara, Eric Melander and Luigi Pascali (2018) “Wars, Local Political Institutions, and Fiscal Capacity: Evidence from Six Centuries of German History” CAGE discussion paper 395.
- [15] Belloni, Alexandre, Chernozhukov, Victor and Christian Hansen (2014) High-Dimensional Methods and Inference on Structural and Treatment Effects. *Journal of Economic Perspectives* 28(2), 29-50.
- [16] Bentzen, Jeanet S, Kaarsen, Nicolai, and Asger M. Wingender (2017) “Irrigation and Autocracy,” *Journal of the European Economic Association*, 15(1), 1-53.
- [17] Besley, Timothy and Torsten Persson (2009) “The Origins of State Capacity: Property Rights, Taxation and Politics,” *American Economic Review*, 99, 1218-1244.
- [18] Besley, Timothy and Torsten Persson (2014) “Why Do Developing Countries Tax So Little?” *Journal of Economic Perspectives*, 28, 99-120.
- [19] Bockstette, Valerie, Areendam Chanda and Louis Putterman, (2002) “States and Markets: The advantage of an Early Start”, *Journal of Economic Growth*, 7, 347-369.
- [20] Bogaard, Amy, Michael Charles, Katheryn C. Twiss, Andrew Fairbairn, Nurcan Yalman, Dragana Filipović, G. Arzu Demirergi, Fusun Ertuğ, Nerissa Russell, and Jennifer Henecke (2009), “Private Pantries and Celebrated Surplus: Storing and Sharing Food at Neolithic Catalhöyük, Central Anatolia,” *Antiquity*, 83, 649–668.
- [21] Boix, Carles (2015) *Political Order and Inequality: Their Foundations and Their Consequences for Human Welfare*, Cambridge University Press.
- [22] Borcan, Oana, Ola Olsson and Louis Putterman, (2018) “State History and Economic Development: Evidence from Six Millennia.” *Journal of Economic Growth*, 23, 1–40.
- [23] Bowles, Samuel and Jung-Kyoo Choi (2019) “The Neolithic Agricultural Revolution and the Origins of Private Property,” *Journal of Political Economy*, 127, pp (NA yet)
- [24] Buonanno P., Durante, R., Prarolo G. and Vanin, P. (2015) “Poor Institutions, Rich Mines: Resource Curse and the Origins of the Sicilian Mafia,” *Economic Journal*, 125, 175-202.
- [25] Bustos, Paula, Caprettini, Bruno and Jacopo Ponticelli (2016) “Agricultural Productivity and Structural Transformation. Evidence from Brazil” *American Economic Review* 106 (6), 1320-65.

- [26] Callaway, Ewen (2014) Domestication: The Birth of Rice. *Nature* 514, 58-59.
- [27] Carneiro, Robert (1970) "A Theory of the Origin of the State," *Science*, 169, 733–738.
- [28] Castaneda-Alvarez, N.P., Khoury, C.K., Achicanoy, H.A., Bernau, V., Dempewolf, H., Eastwood, R.J., Gurarino, L., Harker, R.H., Jarvis, A., Maxted, N., Mueller, J.V., Ramirez-Villegas, J., Sosa, C.C., Struik, P.C., Vincent, H. & Toll. J. (2016) "Global conservation priorities for crop wild relatives," *Nature Plants* 2.
- [29] Cauvin, Jacques, (2000) *The Birth of the Gods and the Origins of Agriculture*, translated by Trevor Watkins, Cambridge University Press.
- [30] Chandler, Tertius (1987) "Four Thousand Years of Urban Growth: An Historical Census;" The Edwin Mellen Press, New York.
- [31] Childe, V. Gordon (1936) *Man Makes Himself*, Watts & Co.
- [32] Chiwona-Karlton, Linley, Chrissie Katundu, James Ngoma, Felistus Chipungu, Jonathan Mkumbira, Sidney Simukoko, and Janice Jiggins. (2002) "Bitter cassava and women, an intriguing response to food security." *LEISA Magazine*, 18, 1–15.
- [33] Conley, Tim (1999), "Estimation with Cross Sectional Dependence," *Journal of Econometrics* 92(1), 1-45.
- [34] Cook, James (1784), *A Voyage to the Pacific Ocean Undertaken by the Command of His Majesty for making Discoveries in the Northern Hemisphere*, volume II, London.
- [35] Dal Bó, Ernesto, Pablo Hernández and Sebastián Mazzuca (2015) "The Paradox of Civilization: Pre-Institutional Sources of Security and Prosperity," NBER working paper 21829.
- [36] De la Sierra, Raul S. (2020), On the Origin of States: Stationary Bandits and Taxation in Eastern Congo. *Journal of Political Economy*. 128(1), 32-74.
- [37] Denham, Tim (2011) "Early Agriculture and Plant Domestication in New Guinea and Island Southeast Asia," *Current Anthropology*, 52, S379-S395.
- [38] Diamond, Jared (1997) *Guns, Germs, and Steel: The Fates of Human Societies*, Norton, New York.
- [39] Dincecco, Mark and Mauricio Prado (2012) "Warfare, Fiscal Capacity, and Performance," *Journal of Economic Growth*, 17, 171-203.
- [40] Dincecco, Mark and Gabriel Katz (2016) "State Capacity and Long-Run Economic Performance," *The Economic Journal*, 126 (590), 189-218.

- [41] Dow Gregory K. and Clyde G. Reed (2013) “The Origins of Inequality: Insiders, Outsiders, Elites, and Commoners,” *Journal of Political Economy*, 121, 609-641.
- [42] Fenske, James (2013) “Does land abundance explain African institutions?” *Economic Journal* 123, 1363-90.
- [43] Fenske James (2014) “Ecology, trade and states in pre-colonial Africa,” *Journal of the European Economic Association*, 12, 612–640.
- [44] Galor Oded and Omer Ozak, (2016), “The Agricultural Origins of Time Preference” *American Economic Review*, 106 (10). 3064–3103
- [45] Gennaioli, Nicola and Ilija Rainer (2007) “The Modern Impact of Pre-Colonial Centralization in Africa.” *Journal of Economic Growth*, 12, 185-234.
- [46] Gennaioli, Nicola and Hans-Joachim Voth (2015) “State Capacity and Military Conflict,” *Review of Economic Studies*, 82(4), 1409-1448.
- [47] Halstead, Paul (1989) “The Economy has a normal Surplus: Economic Stability and Social Change among Early Farming Communities of Thessaly, Greece,” In P. Halstead & J. O’Shea (Eds.), *Bad Year Economics: Cultural Responses to Risk and Uncertainty (New Directions in Archaeology*, pp. Viii-Ix). Cambridge: Cambridge University Press. 68-80.
- [48] Heizer, R.F. and Sturtevant, W.C. (1978) *Handbook of North American Indians*, Smithsonian Institution, Washington, D.C.
- [49] Johnson, Allen W and Earle, Timothy (2000) *The Evolution of Human Societies: from Foraging Group to Agrarian State*. Stanford University Press; Second edition.
- [50] Kemp, Barry J. (2006) *Ancient Egypt: Anatomy of a Civilization*, 2nd edition, Routledge.
- [51] Kuijt, Ian and Bill Finlayson (2009) “Evidence for Food Storage and Predomestication Granaries 11,000 Years Ago in the Jordan valley,” *Proceedings of the National Academy of Science*, 106, 10966-10970.
- [52] La Porta, Rafael, Florencio Lopez-de-Silanes, Andrei Shleifer and Robert W. Vishny. (1998). “Law and Finance,” *Journal of Political Economy* 106, 1113-1155.
- [53] Lang, Hartmut (1998) “CONAN: An Electronic Code-Text Data-Base for Cross- Cultural Studies,” *World Cultures* 9(2):13-56.
- [54] Larson, G. et al. (2014). “Current perspectives and the future of domestication studies.” *Proceedings of the National Academy of Sciences* 111(17), 6139–6146.

- [55] Lenski, Gerhard, E. (1966) *Power and Privilege: A Theory of Stratification*, McGraw-Hill.
- [56] Litina, Anastasia (2014) "The Geographical Origins of Early State Formation," Working Paper.
- [57] Mann, M. (1986), *The Sources of Social Power Vol. I: A History of Power from the Beginning to A.D. 1760*. Cambridge University Press.
- [58] Matranga, Andrea (2017), "The Ant and the Grasshopper: Seasonality and the Invention of Agriculture." Working paper.
- [59] Maurer, Stephan, Jorn-Steffen Pischke and Ferdinand Rauch (2016), "Of Mice and Merchants: Trade and Growth in the Iron Age," mimeo.
- [60] Mayshar, Joram, Omer Moav, and Zvika Neeman, (2017) "Geography, Transparency and Institutions." *American Political Science Review*, 111(3), 622-636.
- [61] McNeill, W. (1992), *The Global Condition: Conquerors, Catastrophes, and Community*. Princeton University Press.
- [62] McNeill, William H. (1999) "How the Potato Changed the World's History", *Social Research*, 66, 67-83.
- [63] Meek, Ronald L. (1976) *Social Science and the Ignoble Savage*, Cambridge University Press.
- [64] Michalopoulos, Stelios and Elias Papaioannou (2013) "Pre-Colonial Ethnic Institutions and Contemporary African Development." *Econometrica*, 81, 113-52.
- [65] Michalopoulos, Stelios and Elias Papaioannou (2014) "National Institutions and Subnational Development in Africa," *Quarterly Journal of Economics*, 129, 151-213.
- [66] Miracle, Marvin P. (1966) *Maize in Tropical Africa*. University of Wisconsin Press.
- [67] Mithen, Steven J. et al. (2011) "An 11,600 Year-old Communal Structure from the Neolithic of Southern Jordan," *Antiquity*, 85, 350-364.
- [68] Modelski, George (2003) "World Cities: -3000 to 2000," *Faros*, Washington
- [69] Mokyr, J. (1985), *Why Ireland Starved: A Quantitative and Analytical History of the Irish Economy, 1800-1850*, London, Allen & Unwin
- [70] Murdock, George P. (1959) *Africa: its Peoples and their Culture History*. New York, McGraw-Hill.

- [71] Murdock, George P. (1967) *Ethnographic atlas*, Pittsburgh, University of Pittsburgh Press.
- [72] Murdock, George P., Douglas R. White (1969) "Standard Cross-Cultural Sample." *Ethnology* 9: 329–369.
- [73] Neves, Eduardo G. and Michael J. Heckenberger (2019) "The Call of the Wild: Rethinking Food Production in Ancient Amazonia," *Annual Review of Anthropology*, 48, 371–388.
- [74] North, Douglass C., John Joseph Wallis and Barry R. Weingast (2009) *Violence and Social Orders: A Conceptual Framework for Interpreting Recorded Human History*, Cambridge University Press.
- [75] Nowak, Ronald (2011) "Walker's mammals of the world," Johns Hopkins University Press, Baltimore
- [76] Nunn, Nathan, (2008) "The Long-Term Effects of Africa's Slave Trades." *Quarterly Journal of Economics* 123(1), 139-176.
- [77] Nunn, Nathan and Nancy Quian, (2011) "The Potato's Contribution to Population and Urbanization: Evidence from an Historical Experiment." *Quarterly Journal of Economics* 126, 593-650.
- [78] Nunn, Nathan and Diego Puga, (2012) "Ruggedness: the Blessing of Bad Geography in Africa," *Review of Economics and Statistics* 94, 20-36.
- [79] Olson, Mancur (1993) "Dictatorship, Democracy, and Development," *American Political Science Review*, 87, 567-576.
- [80] Olsson, Ola and Douglas A. Hibbs Jr. (2005) "Biogeography and long-run Economic Development," *European Economic Review*, 49, 909-938.
- [81] Pearson, Harry W. (1957) "The economy has No Surplus: Critique of a Theory of Development," in *Trade and Market in the Early Empires: Economies in History and Theory*, edited by Karl Polanyi, Conrad Arensberg and Harry W. Pearson, Free Press, 320-341.
- [82] Polanyi, Karl ([1944], 1957) *The Great Transformation: The Political and Economic Origins of Our Time*, Beacon Press.
- [83] Price, T. Douglas and Ofer Bar-Yosef (2010) "Traces of Inequality at the Origins of Agriculture in the Ancient Near East," in *Pathways to Power: New Perspectives on the Emergence of Social Inequality*, edited by T. Douglas Price and Gary Feinman, Springer, 147-168.

- [84] Pryor, Frederic. L. (1985). "The Invention of the Plow," *Comparative Studies in Society and History* 27: 740-744.
- [85] Purugganan, Michael D. and Dorian Q. Fuller (2011) "Archaeological data reveal slow rates of evolution during plant domestication," *Evolution*, 65, 171–183.
- [86] Ramankutty, Navin, Jonathan A. Foley, John Norman and Kevin McSweeney (2002) "The global distribution of cultivable lands: current patterns and sensitivity to possible climate change," *Global Ecology and Biogeography* 11(5), 377-392.
- [87] Reba, Reitsma and Seto (2016) "Spatializing 6,000 years of global urbanization from 3700 BC to AD 2000" *Scientific Data* 3, 160034.
- [88] Richerson, Peter J., Robert Boyd and Robert L. Bettinger (2001) "Was Agriculture Impossible During the Pleistocene but Mandatory During the Holocene? A Climate Change Hypothesis," *American Antiquity*, 66, 387-411.
- [89] Sachs, Jeffrey D., Andrew D. Mellinger, and John L. Gallup (2001) "The Geography of Poverty and Wealth" *Scientific American*, 284, 71-74.
- [90] Sahlins, Marshall (1972) *Stone Age Economics*, de Gruyter.
- [91] Scott, James C. (2009) *The Art of Not Being Governed: An Anarchist History of Upland Southeast Asia*, New Haven: Yale University Press.
- [92] Scott, James C. (2017) *Against the Grain: A Deep History of the Earliest States*. Yale University Press.
- [93] Simmons, Alan H. (2007) *The Neolithic Revolution in the Near East: Transforming the Human Landscape*, University of Arizona Press.
- [94] Smith, Adam (1978) *Lectures on Jurisprudence*, edited by R.L. Meek, D.D. Raphael and P.G. Stein, Oxford University Press.
- [95] Smith, Eric Alden, Monique Borgerhoff Mulder, Samuel Bowles, Michael Gurven, Tom Hertz, and Mary K. Shenk (2010) "Production Systems, Inheritance, and Inequality in Premodern Societies: Conclusions," *Current Anthropology*, 51, 85-94.
- [96] Smith, Michael E. (ed.) (2012) *The comparative Archaeology of Complex Societies*, Cambridge University Press.
- [97] Spolaore, Enrico and Romain Wacziarg (2013) "How Deep Are the Roots of Economic Development?," *Journal of Economic Literature*, 51, 1–45.

- [98] Taylor, Walter W. (1973) "Storage and the Neolithic Revolution," in *Estudios Dedicados a Prof. Dr. Luis Pericot*, Barcelona: Publicaciones Eventuales, 23, 193–197.
- [99] Testart, Alain (1982a) *Les Chasseurs-Cueilleurs ou L'origine des Inégalités*, Societe d'Ethnographie.
- [100] Testart, Alain (1982b) "The Significance of Food Storage among Hunter-Gatherers: Residence patterns, Population densities, and Social Inequalities," *Current Anthropology*, 23, 523-537.
- [101] Testart, Alain (1982c) "Les Tubercules Sont-ils aux Céréales comme la Sauvagerie est à la Civilisation?" *Journal D'agriculture Traditionnelle et de Botanique appliquée*, 29, 349-354.
- [102] Tilly, Charles, *The Formation of National States in Western Europe*, Princeton University Press, 1975.
- [103] Tuden, Arthur and Catherine Marshall (1972) "Political Organization: Cross-Cultural Codes 4," *Ethnology* 11: 436-64.
- [104] Tushingham, Shannon and Robert L. Bettinger (2013) "Why Foragers Choose Acorns Before Salmon: Storage, Mobility, and Risk in Aboriginal California," *Journal of Anthropological Archaeology*, 32, 527-537.
- [105] Vincent, Holly et al., (2019) Modeling of crop wild relative species identifies areas globally for in situ conservation. *Communications Biology* 2(1),1-8.
- [106] Weidmann, Nils.B., Jan K. Rod, and Lars-Erik Cederman, (2010). "Representing Ethnic Groups in Space: a New Dataset," *Journal of Peace Research* 47, 491-9.
- [107] Weil David (2007) "Accounting for the Effect of Health on Economic Growth," *Quarterly Journal of Economics*, 122, 1265-1306.
- [108] Weil, David (2010) "Endemic Diseases and African Economic Growth," *Journal of African Economies*, 19, supplement, 1-29.
- [109] Whitehouse, David and Ruth Whitehouse (1975) *Archaeological Atlas of the World*, Thames and Hudson.
- [110] Wiessner, Polly and Akii Tumu (1998) *Historical Vines: Enga networks of exchange, ritual, and warfare in Papua New Guinea*, Smithsonian Institution Press.
- [111] Willcox, George, and Danielle Stordeur (2012), "Large-scale Cereal Processing before Domestication During the Tenth Millennium cal BC in Northern Syria," *Antiquity*, 86, 99-114.

- [112] Wittfogel, Karl A. (1957) *Oriental Despotism: A Comparative Study of Total Power*, Yale University Press.

Appendix A: Cereals vs. Roots and Tubers

In this appendix we seek to provide evidence in support of our various factual claims on the distinction between cereals and roots/tubers: (i) that reliance on roots and tubers is a major phenomenon in tropical regions; (ii) that roots and tubers are highly productive in the tropics; (iii) that their harvesting is in general non-seasonal; (iv) that after harvest they are significantly more perishable than cereals; and (v) that there exist significant climatic and soil variations in the productivity of cereals and of roots and tubers. (vi) that their moisture content is very high, making them bulky to transport.

Table A.1: Major staple crops produced in regions of the world in 1961

	World 1961		Sub-Sahara 1961		Papua New-Guinea 1961	
	Average Cal Yield (mil Kcal/ha)*	Total Energy Produced (10 ¹² kcal)**	Average Cal Yield (mil Kcal/ha)*	Total Energy Produced (10 ¹² kcal)**	Average Cal Yield (mil Kcal/ha)*	Total Energy Produced (10 ⁹ kcal)**
Rice	6.92	798	4.57	12	13.16	6
Wheat	3.78	772	2.39	6		
Maize	7.09	748	3.66	53	3.34	0
Barley	4.68	255	2.79	3		
Oats	3.19	122	2.86	0		
Sorghum	3.02	139	2.53	29	5.62	0
Rye	3.92	119	0.60	0		
Millet	2.24	97	2.17	24		
Soybean	1.66	40	0.55	0		
Potato	9.41	208	5.14	1	5.21	0
Cassava	11.85	114	9.10	50	15.94	84
Sweet Potato	6.32	84	4.44	3	3.44	254
Yam	8.54	10	8.65	9	19.56	142
Taro	6.63	5	0.54	0	7.51	136
Banana	9.47	19	5.21	3	11.17	276
Plantain	6.37	16	5.65	11		
Total of above		3,545		203		898
Population (mil)		3,083		223		2

The average caloric content of each crop is based on the figures in table E.4. For Soybeans, Taro, Bananas and Plantains the respective figures are 1.47, 1.12, 0.89, and 1.22, based on <http://ndb.nal.usda.gov/ndb/>, accessed Feb 2017. * Calculated on the basis of the caloric content of each crop multiplied by the average land yield, as reported by the FAO (http://faostat3.fao.org/download/Q/*/E accessed Feb 2017). ** Calculated on the basis of the average caloric yield multiplied by the crop area, as reported by the FAO (http://faostat3.fao.org/download/Q/*/E accessed Feb 2015).

Table A1 presents summary data on the main staple crops in sub-Saharan Africa, in Papua New Guinea (PNG) and the world in 1961 – the earliest year for which the Food and Agriculture Orga-

nization (FAO) provides the information.⁴⁹ In relying on relatively recent data, our presumption is that the soil and climatic conditions have not changed significantly since the Neolithic period. Of course, we recognize that the plants that provide most of the calories that humans consume have undergone major modifications since antiquity, and that their availability was greatly impacted by the post-Columbian migration of species between the continents.

(i) The data in Table A1 reveals that roots and tubers provided 37.9 percent of the total calories produced by the main staple crops in sub-Saharan Africa in 1961. We note that cassava alone provided about 45 percent of the total calories produced by these crops in Nigeria in 2013.

(ii) The table reveals further that the average caloric yield of cassava and yam in sub-Saharan Africa (9.10 and 8.65 mil Kcal/Ha) exceeded the comparable world average yield of the three main cereals, rice, maize and wheat (equal to 6.82, 7.09 and 3.56 mil Kcal/Ha, respectively).

(iii) The table reveals that the yield of non-cereals in PNG is generally high.

(iv) The seasonality of cereals is well known. They have to be sown and reaped in a relatively fixed period in the year, and usually once a year. On the other hand, roots and tubers are generally perennial and may be harvested at any time during the year. In fact, cassava can be left intact in the ground for two years. This gives farmers great flexibility around the timing of the harvest, and prevents the need for significant storage. Rees et al. (2012, p. 394) report: “Harvest time [of cassava] ranges from six to 24 months, and roots can be left in the ground until needed, making cassava a very useful food security crop.”⁵⁰

(v) Harvested grains are storable with relatively little loss from one harvest to the next, and even over several years. On the other hand, roots and tubers are in general perishable once out of the ground, though to different degrees. In particular, cassava starts to rot at ambient African temperature within 2-3 days of being harvested. The rapid deterioration is often hastened by abrasions caused by uprooting and transportation. Rees et al. (2012, p. 394) summarize the evidence: “Despite their agronomic advantages over grains, which are the other main staple food crops, root crops are far more perishable. Out of the ground, and at ambient temperatures these root crops have shelf lives that range from a couple of days for cassava... , two to four weeks for sweet potato, to between four and 18 weeks for the natural dormancy of yams...” Cassava’s fast rotting upon harvest can be overcome only by freezing or by laborious processing that turns the

⁴⁹ Given a rough estimate of 1 million calories required per person per year (2740 kcal per day), the columns on total energy produced provide a crude estimate of the population (in millions) whose energy needs could be supported by each crop (ignoring the feeding of animals, seed requirements and wastage). It is evident that (other than in PNG) the total energy produced by the listed major crops could roughly feed the entire population – but in PNG it could feed less than half the population.

⁵⁰ See also Lebot (2009) and Bradshaw (2010).

moist root into dry flour.

(vi) Lebot (2009) lists the optimal annual rainfall for cassava, yams and sweet potato as ranging from 750 to 1500 mm of rain, and the optimal temperature as 20-30 degrees centigrade. This reveals that while these crops are cultivable in the tropics, they cannot be cultivated in temperate climates.

(vii) According to Lebot (pp. vi, 78) the moisture content of cassava is 63% of the weight, and that of sweet potato and yam is 71% and 74% respectively.

By these considerations, even though the potato is biologically a tuber, when it comes to the degree of appropriability, it could be considered a quasi-cereal, since it is cultivable in temperate climates, is seasonal, and is relatively non-perishable upon harvesting.

References

- [1] Bradshaw, J. E. (editor), (2010) *Root and Tuber Crops*, Springer.
- [2] Lebot, Vincent (2009) *Tropical Root and Tuber Crops: Cassava, Sweet Potato, Yams and Aroids*, CABI.
- [3] Rees, Debbie et al. (2012) “Tropical Root Crops,” in *Crop Post-Harvest: Science and Technology – Perishables*, edited by Debbie Rees, Graham Farrell and John Orchard, Blackwell-Wiley, 392-413.

Appendix B: Surplus and appropriation – the role of population

We develop here a simple model to illustrate our Malthusian critique of the surplus theory for explaining the rise of hierarchy following the Neolithic Revolution. In this model, when population size is exogenous, both an increase in the degree of appropriability and a rise in productivity (generating surplus) lead to larger net tax revenue as a share of output. However, when the population is endogenous, according to the Malthusian framework, an increase in appropriability raises the share of net taxes, while a rise in productivity does not.

Denote the total size of the farming population by N . The production function is assumed to be Cobb-Douglas:

$$Y = (AX)^\alpha N^{1-\alpha} = A^\alpha N^{1-\alpha},$$

where A denotes the level of technology, X is the constant size of land which we normalize to one, and $0 < \alpha < 1$.

We assume that the cost of taxing a share τ of total income Y is given by:

$$\frac{Y \cdot C(\tau, m)}{z},$$

where m represents per-capita surplus income. The parameter $z > 0$ represents the degree of appropriability, so that a higher z implies a lower cost of taxation. The function $C(\tau, m)$ is continuous and differentiable, and increasing and convex in the tax rate τ . ($C_1 \geq 0, C_{11} > 0$). In adapting the standard surplus approach, we assume that resistance to tax payment is lower the higher is surplus income. As a result, the cost of taxation is assumed to decrease in surplus income, or $C_2 < 0$. Surplus income is:

$$m = (1 - \tau) \left(\frac{A}{N} \right)^\alpha - s,$$

where s is subsistence income. The share of total net taxes out of total income, denoted by π , is:

$$\pi(\tau, m, z) = \tau - \frac{C(\tau, m)}{z}.$$

The government chooses the tax rate τ to maximize its net revenue $\Pi = \pi Y$. We assume the existence of an interior solution for the tax rate, τ^* , where the first and second order conditions are satisfied. Our aim is to examine how π is affected by changes in productivity A and in the degree of appropriability z .

B1. The case of a fixed population

Given our assumptions, when the population is constant, Y is independent of τ . The optimal tax rate τ^* thus maximizes π and satisfies the first order condition:

$$\left. \frac{1}{z} \frac{dC(\tau, y)}{d\tau} \right|_{\tau=\tau^*} = \frac{C_1(\tau^*, m) - C_2(\tau^*, m) \left(\frac{A}{N}\right)^\alpha}{z} = 1.$$

Consider the effect of an increase in the appropriability parameter z . By the envelope theorem:

$$\frac{d\pi(\tau^*, m, z)}{dz} = \frac{\partial \pi(\tau^*, m, z)}{\partial z} = \frac{C(\tau^*, m)}{z^2} > 0.$$

Consider next the effect of an increase in productivity A . By a similar argument:

$$\frac{d\pi(\tau^*, m, z)}{dA} = \frac{\partial \pi(\tau^*, m, z)}{\partial m} \cdot \frac{dm}{dA} = -\frac{C_2(\tau^*, m)}{z} \cdot \frac{\alpha(m+s)}{A} > 0.$$

Thus, we have:

Proposition B1. *With a fixed population, both an increase in appropriability z and an increase in productivity A raise the share of taxes out of income π .*

B2. The case of Malthusian population

In a Malthusian setting the population size adjusts to keep agents' per capita surplus income m at zero. Thus:

$$N = \frac{(1-\tau)Y}{s}.$$

This implies:

$$Y = A \left(\frac{1-\tau}{s} \right)^{\frac{1-\alpha}{\alpha}} \equiv Y(\tau, A); m \equiv 0.$$

Denote:

$$\pi^*(\tau, z) \equiv \pi(\tau, 0, z) = \tau - \frac{C(\tau, 0)}{z}.$$

In this case, the tax rate has a negative effect on output through its effect on the size of the farming population N .

The optimal tax rate $\tau^* = \tau^*(z, A)$ maximizes $\Pi = \pi^*(\tau, z)Y(\tau, A)$. Our assumptions imply that it is implicitly defined by the first order condition:

$$F(\tau, z, A) \equiv Y(\tau, A) \frac{\partial \pi^*(\tau, z)}{\partial \tau} + \pi^*(\tau, z) \frac{\partial Y(\tau, A)}{\partial \tau} = Y \left(1 - \frac{C_1(\tau, 0)}{z} \right) - \pi^*(\tau, z) Y \frac{1-\alpha}{\alpha(1-\tau)} = 0.$$

Thus, at the optimum τ^* :

$$\frac{\partial \pi^*(\tau, z)}{\partial \tau} = -\frac{\pi^*(\tau, z)}{Y(\tau, A)} \cdot \frac{\partial Y(\tau, A)}{\partial \tau} = \pi^*(\tau, z) \cdot \frac{1 - \alpha}{\alpha(1 - \tau)} > 0.$$

In addition,

$$\frac{d\pi^*(\tau^*(z, A), z)}{dz} = \frac{\partial \pi^*(\tau^*(z, A), z)}{\partial \tau} \frac{d\tau^*(z, A)}{dz} + \frac{\partial \pi^*(\tau^*, z)}{\partial z} = \frac{\partial \pi^*(\tau^*, z)}{\partial \tau} \frac{d\tau^*}{dz} + \frac{C(\tau^*, 0)}{z^2}.$$

To prove that this expression is positive, it is sufficient to prove that $\partial \tau^* / \partial z$ is positive. By the Implicit-Function Theorem, for $F(\tau, z, A)$ defined above:

$$\frac{\partial \tau^*}{\partial z} = - \left. \frac{\partial F / \partial z}{\partial F / \partial \tau} \right|,$$

and by the second-order conditions: $\partial F / \partial \tau < 0$. Thus,

$$\text{sign} \left[\frac{\partial \tau^*}{\partial z} \right] = \text{sign} \left[\frac{\partial F}{\partial z} \right].$$

Now,

$$\frac{\partial F}{\partial z} = Y \cdot \frac{C_1(\tau, 0)}{z^2} + \frac{C(\tau, 0)}{z^2} \cdot Y \cdot \frac{1 - \alpha}{\alpha(1 - \tau)} > 0.$$

Similarly,

$$\frac{d\pi^*(\tau^*(z, A), z)}{dA} = \frac{\partial \pi^*(\tau^*(z, A), z)}{\partial \tau} \frac{d\tau^*(z, A)}{dA}.$$

Once again by the Implicit Function Theorem: $\text{sign} \left[\frac{\partial \tau^*}{\partial A} \right] = \text{sign} \left[\frac{\partial F}{\partial A} \right]$. But

$$\frac{\partial F(\tau, z, A)}{\partial A} = \frac{\partial \pi^*(\tau, z)}{\partial \tau} \cdot \frac{\partial Y(\tau, A)}{\partial A} + \pi^*(\tau, z) \cdot \frac{\partial^2 Y(\tau, A)}{\partial \tau \partial A}.$$

Since $\frac{\partial Y(\tau, A)}{\partial A} = \frac{Y(\tau, A)}{A}$ and $\frac{\partial^2 Y(\tau, A)}{\partial \tau \partial A} = \frac{\frac{\partial Y(\tau, A)}{\partial \tau}}{A}$, we have:

$$\frac{\partial F(\tau, z, A)}{\partial A} = \frac{F(\tau, z, A)}{A}.$$

Since the first order conditions require $F(\tau, z, A) = 0$, it follows that $\frac{\partial \tau^*}{\partial A} = 0$ so that

$$\frac{d\pi^*(\tau^*(z, A), z)}{dA} = 0.$$

Thus, we have:

Proposition B2. *With Malthusian population, an increase in appropriability z raises the share of taxes in the economy π , but an increase in productivity A leaves that share intact.*

Appendix C: Counter-examples in the Ethnographic Atlas

We checked the Ethnographic Atlas database for possible outliers. Out of the total Atlas sample of 1412 societies, only 1259 had data on hierarchy above the local level (variable 33). Out of these 1259 societies, only 959 are coded as reliant on specified food sources (variable 29): 634 relied mostly on cereal grains, 259 on roots and tubers, 82 on fruit trees and 4 on vegetables. The Atlas defines a “large state” as possessing at least four levels of hierarchy above the local community, and defines those with three levels as “states” (Gray, 1998). The following table provides the distribution of these societies by the major crop type.⁵¹

	Total	% Cereal	% Roots & tubers	% Fruit trees	% Other
All societies	979	64.8	26.5	8.4	0.4
Above hierarchy = 4	32	90.6	9.4	0	0
Above hierarchy = 3	82	79.3	13.4	7.3	0

It is evident that societies that rely on grain crops form a significantly larger share of those with higher levels of hierarchy than in the total sample. Still, we were intrigued by the three societies (9.4%) that are coded as “large states” that are reliant on roots and tubers. In this appendix we examine the three apparent exceptions to our statement that all pre-modern large states relied on cereals. Our conclusion is that our statement stands.

1. The first case is identified by Murdock (1981, p. 39) as “Shantung Chinese.” The anthropological data were based on Yang’s (1945) detailed report about the small village of Taitou, apparently pertaining to the 1930s. The province where this village (which no longer exists) was located is now known as Shandong. Shandong, on the eastern edge of China (in between Beijing and Shanghai) is now one of the more prosperous provinces. There is little doubt that the village of Taitou was at the time part of a major state: China. We take issue, however, with the presumption that this state, or even that province, should be classified as based on roots and tubers, as the Ethnographic Atlas has it.

To begin with, Yang reports (1945 p. 16) that the main crops cultivated in the village of Taitou were “wheat, millet, barley, soybean, corn, sweet potatoes, and peanuts.” Thus it is not clear why the village was coded as reliant on the intensive cultivation of roots, even though this may not be entirely wrong. According to Yang, the land of the village was multi-cropped, and sweet potatoes

⁵¹ All of the societies with maximal hierarchies had a specified major crop type, and two societies with above hierarchy coded as 3 had unspecified crop type (thus making the total 84, instead of 82).

were the principal crop during the summer months. More precise information can be obtained from the data collected by Buck (1937). Buck provides detailed statistics from another village, Tsimio, located about forty kilometers from Taitou. According to these data (pp. 73, 75), sweet potatoes supplied 43.3 percent of the total calories consumed in Tsimio, and seeds supplied 54.4 percent of the calories. But Tsimio (and presumably also Taitou) was unrepresentative of Shandong as a whole. In the other eleven villages that entered Buck's sample from the province, one listed the percentage of calories supplied by seeds as 76.0 (with sweet potatoes supplying 20.8 percent), and all of the other ten villages listed seeds as supplying at least 91 percent of the calories consumed. We conclude that the coding of the "shantung Chinese" society as reliant on the intensive cultivation of roots is wrong.

But the problem at hand goes beyond coding. The *Ethnographic Atlas* focuses mostly on cultural issues such as kin relations, with the underlying presumption that the data from one location pertains for the society as a whole. Yet the number of levels of hierarchy presiding over any location depends mostly on the region that encompasses that specific location. Even if Shandong's farming were based solely on roots, it could easily belong to a complex polity whose hierarchy relies on taxing cereals. This is evidently the case for China, whose state relied from the very start on the cultivation of cereals: mostly on wheat in the north (where Shandong is located), and mostly on rice in the south. Furthermore, sweet potatoes reached China only after the Columbian Exchange, while Shandong was already an integral part of the Chinese hierarchical state almost two millennia earlier.

2. The data source for "Byelorussians (White Russian)" society (latitude: 55N, longitude 28E) is identified in *Ethnology*, (vol. 4, April 1965) as "Unpublished ethnographic notes" by Melvin Ember (1954), pertaining to 1910. We were unable to obtain this source, but employed alternative sources about the Belarus.⁵²

Vakar (1956, p. 30) describes the territory as a "low marshy plane, sloping slightly to the south and east. Only in the north and northeast are there elevation of the soil," with the highest point at 345 meters above sea level. "In general ... the soil ... is not very favorable for cultivation. Forest and shallow lakes cover more than one half the whole land. ... Crops are liable to be damaged by humidity rather than droughts. ... fish and potatoes being the staff of life for most Belarussian peasants." This characterization is confirmed by the soil-suitability data according to which the soil suitability for agriculture in Belarus is in the 25th percentile below the average, and

⁵²The following summary is based mostly on Vakar (1956), and *The Great Soviet Encyclopedia* (GSE), vol. 3, 1973.

the productivity of cereals is below average. The fact that potatoes became the main cultivated crop from the 16th century is thus not surprising. In 1913 the land yielded 4.0 million tons of potatoes and only 2.6 million tons of all cereal grains; at the same time, “livestock raising is the leading branch of agriculture in Byelorussia” (GSE, p. 627). Thus, we do not question the Atlas’s coding of Belarus as based on roots and tubers (at least since Columbus).

The multiple rivers that cross Belarus provided the main waterway for early trade between the Black Sea in the south and the Baltic Sea in the north. The area was settled between the 6th and the 8th centuries by East Slav tribes, whose economy was apparently based on trade in forest products like furs, game, honey, beeswax and amber and on primitive (cereal) agriculture. As it turns out, almost throughout its history, the territory of Belarus was ruled from outside by its neighboring states, rather than from within. In the 10-12 centuries it was controlled from Kiev (now in the Ukraine) in the south, under the Grand Duke of the Kievan Rus’ – but several local feudal vassal principalities existed (all centered in the relatively elevated north).⁵³ From the 13th until the 18th centuries, the territory was ruled from the north and the west: first by the Grand Duchy of Lithuania, then by the Polish-Lithuanian Commonwealth, and effectively by Poland. Finally, from 1795 until 1990 Byelorussia was ruled from the east, as part of the Russian Empire and later as a Republic within the USSR. Thus we don’t question the coding of Belarus society as subjected to full state hierarchy, however, we note that as a dependent territory almost throughout its history, its form of agriculture can hardly be considered to have shaped the degree of complexity of the states that controlled it. Indeed, Borcan et al. (2018) assign Belarus a statehood score of 0.5 from 850 until 1950 – where, in their scoring system “Band/tribe is marked by a rule score of 0, paramount chiefdom is assigned 0.75 and fully-fledged state receives the value 1.”

3. The third case concerns the Bubi society on the tropical island of Fernando Po (now called Bioko, and part of the state of Equatorial Guinea). This volcanic island, whose area is about 2,000 square kilometers, is located in the Atlantic Ocean, about 30 kilometers off the coast of present Cameroon. The Bubis’ staples were yams and cocoyams (=taro), in addition to fish and game. Thus, also in this case we do not contest the Atlas identification of this society as reliant on roots and tubers. However, we find problematic the characterization of this society (presumably in 1920) as a large state society.⁵⁴ Indeed, according to the Atlas, the form of agriculture is not extensive,

⁵³Some historians identify the second half of the 11th century as the “first Independent Belorussian State,” under the prince ruling in Polock (=Polacak= Polatsk) – see Vakar (1956) and Plokhly (2006 pp. 12, 46-54). If so, Belarus regained independence only with the breakup of the USSR in 1991.

⁵⁴One indication for the problematic identification of the Bubi society as a “large state” is its lack of urban centers. Of the 32 Atlas societies coded as “large states,” 27 had cities with population exceeding 50,000 people, 3 more had towns with 5,000 or more people, and one (the Siamese) had towns in the range of 1,000-5,000 people. The Bubi

but rather “extensive or shifting agriculture.” Written sources about the island by missionaries and travelers since the 19th century are surprisingly abundant. The Atlas was coded on the basis of several such sources (Ethnology vol. 5, Jan 1966 pp. 132-133). For our purposes of examining mostly the Bubis’ political institutions, we find it expedient to rely mostly on Sundiata (1994, 1996, 2011).

The island of Fernando Po was a Spanish colony, but the British set a station in its northern coast from 1827 until 1943. The colonists (including freed slaves that they brought to the island) settled in the coast, while the native Bubis concentrated in the interior hilly slopes of the island where they managed their own affairs. According to Sundiata (1996 p. 7), in 1912 the island’s Bubi population numbered only 6,800.⁵⁵ The coastal settlers traded with the Bubis by purchasing from them palm oil for export, in exchange for various imported wares (like knives).

The Bubis were divided in the 19th century into 28 districts, with several villages in each district. The districts often fought with one another over women and over imported objects (Sundiata, 1996 pp. 75-79). According to Sundiata (1996 p. 75), “in spite of their fairly simple material culture, the nineteenth-century Bubi had an elaborate system of socioeconomic stratification.” Within chiefdoms, individuals were distinguished as nobles or as commoners, as well as by age groups. The chiefs’ role was mostly in settling disputes, imposing and collecting fines, and leading fights.

The long-standing political fragmentation of the Bubis underwent substantial change in the 1840s. Sundiata (1996, p. 80-82) attributes this development to recurring incidents after the British left the island, whereby foreign interlopers from the coasts raided the Bubis inland. As a result, Moka, the chief of one of the southern districts, was able to form a loose confederacy of the various chiefs and was recognized as “Great Chief,” (or king) with the task of mediating intra-ethnic strife and containing the incursions.⁵⁶ The island under Moka might be considered a hierarchical kingdom.⁵⁷ We doubt that this should be considered a case of a stable state society based on the

society is coded as “missing data,” but in fact had not had villages exceeding 1,000 people.

⁵⁵ An 1837 report by British adventurers states “The native population of Fernando Po may be about five thousand, divided into tribes, that were formerly constantly at war with each other” (Laird and Oldfield, p. 301). That report also notes the Bubis’ “free and independent bearing” and adds that “they still go perfectly naked” (p. 302). The missionary Clarke (1848, p. v) estimates the island’s total population (including the non-Bubis) at “20,000 souls.”

⁵⁶ According to Sundiata (1996 p.80-81), Moka “had seventy wives and concubines” and reinforced his position by exercising magical powers. He refused to be seen by any Europeans and “limited his use of European wares to guns and machetes,” scorning “the use of European cloth, rum, salt and tobacco.”

⁵⁷ Baumann (1888), a German traveler, provides a firsthand report of Moka’s “kingdom” in 1886. He describes Moka’s capital village as consisting of scattered hut complexes, surrounded by fields of yam (p. 103). He also narrates how, on the way there, a Bubi chief asked if he would become their king and resist Moka, telling him how much they resented being subjected to Moka (p. 33).

cultivation of tubers, but rather than an ad hoc confederacy of chiefdoms, motivated primarily by external issues related to warfare and trade (as was the case with several earlier African kingdoms). The fragility of the Bubis' presumed state is evident in that little of it survived into the twentieth century, after cocoa plantations (manned by imported laborers) started to dominate the Island's economy and the Spanish colonial power asserted its rule on the Island.

References

- [1] Baumann, Oscar (1888) *Eine Afrikanische Tropen-Insel: Fernando Póo und die Bube*, Eduard Hölzel.
- [2] Borcan, Oana, Ola Olsson and Louis Putterman, (2018) "State History and Economic Development: Evidence from Six Millennia," Brown University, Department of Economics Working Paper Series.
- [3] Buck, John Lossing (1937) *Land Utilization in China: Statistics*, Shanghai, North-China Daily News.
- [4] Clarke, John (1848), *Introduction to the Fernandian Tongue*.
- [5] Gray, J. Patrick (1998), "Ethnographic Atlas Codebook," *World Cultures*, 10 86-136.
- [6] *The Great Soviet Encyclopedia* (1973) "Byelorussian Soviet Socialist Republic," volume 3, translation of third Edition [1970], New York, Macmillan.
- [7] Laird, Macgregor and R.A.K. Oldfield (1837) *Narrative of Expedition into the Interior of Africa*, London, Richard Bentley.
- [8] Plokhy, Serhii (2006) *The Origins of the Slavic Nations: Premodern Identities in Russia, Ukraine, and Belarus*, Cambridge University Press.
- [9] Sundiata, Ibrahim (1994) "State Formation and Trade: The Rise and Fall of the Bubi Polity, c.1840-1910," *The International Journal of African Historical Studies*, 27, 505-523.
- [10] Sundiata, Ibrahim K. (1996) *From Slaving to Neoslavery: the Bight of Biafra and Fernando Po in the Era of Abolition, 1827-1930*, University of Wisconsin Press.

- [11] Sundiata, Ibrahim (2011) “Moka of Bioko (late 1820s-1899): The Chief Who United a Central African Island” in *The Human Factor in Modern Africa*, edited by Dennis Cordell, Rowman and Littlefield, pp. 47-66.
- [12] Yang, Martin C (1945) *A Chinese Village*, New York, Columbia University Press.

Appendix D: A Model of Cereals and Hierarchy

The basic premise of the model is that regions of the world differ in their productivity of tubers relative to that of cereals. For simplicity we postulate that tubers, unlike cereals, cannot be expropriated by bandits or by tax collectors. We model farmers' choice of what crop to grow in two different regimes: anarchy and hierarchy, and derive conclusions on the circumstances that are conducive for the emergence of hierarchy.

The economy is populated by a measure one of farmers and a measure N of non-farmers. Our main exogenous variable, $\delta > 0$, measures the productivity advantage of cereals over tubers.⁵⁸ The productivity of cereals is normalized to unity. Thus, farmers can grow one unit of cereals, or $1 - \delta$ units of tubers, or any linear combination thereof. Hence, a farmer's output is $\beta + (1 - \beta)(1 - \delta) = (1 - \delta) + \beta\delta$, where $\beta \in [0, 1]$ is the fraction of land allocated to cereals. Output is measured in nutritional units independently of their source.

The income of non-farmers who engage in foraging is assumed to be constant and denoted: $s > 0$. In a state of anarchy, non-farmers can chose to be either foragers or bandits who expropriate crops from farmers. In a state of hierarchy, we assume that some non-farmers are hired by the state to serve as tax collectors, and are paid the wage s . We denote by λ the measure of bandits or tax collectors. N is assumed to be larger than the equilibrium level of λ . We also assume that in order to be viable and to deter bandits, the state has to possess monopoly over the use of force which entails a cost.

Anarchy

Under anarchy, farmers might be raided by bandits. A raided farmer loses his entire cereal crop, but none of his crop of tubers. Farmers are assumed to be risk neutral.⁵⁹ A farmer who faces a raid with probability τ , chooses the fraction of land allocated to cereal β to maximize his expected income I , weighing the productivity advantages of cereals over tubers, δ , against the disadvantage, as measured by the expropriation rate τ :

$$I = (1 - \tau)\beta + (1 - \delta)(1 - \beta) = (1 - \delta) + \beta(\delta - \tau). \quad (\text{D.1})$$

We assume that the rate of expropriation, τ , is a function of the measure of bandits λ : $\tau = \tau(\lambda)$. The function, $\tau(\lambda)$ is strictly increasing and strictly concave, and satisfies: $\tau(0) = 0$, $\lim_{\lambda \searrow 0} \tau'(\lambda) =$

⁵⁸If $\delta \leq 0$ the analysis is trivial: tubers dominate cereals in providing both protection and higher productivity, so that farmers would only grow tubers in equilibrium, and the economy could only be in a state of anarchy.

⁵⁹In Appendix D we show that our results are robust to the introduction of risk aversion.

∞ , $\lim_{\lambda \nearrow \infty} \tau'(\lambda) = 0$ with $\lim_{\lambda \nearrow \infty} \tau(\lambda) = \bar{\tau} \leq 1$, with corresponding properties for the inverse function $\lambda(\tau)$. Bandits are uncoordinated (if they coordinate they become the ruling elite and the equilibrium becomes identical to that under hierarchy). Thus each bandit's expected income π is given by the total amount of confiscated cereals divided by the measure of bandits: $\pi = \tau(\lambda)\beta/\lambda$.

Definition. *Equilibrium consists of a pair (β, τ) such that:*

1. β maximizes farmers' income I , given the confiscation rate τ ;
2. given β , non-farmers are indifferent between being foragers or bandits, so that $\pi = s$.⁶⁰

Using the inverse relation $\lambda(\tau)$, the last condition can be restated as requiring: $\tau\beta/\lambda(\tau) = s$.

Define now a threshold rate δ_A by the implicit relationship:⁶¹

$$\frac{\delta_A}{\lambda(\delta_A)} = s.$$

Proposition 1. *The economy under anarchy has a unique equilibrium (β_A, τ_A) that is given by:*

$$(\beta_A, \tau_A) = \begin{cases} \left(\frac{\lambda(\delta)s}{\delta}, \delta \right) & \text{if } \delta < \delta_A \\ (1, \delta_A) & \text{if } \delta \geq \delta_A \end{cases}.$$

That is, the equilibrium confiscation rate τ_A equal δ if tubers are grown, and tubers are not grown if $\delta > \delta_A$.

Proof. If $\delta > 0$, an equilibrium with no cereals ($\beta_A = 0$) can be ruled out since in that case $\pi = 0$, leading to $\lambda = 0$ and $\tau = 0$, which would lead to $\beta = 1$, a contradiction. This implies that the equilibrium can either be one with cereals only ($\beta_A = 1$) or mixed ($0 < \beta_A < 1$), where both crops are cultivated.

If $\delta \geq \delta_A$, farmers cultivate only cereals ($\beta_A = 1$), even though this entails a maximal confiscation rate $\tau_A = \delta_A$ and a corresponding maximal number of bandits, $\lambda(\delta_A)$.

Our assumptions on $\tau(\cdot)$ imply that the confiscation rate, $\tau(\lambda)/\lambda$, is monotonically decreasing in τ , from infinity towards zero. Thus, when $\delta < \delta_A$, both cereals and tubers are cultivated and we have: $\delta/\lambda(\delta) > \delta_A/\lambda(\delta_A) = s$. Hence, there exists a unique $\beta_A \in (0, 1]$ such that $\pi_A \equiv \delta\beta_A/\lambda(\delta) =$

⁶⁰ Micro-foundation for the shape of $\tau(\lambda)$ can be obtained by assuming that banditry is time consuming and that bandits are not coordinated, so that when their number increase, the probability of raiding the same farmers increases, and the marginal total loot declines.

⁶¹ We use the subscript A to denote parameters and equilibrium values in a regime of anarchy, and similarly use the subscript H in a state of hierarchy.

s . The last condition, in conjunction with the condition $\tau_A = \delta$, defines the equilibrium combination (β_A, τ_A) . ■

Income distribution. It follows from Proposition 1 that if cereals' productivity advantage is low ($\delta < \delta_A$), β_A , τ_A and $\lambda_A = \lambda(\tau_A)$ are strictly increasing in δ and all tend to zero when δ decreases towards zero. As a result, also the total expected amount of cereals confiscated by bandits, $\tau_A \beta_A$, strictly increase in δ . In that range, farmers' income equals $1 - \delta$, and thus decreases in δ . On the other hand, when the productivity advantage of cereals exceeds the threshold δ_A , all these variables become independent of δ , with farmers income equaling $1 - \delta_A$. In these two ranges combined, proposition 1 thus implies that $\tau_A \beta_A$, τ_A and λ_A all weakly increase in δ . In turn, even though bandits' welfare is equal to s independently of the value of δ , farmers' welfare weakly decreases with δ .

The effect of the reservation income s . The smaller is s , the larger the incentive for foragers to engage in banditry. This implies a higher threshold δ_A , meaning that farmers will raise tubers in a wider range of δ . Thus, for values of $\delta > \delta_A$, a lower s reduces farmers' income. However, for $\delta < \delta_A$, a smaller s has no effect on farmers income, or on τ and therefore on λ ; it will reduce however the equilibrium value of β .

Two sources of inefficiency. Denote by Y_0 the maximal possible level of output in the economy, when all farmers cultivate only the more productive cereals (assuming $\delta > 0$) and all non-farmers engage in foraging. This maximal output level is: $Y_0 = 1 + Ns$. We can observe that the equilibrium (β_A, τ_A) introduces two deviations from this maximal output level: the first is due to farmers growing tubers (if $\delta < \delta_A$); and the other is due to the forgone output by banditry. Thus, equilibrium output is:

$$Y = Y_0 - (1 - \beta_A) \delta - s \lambda(\tau_A).$$

Inspection of the equilibrium values (β_A, τ_A) reveals that for large values of δ , the only distortion is the loss of output due to bandits being unproductive: $s \lambda_A = s \lambda(\tau_A)$, which equals the threshold level δ_A . For small values of δ , tubers are cultivated, $\tau_A = \delta$, and farmers are indifferent between the two crops. From the fact that expected revenue per-bandit is equal to $\tau_A \beta_A / \lambda(\tau_A) = s$ it follows that $s \lambda(\tau_A) = \tau_A \beta_A$. Thus we obtain:

Corollary 1. *The output loss $(Y_0 - Y)$ due to an anarchy regime is:*

$$(1 - \beta_A) \delta + \lambda_A s = \begin{cases} \delta & \text{if } \delta < \delta_A \\ \delta_A & \text{if } \delta \geq \delta_A \end{cases}.$$

Hierarchy

We assume that in a state of hierarchy the elite (the state) chooses its tax policy to maximize the revenue net of the cost of tax collection. In order to facilitate comparison between the two regimes, we assume that the state has access to the same expropriation technology as bandits. Namely, the state cannot tax tubers, and if it employs a measure λ of tax collectors (hired from among the potential foragers) at cost s per tax collector, it can generate revenue of $\tau(\lambda)\beta$ from the farming sector. Adopting Weber's definition of the state, we also assume that a state has to have a monopoly power over the use of force, and thus be able to deter bandits. This deterrence power, we assume, entails a fixed cost $G_0 > 0$.

A key advantage that a state has is that it is farsighted and organized, and can thus commit not to expropriate farmers beyond a certain tax rate.⁶² That is, the state selects a tax rate τ , and hires $\lambda(\tau)$ tax-collectors at cost $s\lambda(\tau)$, to maximize its net revenue, subject to the constraint that farmers respond to the tax rate:

$$\max_{\tau \geq 0} R(\tau) = \tau\beta - s\lambda(\tau),$$

subject to

$$\beta = \arg \max_{\beta' \in [0,1]} \{(1 - \delta) + \beta'(\delta - \tau)\}.$$

It is evident that $\beta = 0$ if $\tau > \delta$. This implies that in the optimum $\tau \leq \delta$. In addition $\beta = 1$ if $\tau < \delta$. Assuming that $\beta = 1$ when $\tau = \delta$, the state's problem becomes:

$$\max_{\tau} [\tau - s\lambda(\tau)], \text{ subject to } \tau \leq \delta.$$

The optimal tax rate under hierarchy is therefore: $\tau_H(\delta) = \min\{\delta, \delta_H\}$, where δ_H is the parameter that solves $s\lambda'(\delta_H) = 1$. Thus, at the high range of tubers' productivity, where $\delta < \delta_H$, $\tau_H = \delta$, the net revenue is $R(\tau_H(\delta)) = \delta - s\lambda(\delta)$, which is increasing in δ . Our assumption that the state has to sustain deterrence against bandits at a fixed cost $G_0 > 0$ sets a lower limit on the

⁶² Another difference between bandits and the state is that bandits confiscate a farmer's entire cereal crop with probability τ , while an organized hierarchy taxes farmers at the rate τ with certainty. If farmers are risk neutral, as assumed here, this difference is unimportant.

net revenue for the state to be viable. Clearly if G_0 exceeds the maximal revenue $R(\delta_H)$, the state is not viable. We thus assume that $R(\delta_H) > G_0$. Under this assumption we can define a viability threshold $\underline{\delta} < \delta_H$, such that: $R(\underline{\delta}) = G_0$, so that the state is viable when $\delta > \underline{\delta}$.

Proposition 2. (i) If δ is small ($\delta < \underline{\delta}$), a state cannot exist. (ii) In the intermediate range where $\underline{\delta} \leq \delta < \delta_H$, the optimal tax rate is $\tau_H = \delta$. (iii) If δ is large ($\delta \geq \delta_H$), then the optimal tax rate is equal to δ_H .

Income distribution. Under hierarchy, farmers grow only cereals. Thus, their income is $1 - \tau_H = 1 - \min\{\delta, \delta_H\}$, which is weakly decreasing in the cereal productivity advantage over tubers δ . Total tax receipts equals τ_H , and the net tax revenue to the elite is: $\tau_H - s\lambda(\tau_H) - G_0$. Both increase in δ up to the threshold δ_H , after which they remain constant.

Output Loss. Analogously to the case of anarchy, we obtain here:

$$Y_0 - Y = (1 - \beta_H) \delta + s\lambda(\tau_H) + G_0 \text{ and since } \beta_H = 1, Y_0 - Y = s\lambda(\tau_H) + G_0. \text{ Thus:}$$

Corollary 2. The output loss due to hierarchy is:

$$Y_0 - Y = \begin{cases} s\lambda(\delta) + G_0 & \text{if } \delta < \delta_H \\ s\lambda(\delta_H) + G_0 & \text{if } \delta \geq \delta_H \end{cases}.$$

Anarchy vs. Hierarchy

As explained in the previous sub-section, a state can only exist if tubers are sufficiently unattractive to farmers, that is, if the productivity advantage of cereals, δ , is above the threshold $\underline{\delta}$. The comparison between the regimes of anarchy and hierarchy depends on the relationship between the thresholds δ_A , δ_H and $\underline{\delta}$.

These threshold levels satisfy: $\delta_A > \delta_H > \underline{\delta}$. Since δ_H is defined by $\tau'(\lambda(\delta_H)) = s$, and δ_A is defined by $\delta_A/\lambda(\delta_A) = s$, $\delta_A > \delta_H$ follows from the strict concavity of $\tau(\lambda)$.

Proposition 3. For $\delta > \underline{\delta}$ the state is viable and:

(i) Hierarchy weakly Pareto dominates anarchy. (ii) The economy is more productive under hierarchy than under anarchy.

Proof. (i) Because the function $\tau(\cdot)$ is strictly concave, the marginal productivity of tax collectors (or bandits) is lower than the average productivity: $\tau'(\lambda) < \tau(\lambda)/\lambda$ and $\tau'(\lambda(\tau)) < \tau/\lambda(\tau)$.

Recall that, $\lambda(\delta_H)$ is defined by $\tau'(\lambda(\delta_H)) = s$ and $\lambda(\delta_A)$ is defined by $\delta_A/\lambda(\delta_A) = s$. It therefore follows from the concavity of $\tau(\cdot)$ that $\delta_H < \delta_A$ and $\lambda(\delta_H) < \lambda(\delta_A)$.

Non-farmers earn the same income s irrespective of the regime. Suppose that $\delta > \underline{\delta}$. On the other hand, the implied tax rate on farmers under anarchy is larger than or equal than the tax rate under hierarchy. In the range where $\underline{\delta} \leq \delta \leq \delta_H$, the tax rate under both anarchy and hierarchy is δ ; in the range $\delta_H \leq \delta < \delta_A$ the tax rate under anarchy δ is higher than the tax rate under hierarchy δ_H and in the range $\delta_A \leq \delta$ the tax rate under anarchy is δ_A , whereas under hierarchy it is lower δ_H . Hence, farmers are weakly better off in all cases under hierarchy than under anarchy. Finally, when $\delta > \underline{\delta}$, a hierarchy generates an additional surplus to the elite, since by construction: $\tau - s\lambda(\tau) - G_0 > 0$. ■

(ii) From corollaries 1 and 2 we obtain that the difference between total output under hierarchy to that under anarchy is equal to:

$$Y_H(\delta) - Y_A(\delta) = \begin{cases} \delta - s\lambda(\delta) - G_0 & \text{if } \delta \in [\underline{\delta}, \delta_H] \\ \delta - s\lambda(\delta_H) - G_0 & \text{if } \delta \in (\delta_H, \delta_A] \\ \delta_A - s\lambda(\delta_H) - G_0 & \text{if } \delta > \delta_A \end{cases} .$$

When $\delta = \underline{\delta}$, by the definition of $\underline{\delta}$, $R(\underline{\delta}) = \underline{\delta} - s\lambda(\underline{\delta}) = G_0$, the output gap between the two regimes is zero. When $\underline{\delta} \leq \delta \leq \delta_A$, the output gap equals the rent enjoyed by the elite, which is increasing in δ . ■

The total output under hierarchy is weakly higher for two reasons. (1) Under hierarchy farmers cultivate only cereals. Thus they do not resort to self-protection through the cultivation of the less productive tubers, as they do (when $\delta < \delta_A$) under anarchy. (2) The state employs (weakly) fewer tax collectors than the scale of foragers who engage in banditry under anarchy, since with the state their marginal product is higher or equals their cost s , whereas under anarchy it is their average product that equal s .

The main predictions of the analysis

1. Farmers may choose to grow tubers even when tubers are less productive for the purpose of self-protection against appropriation by bandits or by tax collectors.
2. If the productivity advantage of cereals over tubers is sufficiently small, $\delta < \underline{\delta}$, a state cannot exist. This result illustrates our claim that it isn't low productivity that hinders the development of hierarchy and related institutions, but rather relatively high productivity of

crops that are hard to expropriate. A hierarchy could emerge if the productivity advantage of cereals is sufficiently high: $\delta > \underline{\delta}$.

3. Whenever it exists, even a non-benevolent state hierarchy that monopolizes coercive force dominates anarchy from an efficiency point of view. This follows from our assumption that the state can commit to a tax rate that maximizes its net revenue, and that consequently farmers cultivate only the more efficient cereals.
4. If cereals are sufficiently more productive than tubers, a state of hierarchy is a possible outcome but anarchy can persist. A reasonable interpretation for a transition from anarchy to hierarchy, is that coordination is required, either within the emerging state, by the farmers who seek protection, or by the outsider bandits under the leadership of the would be elite. The coordination problem could explain the lag of several millennia between the adoption of cereal cultivation, for instance in the fertile crescent, and the emergence of early states (Scott, 2017).

Example

A simple example helps to illustrate our results diagrammatically. Consider the following specification for the expropriation function: $\tau(\lambda) = \rho\sqrt{\lambda}$, with $\rho \in (0, 1)$.

In this case, $\delta_A = \rho^2/s$ and $\delta_H = \alpha^2/2s$.⁶³ The equilibrium under anarchy is given by

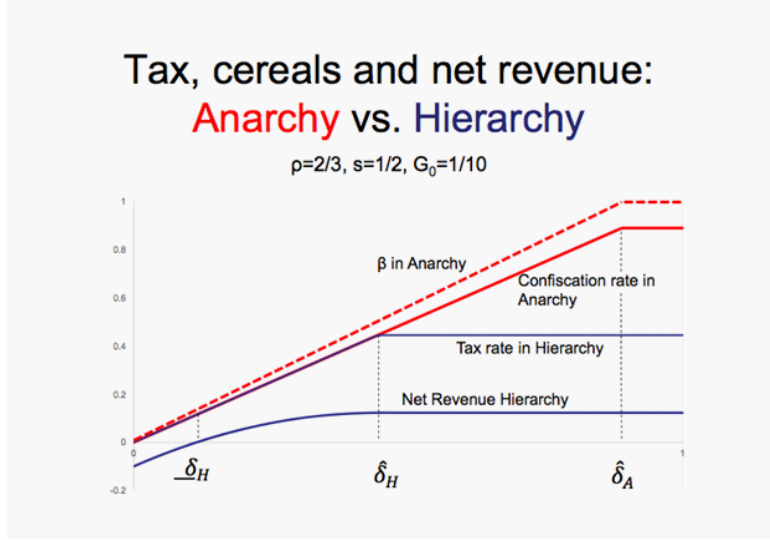
$$(\beta_A, \tau_A) = \begin{cases} \left(\frac{s\delta}{\rho^2}, \delta \right) & \text{if } \delta < \delta_A \\ \left(1, \frac{\rho^2}{s} \right) & \text{if } \delta \geq \delta_A \end{cases}.$$

For $\underline{\delta} \leq \delta \leq \delta_H$ a state sets a tax rate equal to δ and generates net tax revenue: $R(\delta) = \delta - s \left(\frac{\delta}{\rho} \right)^2$, which increases in δ up to the point where $\delta = \delta_H$ after which $R(\delta) = R(\delta_H)$. Figure D1 presents the comparison between anarchy and hierarchy with respect to the tax rate and the production of cereals, as a function of δ . It also presents the net revenue of the elite in a regime of hierarchy.

Risk averse farmers. We illustrate here the robustness of the model's qualitative predictions when farmers are risk averse. The results are in a sense even stronger, given that risk-averse farmers under anarchy seek more protection by choosing a smaller share of cereals. Farmers' risk aversion

⁶³The lower limit for state existence, $\underline{\delta} > 0$, is implicitly defined by the quadratic equation: $\underline{\delta} - s \left(\frac{\underline{\delta}}{\rho} \right)^2 = G_0$, where to have any solution we require that : $G_0 \leq \rho^2/4s$.

Figure D.1: Tax, cereals and net revenue: Anarchy vs. Hierarchy



does not affect the analysis of the model under a regime of hierarchy since in this case the tax rate that the state imposes is certain. We chose to illustrate the case of anarchy with risk-averse farmers by examining a case where a simple analytic solution can be obtained. For that purpose, we employ the specification of the expropriation function, $\tau(\lambda) = \rho\sqrt{\lambda}$, as in the model's example, and consider the case where farmers have a log-utility function: $u(I) = \log(I)$. Farmers under anarchy thus chose $\beta \geq 0$ to maximize the expected utility:

$$U(I) = (1 - \tau) \log(\beta + (1 - \delta)(1 - \beta)) + \tau \log(1 - \delta)(1 - \beta).$$

The solution is

$$\beta_A = \max \left\{ \frac{\delta - \tau}{\delta}, 0 \right\}.$$

Non-farmers' freedom to enter banditry implies: $s = \tau\beta/\lambda(\tau)$. And thus:

$$\tau_A = \frac{\rho^2 \beta_A}{s}.$$

Solving for the equilibrium values of (β_A, τ_A) yields (when $\beta_A > 0$):

$$\beta_A = \frac{s\delta}{\rho^2 + s\delta}; \quad \tau_A = \frac{\rho^2 \delta}{\rho^2 + s\delta}.$$

Inspection of the equilibrium values of (β_A, τ_A) reveals that as δ tends to zero, both β_A and τ_A tend to zero. As δ increases towards one, τ_A approaches $\rho^2/(\rho^2 + s)$ and β_A approaches $s/(\rho^2 + s)$. This implies that even in the limit, when the productivity of tubers approaches zero, they are still grown by farmers.

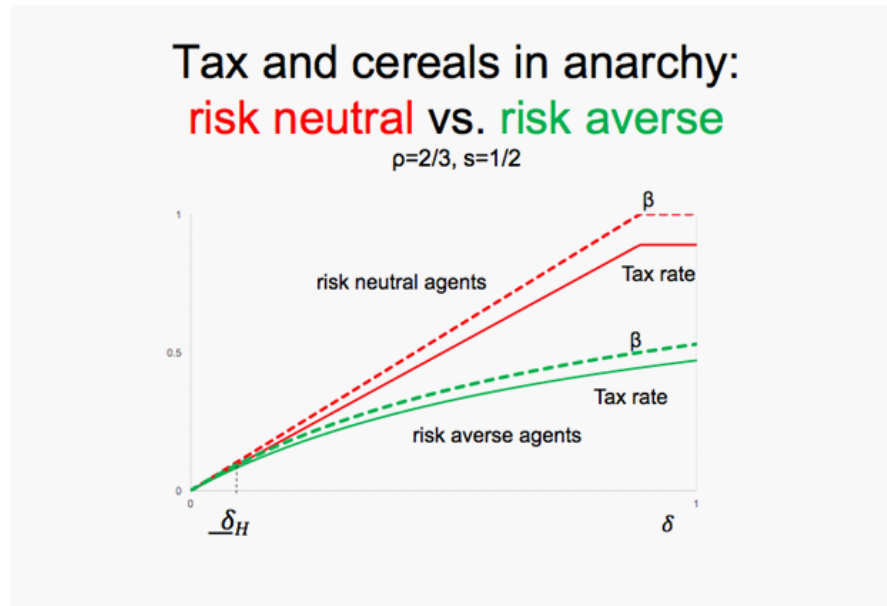
Compared to the model with risk neutrality (in the preceding sub-section), the introduction of risk aversion implies that farmers reduce the cultivation of cereals β_A , and increase the share of land devoted to tubers as a device for self-insurance. Consequently the confiscation rate τ_A is lower, and the measure of banditry λ_A is smaller as well.

While the former effect tends to increase overall inefficiency, the total efficiency effect of introducing risk aversion in a regime of anarchy is positive. To recall from corollary 1, under risk neutrality the overall inefficiency $(1 - \beta_A) \delta + s \lambda_A$ is equal to δ . This is smaller than the inefficiency under risk aversion, which under our specification is equal to $(1 - \beta_A) \delta + \lambda_A s = \delta - \beta_A (\delta - \tau_A) < \delta$. Correspondingly, the expected income of each farmer under anarchy is also higher under risk aversion, because

$(1 - \tau_A) (\beta_A + (1 - \delta) (1 - \beta_A)) + \tau_A (1 - \delta) (1 - \beta_A) = 1 - \delta + (\delta - \tau_A) \beta_A$ is equal to $1 - \delta$ under risk neutrality, but is strictly larger under risk aversion because under risk aversion $\tau_A < \delta$. The reason for this is that under risk neutrality farmers in a mixed equilibrium are indifferent between growing cereals and tubers and so derive an identical income of $1 - \delta$. In contrast, under risk aversion, farmers derive a strictly larger expected income from cereals to compensate for the risk associated with cereals, which pushes their expected income higher.⁶⁴ The figure illustrates the difference between the two types of equilibrium: the case of risk neutral farmers and risk averse farmers.

⁶⁴This implies that risk neutral farmers would benefit if they could commit to grow less cereals in equilibrium, which we assume they cannot. The problem is that when a farmer decides how much cereal to grow, he ignores the negative externality this imposes on other farmers through contributing to the measure of bandits.

Figure D.2: Output: Anarchy vs. Hierarchy



Appendix E: Wild relatives of domesticated crops

In this appendix, we report the list of the primary wild relatives of the most relevant domesticated cereals, roots and tubers. An asterisks indicates that data on the potential distribution of the wild relative are not available.

BARLEY

Hordeum vulgare spontaneum.

CASSAVA

Manihot esculenta, *Manihot esculenta subsp. flabellifolia**, *Manihot esculenta subsp. peruviana*, *Manihot pruinosa*.

MAIZE

*Zea mays subsp. huehuetenangensis**, *Zea mays subsp. mexicana*, *Zea mays subsp. parviglumis**.

FOXTAIL MILLET

Setaria italica, *Setaria viridis*.

OAT

Avena byzantina, *Avena fatua*, *Avena hybrida*, *Avena occidentalis*.

POTATO

Solanum acaule, *Solanum berthaultii*, *Solanum brevicaulle*, *Solanum candolleanum*, *Solanum coelestipetalum**, *Solanum infundibuliforme*, *Solanum marinasense**, *Solanum okadae*, *Solanum ortegae**, *Solanum vernei*.

RICE

Oryza barthii, *Oryza glaberrima*, *Oryza glumipatula*, *Oryza longistaminata*, *Oryza meridionalis*, *Oryza nivara*, *Oryza rufipogon*.

RICE BEAN

Vigna umbellata var. *gracilis*.

AFRICAN RICE

Oryza barthii, *Oryza glaberrima*, *Oryza glumipatula*, *Oryza longistaminata*, *Oryza meridionalis*, *Oryza nivara*, *Oryza rufipogon*,

RYE

*Secale cereale subsp. afghanicum**, *Secale cereale subsp. ancestrale*, *Secale cereale subsp. dighoricum**, *Secale cereale subsp. segetale*.

SORGHUM

*Sorghum bicolor**, *Sorghum bicolor* subsp.*verticilliflorum*, *Sorghum propinquum*.

WHEAT

Triticum aestivum subsp. *tibeticum**, *Triticum monococcum*, *Triticum monococcum* subsp.*aegilopoides*, *Triticum timopheevii*, *Triticum timopheevii* subsp.*armeniicum*, *Triticum turgidum* subsp.*dicoccoides*, *Triticum turgidum* subsp.*paleocolchicum*, *Triticum urartu*.

LAGOS YAM

Dioscorea abyssinica, *Dioscorea burkilliana*, *Dioscorea cayenensis*, *Dioscorea cayenensis* subsp.*rotundata*, *Dioscorea liebrechtsiana*, *Dioscorea minutiflora*, *Dioscorea praehensilis*, *Dioscorea rotundata**, *Dioscorea smilacifolia*.

WATER YAM

Dioscorea nummularia, *Dioscorea transversa*.

WHITE GUINEA YAM

Dioscorea abyssinica, *Dioscorea cayenensi*, *Dioscorea cayenensis* subsp.*rotundata*, *Dioscorea liebrechtsiana*, *Dioscorea praehensilis*, *Dioscorea rotundata**.

Appendix F: Additional data tables and figures

F.1 Descriptive statistics

Table F.1: Descriptive Statistics: societies in Ethnoatlas

	SOURCE	Mean	p50	SDev	Min	Max	N
<i>Hierarchy</i>	Ethnoatlas	1.89	2.00	1.04	1.00	5.00	1,059
<i>CerMain</i>	Ethnoatlas	0.54	1.00	0.50	0.00	1.00	1,092
Dependence on agriculture	Ethnoatlas	0.45	0.50	0.27	0.03	0.93	1,178
Dependence on herding > 5%	Ethnoatlas	0.66	0.00	0.47	0.00	1.00	1,178
Herding main animals: ruminants	Ethnoatlas	0.51	1.00	0.50	0.00	1.00	1,179
Animals in cultivation	Ethnoatlas	0.14	0.00	0.35	0.00	1.00	1,086
Farming surplus	Tuden and Marshall (1972)	0.49	0.00	0.50	0.00	1.00	162
Tax burden	Ross (1983)	1.11	1.00	0.90	0.00	2.00	66
Population density (categorical)	Pryor (1985)	3.83	4.00	1.57	2.00	7.00	168
<i>LandProd</i> (std)	authors	0.00	0.23	1.00	-1.92	2.66	1,179
<i>CerAdv</i> (std)	authors	0.00	-0.13	1.00	-1.73	4.16	1,179
Precipitation (std)	FAO-GAEZ	0.00	-0.13	1.00	-1.39	10.65	1,179
Temperature (std)	FAO-GAEZ	0.00	0.37	1.00	-2.57	1.32	1,179
Elevation (std)	GDEM	0.00	0.17	1.00	-9.24	3.58	1,179
Ruggedness (std)	GDEMs	0.00	-0.35	1.00	-0.90	6.41	1,179
Absolute Latitude (std)	Ethnoatlas	0.00	-0.43	1.00	-1.21	3.36	1,179
Distance to major river (std)	Fenske (2013)	0.00	-0.63	1.00	-0.63	1.58	1,179
Distance to coast (std)	Fenske (2013)	0.00	-0.30	1.00	-1.11	3.14	1,179
Pct malaria	MAP	0.17	0.06	0.21	0.00	0.69	1,179
Historical population density (std)	HYDE	0.00	-0.23	1.00	-0.30	25.85	1,179
Plow Advantage (std)	FAO-GAEZ	-0.00	0.31	1.00	-2.83	2.61	1,179
Irrigation Potential (std)	authors	0.00	-0.54	1.00	-0.54	2.45	1,045
% Fertile land	Ramankutty et al (2002)	-0.00	-0.03	1.00	-1.43	2.53	1,134
Caloric Suitability Index (std)	Galor and Ozak (2015)	0.00	0.28	1.00	-1.95	2.63	1,179

FAO GAEZ v3 database downloaded on 15/01/2016. std - a standardized variable that has been rescaled to have a mean of zero and a standard deviation of one.

Table F.2: Descriptive Statistics: Countries X 50 years

	SOURCE	Mean	p50	SDev	Min	Max	N
<i>Hierarchy</i>	Borcan et al. (2018)	0.72	1.00	0.45	0.00	1.00	2,869
<i>LandProd</i> (std)	authors	0.00	0.35	1.00	-1.64	2.69	2,959
<i>CerAdv</i> (std)	authors	0.00	-0.00	1.00	-1.49	3.12	2,959
WR of cereals	authors	0.49	0.00	0.50	0.00	1.00	2,959
WR of roots and tubers	authors	0.05	0.00	0.22	0.00	1.00	2,959
WR of cereals, roots and tubers	authors	0.34	0.00	0.47	0.00	1.00	2,959
Precipitation (std)	FAO-GAEZ	0.00	-0.29	1.00	-1.38	2.89	2,940
Temperature (std)	FAO-GAEZ	0.00	0.20	1.00	-2.68	1.52	2,884
Elevation (std)	GDEM	0.00	-0.33	1.00	-1.10	4.65	2,845
Ruggedness (std)	GDEM	0.00	-0.31	1.00	-1.12	4.25	2,959
Absolute latitude (std)	Nunn and Puga (2012)	0.00	-0.17	1.00	-1.51	2.18	2,959
Legal origin: English common law	La Porta et al. (1999)	0.27	0.00	0.44	0.00	1.00	2,959
Legal origin: French civil law	La Porta et al. (1999)	0.45	0.00	0.50	0.00	1.00	2,959
Legal origin: Socialist law	La Porta et al. (1999)	0.22	0.00	0.41	0.00	1.00	2,959
Legal origin: German civil law	La Porta et al. (1999)	0.03	0.00	0.18	0.00	1.00	2,959
Legal origin: Scandinavian law	La Porta et al. (1999)	0.03	0.00	0.18	0.00	1.00	2,959
Population density 1500 (std)	Acemoglu et al. (2002)	0.00	-0.05	1.00	-2.96	2.78	2,959
Mortality of early settlers (std)	Acemoglu et al. (2002)	0.00	-0.11	1.00	-2.91	2.56	1,519
Slaves exported (std)	Nunn (2008)	0.00	-0.26	1.00	-0.26	9.01	2,959
Genetic diversity (std)	Ashraf and Galor (2013)	0.00	0.24	1.00	-3.66	1.74	2,675
Hist. domesticable animals/area	Ashraf and Galor (2011)	2.10	0.04	14.79	0.00	168.00	2,941
Hist. domesticable ruminants/area	authors	0.85	0.01	6.33	0.00	72.00	2,941
Distance to major river (std)	www.pdx.edu/econ/	0.00	-0.29	1.00	-0.89	7.63	2,845
Distance to coast (std)	www.pdx.edu/econ/	0.00	-0.41	1.00	-0.75	4.48	2,845
Pct malaria	MAP	0.65	0.94	0.41	0.00	1.00	2,883
% country with tropical climate (std)	Nunn and Puga (2012)	0.35	0.00	0.43	0.00	1.00	2,959
Potential irrigation	Bentzen et al. (2017)	0.18	0.02	0.30	0.00	1.00	2,921
Caloric suitability index (std)	Galor and Ozak (2015)	0.00	0.29	1.00	-1.82	2.93	2,959

FAO GAEZ v3 database downloaded on 15/01/2016. std - a standardized variable that has been rescaled to have a mean of zero and a standard deviation of one.

Table F.3: Descriptive Statistics: 1x1 decimal degree pixel

	SOURCE	Mean	p50	SDev	Min	Max	N
Panel A: <i>cross-sectional data-15,927 raster points</i>							
City founded <AD 450 (dummy)	Reba et al. (2016)	0.01	0.00	0.08	0.00	1.00	15,927
City founded <500 BC (dummy)	Reba et al. (2016)	0.00	0.00	0.06	0.00	1.00	15,927
City founded <AD 400 (dummy)	DeGroff (2009)	0.05	0.00	0.22	0.00	1.00	15,927
Archaeological sites	ANCIENTLOCATIONS.NET	0.24	0.00	2.58	0.00	138.00	15,927
Pyramids or Mastaba	MEGALITHIC.CO.UK	0.01	0.00	0.75	0.00	87.00	15,927
Temples	MEGALITHIC.CO.UK	0.04	0.00	0.64	0.00	46.00	15,927
Mines	MEGALITHIC.CO.UK	0.01	0.00	0.21	0.00	22.00	15,927
Palaces	MEGALITHIC.CO.UK	0.00	0.00	0.06	0.00	5.00	15,927
Sculptured Stones	MEGALITHIC.CO.UK	0.02	0.00	0.83	0.00	101.00	15,927
Standing Stones	MEGALITHIC.CO.UK	0.04	0.00	0.71	0.00	45.00	15,927
<i>LandProd</i> (std)	authors	-0.70	-1.28	1.26	-1.78	3.76	15,927
<i>CerAdv</i> (std)	authors	-0.93	-1.40	1.03	-2.67	2.96	15,927
WR of cereals	authors	0.19	0.00	0.39	0.00	1.00	15,927
WR of roots and tubers	authors	0.08	0.00	0.27	0.00	1.00	15,927
WR of cereals, roots and tubers	authors	0.05	0.00	0.23	0.00	1.00	15,927
<i>DistanceCer</i>	authors	28.71	24.72	22.78	0.00	270.26	15,927
<i>DistanceAgr</i>	authors	19.02	15.51	17.57	0.00	234.51	15,927
Precipitation (std)	FAO-GAEZ	0.00	-0.32	1.00	-1.06	9.24	15,862
Temperature (std)	FAO-GAEZ	0.00	-0.32	1.00	-1.33	1.81	15,833
Elevation (std)	GDEM	-0.00	-0.34	1.00	-0.87	6.01	15,927
Ruggedness (std)	GDEM	-0.00	0.35	1.00	-3.38	1.10	15,927
Absolute Latitude	authors	40.52	41.50	22.20	0.50	83.50	15,927
Irrigation Potential (std)	authors	0.00	0.67	1.00	0.67	1.74	8,214
Plow Advantage (std)	authors	0.00	0.05	1.00	-3.00	3.44	15,927
Population density 1995 (std)	FAO-GAEZ	0.00	-0.59	1.00	-0.76	3.58	15,861
Panel B: <i>panel data-15,927 raster points with observations either before or after Neolithic transition</i>							
Prehistoric archaeological sites	D. and R. Whitehouse (1975)	0.11	0.00	0.71	0.00	23.00	31,854
Prehistoric settlements	D. and R. Whitehouse (1975)	0.08	0.00	0.59	0.00	21.00	31,854

FAO GAEZ v3 database downloaded on 15/01/2016. std - a standardized variable that has been rescaled to have a mean of zero and a standard deviation of one.

Table F.4: Caloric content of cereals, roots and tubers

Crop	Energy	Crop	Energy
Barley	3.52	Sorghum	3.39
Buckwheat	3.43	Sweet Potato	0.86
Cassava	1.6	Wetland Rice	3.7
Foxtail Millet	3.78	Wheat	3.47
Indigo Rice	3.7	White Potato	0.77
Maize	3.65	Yams	1.18
Oat	2.46	Sorghum	3.39
Rye	3.38		

Values are in kilo calories per 100g. Source: Galor and Ozak (2015) and USDA Nutrient Database for Standard Reference (R25). The data source in table A1 is different, and therefore the caloric content reported there is slightly different as well.

Table F.5: Pairwise correlations of the main variables used in the empirical analysis on the societies in the Ethnoatlas

Variables	<i>Hier.</i>	<i>CerMain</i>	Dep. agric.	Farm. surp.	H. Pop dens.	<i>LandProd</i>	<i>CerAdv</i>	% Fertile land	Caloric suit. ind.
<i>Hierarchy</i>	1.0								
<i>CerMain</i>	0.3	1.0							
Dependence agriculture	0.4	0.5	1.0						
Farming surplus	0.6	0.4	0.3	1.0					
Hist Pop density (Pryor)	0.6	0.5	0.7	0.4	1.0				
<i>LandProd</i>	0.2	0.3	0.4	0.2	0.3	1.0			
<i>CerAdv</i>	0.2	0.4	0.3	0.3	0.2	0.8	1.0		
% Fertile land	0.2	0.2	0.3	0.2	0.3	0.4	0.5	1.0	
Caloric suitability index	0.2	0.3	0.5	0.2	0.3	1.0	0.8	0.5	1.0

F.2 Pre-industrial societies: the choice of the crop

We formally analyze farmers' choice in the model presented in Appendix D. Farmers decide whether to cultivate cereals rather than roots or tubers based on two features of the environment: the productivity advantage of cereals over roots and tubers, which is exogenous, and the greater vulnerability of cereals to appropriation by bandits or the state, which is determined endogenously. To capture the response of the crop choice to geography, we run the following regression:

$$Y_i = \alpha CerAdv_i + X_i' \beta + \varepsilon_i, \quad (F.1)$$

where the dependent variable Y_i is either a dummy variable that identifies societies choosing cereals as the main crop ($CerMain_i$) or a measure of the reliance of these societies on agriculture for their diet ($AgrRely_i$); $CerAdv_i$ is the caloric advantage of cereals in the land of society i (the difference between the maximum potential caloric yield of cereals and of roots or tubers), and X_i' is a vector of control variables. The caloric advantage of cereals is constructed using the procedure detailed in section 2.1.5, under the assumption that the Columbian exchange was completed and every continent had potential access to every crop.⁶⁵

Results are reported in Table F.6. Column 1 reports the bivariate relationship between cereal advantage and the decision to cultivate cereals as the main crop, without any controls. The association is positive and statistically significant. An increase in the productivity advantage of cereals over roots and tubers by one standard deviation is associated with an increase in the probability of growing cereals as the main crop by about 20 percent. Moreover, variation in this regressor alone is able to explain 13 percent of the variation in the dependent variable. Column 2 reports the results when adding as a control variable land productivity (the highest potential caloric yield across all 11 cereals and 4 roots/tubers), to address the concern that the productivity advantage of cereals might reflect land productivity. The impact of the productivity advantage of cereals is unchanged, land productivity doesn't have any significant impact on crop choice and the R^2 of the regression is practically unchanged. Results hold with continental fixed effects (in column 3), and when resorting to logistic estimation to account for the binary nature of the dependent variable (columns 4 and 5).

The main results survive when controlling sequentially for: precipitation, temperature, elevation, and ruggedness, which are the main factors affecting crop productivity in the GAEZ dataset (Table F.7). We also control for: geographical isolation (proxied by the distance to the nearest

⁶⁵This is a reasonable assumption as the great majority of the societies in the Ethnoatlas are captured between the end of the 19th century and the beginning of the 20th century.

major river or coast), historical and current population density, evidence of intensive irrigation, and the productivity advantage from using the plow (Table F.8). In all cases, the qualitative results on the effect of cereal productivity advantage over roots and tubers are almost unaffected: the coefficients vary from 0.250 to 0.261 and are always statistically significant at the 1 percent confidence level.

Table F.6: Potential Crop Yields, Choice of Crops and Reliance on Agriculture

Dependent variable is:								
Major crop is cereal grains (<i>CerMain</i> dummy)					Reliance on agriculture (<i>AgrRely</i>)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	OLS	OLS	Logit	Logit	OLS	OLS	OLS
<i>CerAdv</i>	0.205 (0.028)*** (0.024)*** (0.029)***	0.210 (0.086)** (0.050)*** (0.063)***	0.253 (0.072)*** (0.047)*** (0.059)***	1.150 (0.416)*** (0.276)*** (0.339)***	1.617 (0.431)*** (0.316)*** (0.380)***	0.081 (0.019)*** (0.016)*** (0.022)***	-0.098 (0.050)** (0.023)*** (0.029)***	-0.046 (0.021)** (0.019)** (0.022)**
<i>LandProd</i>		-0.007 (0.097) (0.063) (0.083)	-0.137 (0.079)* (0.056)** (0.069)**	-0.119 (0.427) (0.292) (0.384)	-0.896 (0.451)** (0.337)** (0.407)**		0.230 (0.0660)*** (0.033)*** (0.046)***	0.128 (0.031)*** (0.029)*** (0.035)***
CONT. FE	NO	NO	YES	NO	YES	NO	NO	YES
Ave marg. effect of <i>CerAdv</i>				0.282*** (0.416)*** (0.066)*** (0.081)***	0.385*** (0.431)*** (0.075)*** (0.092)***			
r ²	0.609	0.609	0.711			0.0733	0.235	0.387
pseudo r ²				0.109	0.258			
N	982	982	982	982	982	1063	1063	1063

The table reports cross-sectional OLS and Logit estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors in curly brackets are clustered at the country levels; standard errors in square brackets and round brackets are adjusted for spatial correlation using the methodology in Conley (1999) with a distance cut-off of respectively 5 and 10 decimal degrees. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

In the last three columns of Table F.6 the reliance of the society on agriculture is the dependent variable. As reasonably expected, land productivity increases the probability of reliance on farming. Interestingly, the productivity advantage of cereals has a negative effect on farming, when controlling for land productivity. That is, for the same level of land productivity (measured in calories), cultivating roots and tubers is more rewarding than cultivating cereals. This is consistent with our thesis, as cereals are more vulnerable to expropriation.

Table F.7: Potential Crop Yields and Choice of Crops. Robustness checks: Controlling for Geography.

	Dep. Variable: Major crop is a cereal grain (dummy)			
	(1)	(2)	(3)	(4)
<i>CerAdv</i>	0.214*** (0.057)	0.274*** (0.054)	0.248*** (0.059)	0.250*** (0.059)
<i>LandProd</i>	-0.088 (0.067)	-0.174*** (0.064)	-0.132* (0.069)	-0.133* (0.070)
Precipitation	-0.058* (0.031)			
Temperature		0.066** (0.033)		
Elevation			0.030* (0.017)	
Ruggedness				0.012 (0.026)
CONTINENT FE	YES	YES	YES	YES
r ²	0.367	0.364	0.362	0.359
N	982	982	982	982

The table reports cross-sectional OLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.8: Potential Crop Yields and Choice of Crops. Robustness checks: Controlling for Isolation, Population Density and the Plow.

	Dep. Variable: Major crop is a cereal grain (dummy)						
	(1)	(2)	(5)	(6)		(7)	(8)
<i>CerAdv</i>	0.252*** (0.058)	0.250*** (0.060)	0.253*** (0.059)	0.261*** (0.071)	0.254*** (0.059)	0.253*** (0.058)	0.257*** (0.049)
<i>LandProd</i>	-0.137** (0.067)	-0.135* (0.069)	-0.136** (0.069)	-0.222*** (0.086)	-0.139** (0.069)	-0.204*** (0.060)	-0.205*** (0.061)
Major River	-0.028* (0.017)						
Distance Coast		0.016 (0.035)					
Hist Pop Dens (HYDE)			-0.015 (0.016)				
Hist Pop Dens (PRYOR)				0.206*** (0.038)			
Pop Dens 1995					-0.004 (0.025)		
Irrigation						0.191*** (0.049)	
Plow Advantage							-0.148*** (0.033)
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
r ²	0.362	0.360	0.360	0.383	0.348	0.396	0.398
N	982	982	982	144	966	800	982

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F3. Pre-industrial societies: reduced-form estimates

Table F.9: Cereals and Hierarchy - Reduced Form

Jurisdictional Hierarchy Beyond Local Community	Dependent variable is:		
	(1)	(2)	(3)
	OLS	OLS	OLS
<i>CerAdv</i>		0.138 {0.063}** [0.049]*** (0.057)**	0.274 {0.106}*** [0.096]*** (0.107)**
<i>LandProd</i>	0.087 {0.089} [0.054] (0.064)		-0.188 {0.123} [0.102]* (0.108)*
CONTINENT FE	YES	YES	YES
r ²	0.821	0.824	0.825
N	952	952	952

The table reports cross-sectional OLS estimates. The unit of observation is the society in Murdock's Ethnoatlas. Standard errors in curly brackets are clustered at the country levels; standard errors in square brackets and round brackets are adjusted for spatial correlation using the methodology in Conley (1999) with a distance cut-off of respectively 5 and 10 decimal degrees. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.10: Cereals and Hierarchy - Reduced Form using ordered logit and generalized ordered logit

	Dependent variable: Jurisdictional Hierarchy Beyond Local Community				
	Order Logit	Generalized logit			
		Hierarchy<=1 vs Hierarchy>1	H<=2 vs H>2	H<=3 vs H>3	H<=4 vs H>4
<i>CerAdv</i>	0.495*** (0.149)	0.327 (0.2442)	0.542** (0.2363)	0.674** (0.3114)	0.841** (0.4132)
<i>LandProd</i>	-0.224 (0.178)	0.0596 (0.2352)	-0.392 (0.2916)	-0.485 (0.3595)	-0.597 (0.4991)

The table reports the estimates from both an ordered logit and a generalized ordered logit. All regressions include continent fixed effects. The unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.4 Pre-industrial societies: robustness checks on the 2SLS estimates

In this subsection, we introduce a battery of robustness checks on the results presented in Table 1 in the main text.

In Table F.11, we control sequentially for precipitation, temperature, elevation, ruggedness, latitude/longitude – the main factors affecting crop productivity. The estimated coefficient on *CerMain* is always positive and close to 1. In two cases, however, the coefficient is estimated less precisely and is barely below statistical significance. Table F.12 addresses other potential channels, through which the cultivation of cereals might affect economic development and hierarchical complexity. Since cereals can be stored and transported, their cultivation facilitates trade, which could give rise to leadership and taxation. In an attempt to mitigate the possibility that the effect of cereals on hierarchy is only a result of trade, we control for geographical isolation (proxied by distance from a major river and distance from the coast). There is some evidence that close proximity to a river is associated with more complex hierarchies, but the impact of cereals on hierarchy is unaffected. We control also for historical population density,⁶⁶ which is associated in several theories with increased hierarchy (and with economic and technological development). Cereals still have a large impact on hierarchy when adding these controls. The data, as one would expect, reveal a positive correlation between population density and hierarchical complexity. We control for evidence of agriculture based on intensive irrigation. This could be an important potential confounder as Bentzen et al. (2017) provide evidence of a causal impact of irrigation potential on autocracy. Our estimates confirm the validity of their results: societies that practice intensive irrigation are characterized by relatively more complex hierarchies. Our results on the impact of cereals are unchanged. Finally, results are robust to the inclusion of the productivity advantage of the plow, a variable that Alesina, Giuliano and Nunn (2013) have identified as an important determinant of gender roles.

The qualitative results are maintained (the coefficient varies between 0.750 and 1.471) when using ethnic boundaries as defined by Fenske (2013) to extract data on crops productivity (Table F.14), when the sample includes societies living in desertic soils (Table F.13), or when using either the Ramankutty et al. index of fertile land or the Galor and Ozak index of caloric suitability as alternative measures of land productivity (Table F.15).

⁶⁶We use two different proxies for historical population density. The first one, HYDE, is based on historical reconstruction at the raster level and is available for the entire sample of societies in the Ethnoatlas. Because historical population reconstruction is unavoidably inexact, we also show the robustness of our results when using data from Pryor (1985), which are available for a much smaller sample (only 144 societies) and are based on completely different sources.

In Table F.16, we control for animal husbandry and animal use in agriculture. Throughout the table, the estimates on the cereal coefficient are practically unaffected by the addition of these controls, reassuring that cereal advantage is not picking up an effect coming from an unobservable variable correlated with animal domestication. We start in column 1 by adding a dummy for each disconnected continent. In columns 2 and 3, we add a dummies that identify societies relying on herding for more than 5% of their diet. The higher the reliance of a society on herding, the higher is the level of hierarchy. This association is in line with our thesis as domesticated animals, one can reasonably argue, are appropriable. In columns 4 and 5, we control for animal husbandry based on ruminants. Finally, in columns 5 and 6, we control for the use of draft animals in agriculture. In line with Diamond's (1997) geographical hypothesis, in particular the role of draft animals in the emergence of complex hierarchy, societies that use draft animals in farming are on average one step higher in the hierarchy ladder.

Table F.11: Cereals and Hierarchy - 2SLS. Robustness checks: Controlling for Geography.

	Dependent variable: Jurisdictional Hierarchy Beyond Local Community				
	(1)	(2)	(3)	(4)	(5)
	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.911 (0.624)	0.750* (0.407)	1.102** (0.553)	1.071** (0.545)	0.697 (0.447)
<i>LandProd</i>	-0.008 (0.081)	0.051 (0.062)	-0.045 (0.074)	-0.039 (0.075)	0.059 (0.072)
Precipitation	-0.001 (0.001)				
Temperature		-0.248*** (0.072)			
Elevation			-0.069* (0.039)		
Ruggedness				-0.008 (0.050)	
Absolute Latitude					0.287*** (0.083)
Latitude					-0.005 (0.004)
Longitude					-0.003*** (0.001)
CONTINENT FE	YES	YES	YES	YES	YES
N	952	952	952	952	952
F excl instrum.	15.20	26.26	19.26	19.13	21.41

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.12: Cereals and Hierarchy - 2SLS. Robustness checks: Controlling for Isolation, Population Density and the Plow.

	Dep. variable: Jurisdictional Hierarchy Beyond Local Community					
	(1)	(2)	(3)	(4)	(5)	(6)
	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	1.073** (0.519)	1.078** (0.545)	1.021** (0.504)	1.471* (0.811)	1.022* (0.552)	1.178*** (0.452)
<i>LandProd</i>	-0.040 (0.070)	-0.038 (0.071)	-0.056 (0.071)	0.006 (0.119)	0.012 (0.078)	0.080 (0.066)
Major River	0.122*** (0.038)					
Distance to Coast		-0.024 (0.058)				
Hist Population Dens (HYDE)			0.211** (0.102)			
Hist Population Dens (PRYOR)				0.276 (0.192)		
Irrigation Potential					0.445* (0.242)	
Plow Advantage						0.297*** (0.101)
Plow Advantage sqr						-0.061* (0.037)
CONTINENT FE	YES	YES	YES	YES	YES	YES
N	952	952	952	142	770	952
F excl instrum.	20.52	18.79	20.03	13.24	18.86	30.83

The table reports cross-sectional OLS and 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.13: Cereals and Hierarchy - 2SLS. Robustness checks: Sample Including Societies Living in Desertic Soils.

Dependent variable: Jurisdictional Hierarchy Beyond Local Community								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	2SLS	2SLS	2SLS	OLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.712*** (0.122)	1.200*** (0.403)	0.831 (0.591)	0.999** (0.432)	0.313*** (0.111)	0.839* (0.432)	1.180** (0.541)	1.092** (0.482)
<i>LandProd</i>			0.067 (0.100)				-0.049 (0.056)	
DEPENDENCE ON AGRICULTURE				0.327 (0.488)				-0.513 (0.568)
CONTINENT FE	NO	NO	NO	NO	YES	YES	YES	YES
N	1059	1059	1059	1059	1059	1059	1059	1059
R-squared	0.793				0.825			
F excl instrum.		44.31	9.930	326.9		28.73	20.50	21.21

The table reports cross-sectional OLS and 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. All societies included in the Ethnoatlas, for which the relevant data are available, are included in the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.14: Cereals and Hierarchy - 2SLS. Robustness checks: Potential Calorie Yields Refer to Ethnic Boundaries in Fenske (2013)

Dependent variable: Jurisdictional Hierarchy Beyond Local Community								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	2SLS	2SLS	2SLS	OLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.707*** (0.131)	1.105*** (0.364)	0.755 (0.483)	0.873** (0.413)	0.304** (0.120)	0.840** (0.395)	0.897** (0.435)	0.899** (0.440)
<i>LandProd</i>			0.104 (0.099)				-0.015 (0.060)	
DEPENDENCE ON AGRICULTURE				0.567 (0.521)				-0.226 (0.893)
CONTINENT FE	NO	NO	NO	NO	YES	YES	YES	YES
N	952	941	941	941	952	941	941	941
R-squared	0.793				0.824			
F excl instrum.		46.58	14.79	230.3		35.94	24.45	16.65

The table reports cross-sectional OLS and 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.15: Cereals and Hierarchy - 2SLS. Robustness checks: Controlling for Alternative Measures of Land Suitability for Agriculture

	Dependent variable:			
	Jurisdictional	Hierarchy	Beyond Local	Community
	(1)	(2)	(3)	(4)
	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	1.009*** (0.372)	0.723 (0.478)	0.867 (0.636)	1.121* (0.585)
% fertile land (Ramankutty et al. 2002)	0.073 (0.061)	0.057 (0.054)		
Caloric Suitability Index (Galer and Ozak, 2016)			0.081 (0.138)	-0.049 (0.078)
CONTINENT FE	NO	YES	NO	YES
N	952	952	952	952
F excl instrum.	34.37	28.79	8.847	17.34

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. All societies included in the Ethnoatlas, for which the relevant data are available, are included in the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.16: Cereals and Hierarchy - 2SLS. Robustness checks: the role of domesticated animals

	Dependent variable:						
	Jurisdictional	Hierarchy	Beyond Local	Community			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.820* (0.440)	0.954** (0.442)	0.888** (0.434)	0.727 (0.483)	0.772* (0.453)	1.037*** (0.313)	0.774** (0.390)
America	-0.200 (0.194)						
Eurasia and Africa	0.367 (0.325)						
Dependent on animal husbandry		-0.395* (0.205)	-0.044 (0.222)				
Presence ruminants				0.584** (0.231)	0.435* (0.226)		
Animals Cultivation						1.001*** (0.207)	1.043*** (0.215)
CONTINENT FE	NO	NO	YES	NO	YES	NO	YES
N	952	952	952	949	949	949	949
F excl instrum.	28.79	28.82	32.90	20.88	26.27	46.51	34.27

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Standard errors (in parentheses) are adjusted for spatial correlation using Conley's (1999) method. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.5 Pre-industrial societies: the role of surplus and taxation

Table F.17: Cereals, Surplus, and Taxation -Reduced Form

	Dependent variable is:							
	Existence of a farming surplus				Tax burden			
	(1) OLS	(2) OLS	(3) OLS	(4) Logit	(5) OLS	(6) OLS	(7) OLS	(8) Ord Logit
<i>CerAdv</i>	0.141*** (0.0319)	0.241*** (0.0681)	0.202*** (0.0742)	1.128*** (0.363)	0.223* (0.116)	0.433** (0.183)	0.257 (0.175)	1.038** (0.494)
<i>LandProd</i>		-0.132 (0.0870)	-0.0985 (0.0985)	-0.601 (0.392)		-0.293 (0.238)	-0.206 (0.231)	-0.654 (0.536)
CONTINENT FE	NO	NO	YES	NO	NO	NO	YES	NO
N	140	140	140	140	56	56	56	56
r2		0.0757	0.0911	0.157		0.0533	0.0781	0.341

The table reports cross-sectional OLS (columns 1-3 and 5-7), Logit (column 4) and ordered logit (column 8) estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.18: Cereals, Surplus, and Taxation - OLS and 2SLS

	Dependent variable is:							
	Existence of a farming surplus				Tax burden			
	(1) OLS	(2) 2SLS	(3) 2SLS	(4) 2SLS	(5) OLS	(6) 2SLS	(7) 2SLS	(8) 2SLS
<i>CerMain</i>	0.359*** (0.0791)	0.940*** (0.260)	0.846*** (0.273)	0.846*** (0.275)	0.853*** (0.209)	0.846** (0.406)	1.018** (0.471)	1.173 (0.850)
<i>LandProd</i>			0.0186 (0.0626)				-0.0634 (0.158)	
DEPENDENCE ON AGRICULTURE				0.191 (0.663)				-0.584 (1.574)
CONTINENT FE	NO	NO	YES	NO	NO	NO	YES	NO
N	139	139	139	139	56	56	56	56
F excl instrum.		16.08	17.37	5.486		22.58	15.11	3.184
r2	0.128				0.233			

The table reports cross-sectional OLS and 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.19: Cereals and Surplus - 2SLS. Robustness checks: Controlling for Geography.

	Dependent variable: Existence of a farming surplus				
	(1)	(2)	(3)	(4)	(5)
	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.774** (0.375)	0.764*** (0.261)	0.921*** (0.301)	0.930*** (0.315)	0.681** (0.267)
<i>LandProd</i>	0.0334 (0.0793)	0.0387 (0.0686)	0.00222 (0.0677)	-0.0215 (0.0811)	0.0534 (0.0637)
Precipitation	-0.0334 (0.0762)				
Temperature		-0.0283 (0.0479)			
Elevation			-0.106*** (0.0372)		
Ruggedness				-0.111 (0.0723)	
Abs Latitude					0.0516 (0.0472)
N	139	139	139	139	139
F excl instrum.	10.41	19.42	15.50	14.83	15.68

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.20: Cereals and Surplus - 2SLS. Robustness checks: Controlling for Isolation, Population Density, Potential for Irrigation, and the Plow.

	Dependent variable: Existence of a farming surplus						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.823*** (0.277)	0.851*** (0.275)	0.820*** (0.300)	0.848*** (0.288)	0.916*** (0.314)	0.646*** (0.245)	0.822*** (0.237)
<i>LandProd</i>	0.0215 (0.0625)	0.0191 (0.0626)	0.0132 (0.0589)	0.0208 (0.0530)	0.0117 (0.0616)	0.101* (0.0540)	0.0265 (0.0574)
Major River	0.0368 (0.0414)						
Distance to Coast		-0.0149 (0.0447)					
Hist Pop Dens (HYDE)			0.0291 (0.0379)				
Hist Pop Dens (PRYOR)				-0.00808 (0.0840)			
Pop Density 1995					0.00145 (0.0357)		
Irrigation Potential						0.0291 (0.128)	
Plow Advantage							0.0129 (0.0489)
N	139	139	139	139	137	111	139
F excl instrum.	15.86	17.09	13.35	17.91	12.99	23.25	22.07

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.21: Cereals and Taxation- 2SLS. Robustness checks: Controlling for Geography.

	Dependent variable: Tax Burden				
	(1)	(2)	(3)	(4)	(5)
	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.808 (0.577)	1.056** (0.495)	1.019** (0.473)	1.014** (0.501)	1.132** (0.571)
<i>LandProd</i>	0.0487 (0.214)	-0.0814 (0.190)	-0.0667 (0.164)	-0.0603 (0.178)	-0.112 (0.205)
Precipitation	-0.180 (0.167)				
Temperature		0.0232 (0.128)			
Elevation			-0.0488 (0.250)		
Ruggedness				0.00906 (0.116)	
Abs Latitude					-0.0593 (0.150)
N	56	56	56	56	56
F excl instrum.	10.10	14.75	15.43	13.08	8.491

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.22: Cereals and Taxation - 2SLS. Robustness checks: Controlling for Isolation, Population Density, Potential for Irrigation, and the Plow.

	Dependent variable: Tax Burden						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
<i>CerMain</i>	0.999** (0.470)	0.993** (0.438)	0.380 (0.591)	0.00327 (0.536)	0.627 (0.492)	0.689 (0.659)	1.102** (0.483)
<i>LandProd</i>	-0.0530 (0.158)	-0.0573 (0.152)	-0.0998 (0.154)	-0.174 (0.138)	-0.158 (0.151)	-0.0175 (0.163)	-0.108 (0.160)
Major River	0.101 (0.122)						
Distance to Coast		-0.0310 (0.0998)					
Hist Pop Dens (HYDE)			0.291** (0.119)				
Hist Pop Dens (PRYOR)				0.641*** (0.166)			
Pop Density 1995					0.227*** (0.0852)		
Irrigation						0.329 (0.328)	
Plow Advantage							-0.0601 (0.0974)
N	56	56	56	56	56	42	56
F excl instrum.	13.15	18.51	9.980	4.693	11.48	8.747	14.18

The table reports cross-sectional 2SLS estimates and the unit of observation is the society in Murdock's Ethnoatlas. Societies that live on lands that are suitable for neither cereals nor roots and tubers are excluded from the sample. Robust standard errors in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.6 Country-level data: robustness checks

Table F.23: Cereals and Hierarchy in Classical Antiquity - Cross-sectional regressions. Further robustness checks 1

	Dep. Variable: Hierarchy Index in AD 450						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>WR Cer</i>	0.460*** (0.104)	0.414*** (0.104)	0.500** (0.195)	0.467*** (0.124)	0.499*** (0.125)	0.417*** (0.135)	0.449*** (0.117)
<i>WR RT</i>	0.0887 (0.175)	0.197 (0.164)	0.152 (0.186)	0.181 (0.173)	0.123 (0.174)	0.148 (0.176)	0.172 (0.180)
<i>WR Cer and RT</i>	0.0586 (0.101)	0.0569 (0.0983)	-0.0283 (0.152)	0.0573 (0.113)	0.0806 (0.118)	0.0265 (0.124)	0.0534 (0.112)
Controls:							
Legal Origin	YES	NO	NO	NO	NO	NO	NO
Pop. Dens. 1500	NO	YES	NO	NO	NO	NO	NO
Settlers Mortality	NO	NO	YES	NO	NO	NO	NO
Slave Exports	NO	NO	NO	YES	NO	NO	NO
Genetic Diversity	NO	NO	NO	NO	YES	NO	NO
Distance River	NO	NO	NO	NO	NO	YES	NO
Distance Coast	NO	NO	NO	NO	NO	NO	YES
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
N	151	151	79	151	137	145	145
r2	0.509	0.467	0.568	0.409	0.418	0.396	0.429

The table reports cross-sectional OLS estimates and the unit of observation is the territory delimited by modern-country borders. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.24: Cereals and Hierarchy in Classical Antiquity - Cross-sectional regressions. Further robustness checks 2

	Dep. Variable: Hierarchy Index in AD 450				
	(1)	(2)	(3)	(4)	(5)
<i>WR Cer</i>	0.457*** (0.132)	0.412*** (0.130)	0.437*** (0.127)	0.436*** (0.127)	0.456*** (0.125)
<i>WR RT</i>	0.204 (0.177)	0.117 (0.188)	0.171 (0.173)	0.155 (0.171)	0.174 (0.177)
<i>WR Cer and RT</i>	0.0440 (0.127)	0.0787 (0.114)	0.0372 (0.114)	0.0361 (0.114)	0.102 (0.112)
Controls:					
Pct Malaria	YES	NO	NO	NO	NO
Tropical Land	NO	YES	NO	NO	NO
Domesticable Animals	NO	NO	YES	NO	NO
Domesticable Ruminants	NO	NO	NO	YES	NO
Potential Irrigation	NO	NO	NO	NO	YES
CONTINENT FE	YES	YES	YES	YES	YES
N	147	151	151	151	149
r2	0.412	0.417	0.411	0.412	0.409

The table reports cross-sectional OLS estimates and the unit of observation is the territory delimited by modern-country borders. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.25: Cereals and Hierarchy in Classical Antiquity - Cross-sectional regressions. Robustness Checks: Excluding Continents one-by-one

	Dep. Variable: Hierarchy Index in AD 450				
	(1)	(2)	(3)	(4)	(5)
<i>WR Cer</i>	0.481*** (0.143)	0.398*** (0.133)	0.493*** (0.140)	0.510*** (0.168)	0.465*** (0.124)
<i>WR RT</i>	0.179 (0.175)	0.170 (0.172)	0.174 (0.174)	0 (.)	0.182 (0.173)
<i>WR Cer and RT</i>	0.0429 (0.133)	0.0573 (0.104)	0.0164 (0.120)	0.141 (0.180)	0.0623 (0.114)
CONTINENT FE	YES	YES	YES	YES	YES
Sample excludes	Africa	Asia	Europe	America	Oceania
r2	0.371	0.206	0.538	0.410	0.401
N	103	113	116	124	148

The table reports cross-sectional OLS estimates and the unit of observation is the territory delimited by modern-country borders. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.26: Cereals and Hierarchy in Classical Antiquity - Cross-sectional regressions

	Dep. Variable: Hierarchy Index in AD 450					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>CerAdv</i>	0.245** (0.0993)	0.355*** (0.118)	0.160 (0.102)	0.278** (0.108)	0.198* (0.111)	0.187* (0.101)
<i>LandProd</i>	-0.262*** (0.0986)	-0.400*** (0.123)	-0.154 (0.104)	-0.312*** (0.107)	-0.214* (0.112)	-0.210** (0.0984)
Controls:						
Abs Latitude	NO	YES	NO	NO	NO	NO
Precipitation	NO	NO	YES	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO
Elevation	NO	NO	NO	NO	YES	NO
Ruggedness	NO	NO	NO	NO	NO	YES
CONTINENT FE	YES	YES	YES	YES	YES	YES
r ²	0.328	0.343	0.336	0.350	0.326	0.353
N	151	151	150	148	145	151

The table reports cross-sectional OLS estimates and the unit of observation is the territory delimited by modern-country borders. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.27: Cereals and Hierarchy - Panel Regressions - Robustness Checks 1

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>CerAdv</i>	0.160* (0.0892)	0.127 (0.0843)	0.206* (0.116)	0.274*** (0.0833)	0.162* (0.0831)	0.245*** (0.0928)	0.258*** (0.0957)
<i>LandProd</i>	-0.0507 (0.133)	0.0471 (0.132)	-0.261 (0.192)	-0.176 (0.143)	-0.00456 (0.148)	-0.121 (0.151)	-0.133 (0.151)
Controls (xYear):							
Legal Origin	YES	NO	NO	NO	NO	NO	NO
Population Dens. 1500	NO	YES	NO	NO	NO	NO	NO
Settlers Mortality	NO	NO	YES	NO	NO	NO	NO
Slave Exports	NO	NO	NO	YES	NO	NO	NO
Genetic Diversity	NO	NO	NO	NO	YES	NO	NO
Distance River	NO	NO	NO	NO	NO	YES	NO
Distance Coast	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES	YES	YES	YES
r ²	0.699	0.714	0.707	0.683	0.682	0.678	0.679
N	2869	2869	1501	2869	2603	2755	2755

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.28: Cereals and Hierarchy - Panel Regressions - Robustness Checks 2

	Dep. Variable: Hierarchy Index				
	(1)	(2)	(3)	(4)	(5)
<i>CerAdv</i>	0.273*** (0.0840)	0.254*** (0.0675)	0.273*** (0.0835)	0.278*** (0.0833)	0.452*** (0.0920)
<i>LandProd</i>	-0.199 (0.145)	-0.211** (0.102)	-0.162 (0.141)	-0.165 (0.139)	-0.444*** (0.154)
Controls (xYear):					
Pct Malaria	YES	NO	NO	NO	NO
Tropical Land	NO	YES	NO	NO	NO
Domesticable Animals	NO	NO	YES	NO	NO
Domesticable Ruminants	NO	NO	NO	YES	NO
Irrigation Potential	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES	YES
r2	2793	2869	2869	2869	2831
N	0.681	0.744	0.683	0.683	0.715

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.29: Cereals and Hierarchy - Panel Regressions. Robustness checks: Excluding Colonies

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>CerAdv</i>	0.128* (0.0660)	0.186** (0.0786)	0.230*** (0.0735)	0.162** (0.0816)	0.182** (0.0857)	0.178** (0.0788)	0.135* (0.0772)
<i>LandProd</i>		-0.111 (0.136)	-0.179 (0.131)	-0.0997 (0.136)	-0.0884 (0.138)	-0.0879 (0.134)	-0.115 (0.119)
Controls (x Year FE):							
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
YEAR FE	YES	YES	YES	YES	YES	YES	YES
r2	0.773	0.774	0.789	0.774	0.770	0.777	0.786
N	2414	2414	2398	2365	2329	2414	2414

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. The sample excludes those cells 50yearsXcountry in which countries were either colonies or protectorates. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.30: Cereals and Hierarchy - Panel Regressions. Robustness checks: a Different Measure of Hierarchy

	Dep. Variable: Government above tribal level						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>CerAdv</i>	0.188*** (0.0683)	0.270*** (0.0835)	0.280*** (0.0758)	0.235*** (0.0855)	0.252*** (0.0890)	0.259*** (0.0840)	0.192** (0.0791)
<i>LandProd</i>		-0.159 (0.140)	-0.189 (0.131)	-0.150 (0.138)	-0.110 (0.142)	-0.145 (0.138)	-0.161 (0.122)
Controls (x Year FE):							
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
YEAR FE	YES	YES	YES	YES	YES	YES	YES
r2	0.672	0.674	0.707	0.677	0.673	0.677	0.699
N	2869	2869	2850	2812	2755	2869	2869

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. The dependent variable is a dummy that identifies those territories characterized by a supra-tribal government. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.31: Cereals and Hierarchy - Panel Regressions. Robustness checks: a Different Measure of Soil Suitability for Agriculture

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>CerAdv</i>	0.189*** (0.0683)	0.272*** (0.0834)	0.282*** (0.0760)	0.240*** (0.0857)	0.255*** (0.0889)	0.261*** (0.0839)	0.197** (0.0795)
Caloric Suitability Index		-0.163 (0.141)	-0.193 (0.131)	-0.152 (0.139)	-0.115 (0.142)	-0.148 (0.138)	-0.165 (0.123)
Controls (x Year FE):							
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES	YES	YES	YES
r ²	0.680	0.682	0.716	0.684	0.681	0.686	0.705
N	2869	2869	2850	2812	2755	2869	2869

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.32: Cereals and Hierarchy - Panel Regressions. Robustness Checks: Excluding Years 1500-1750

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>CerAdv</i>	0.198*** (0.0720)	0.272*** (0.0889)	0.282*** (0.0811)	0.235*** (0.0912)	0.249*** (0.0946)	0.260*** (0.0892)	0.190** (0.0846)
<i>LandProd</i>		-0.145 (0.149)	-0.176 (0.140)	-0.140 (0.146)	-0.0889 (0.150)	-0.130 (0.146)	-0.148 (0.129)
Controls (x Year FE):							
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
YEAR FE	YES	YES	YES	YES	YES	YES	YES
r ²	0.711	0.712	0.743	0.715	0.711	0.716	0.735
N	2416	2416	2400	2368	2320	2416	2416

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. The years 1500-1750 are excluded from the regression. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.33: Cereals and Hierarchy - Panel Regressions. Robustness Checks: Excluding Years 1500-1750

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Change CerAdv * LinearTrend</i>	-0.0000601 (0.0000691)	-0.000138 (0.000135)	-0.000145 (0.000137)	-0.000153 (0.000138)	-0.000169 (0.000134)	-0.000138 (0.000135)	-0.000138 (0.000135)
<i>Change LandProd * LinearTrend</i>		0.000142 (0.000191)	0.000152 (0.000193)	0.000164 (0.000194)	0.000196 (0.000187)	0.000142 (0.000191)	0.000142 (0.000191)
Controls (x Year FE):							
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
YEAR FE	YES	YES	YES	YES	YES	YES	YES
r2	0.887	0.887	0.887	0.885	0.888	0.887	0.887
N	1661	1661	1650	1628	1595	1661	1661

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. The years 1500-1750 are excluded from the regression. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.34: Cereals and Hierarchy - Panel Regressions 2

	Dep. Variable: Hierarchy Index						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Change CerAdv* 1050</i>	-0.00422 (0.00332)	-0.00799 (0.00573)	-0.00811 (0.00581)	-0.00825 (0.00592)	-0.00810 (0.00581)	-0.00799 (0.00573)	-0.00799 (0.00573)
<i>Change CerAdv* 1100</i>	0.0135 (0.0180)	0.0119 (0.0205)	0.0118 (0.0205)	0.0117 (0.0206)	0.0131 (0.0218)	0.0119 (0.0205)	0.0119 (0.0205)
<i>Change CerAdv* 1150</i>	0.0172 (0.0185)	0.0305 (0.0278)	0.0303 (0.0278)	0.0302 (0.0279)	0.0334 (0.0300)	0.0305 (0.0278)	0.0305 (0.0278)
<i>Change CerAdv* 1200</i>	0.0164 (0.0186)	0.0298 (0.0281)	0.0294 (0.0281)	0.0292 (0.0281)	0.0334 (0.0303)	0.0298 (0.0281)	0.0298 (0.0281)
<i>Change CerAdv* 1250</i>	0.00759 (0.0194)	0.0200 (0.0299)	0.0191 (0.0299)	0.0182 (0.0300)	0.0252 (0.0323)	0.0200 (0.0299)	0.0200 (0.0299)
<i>Change CerAdv* 1300</i>	-0.0253 (0.0254)	-0.0404 (0.0440)	-0.0425 (0.0443)	-0.0448 (0.0448)	-0.0350 (0.0462)	-0.0404 (0.0440)	-0.0404 (0.0440)
<i>Change CerAdv* 1350</i>	-0.0234 (0.0254)	-0.0388 (0.0440)	-0.0409 (0.0443)	-0.0431 (0.0448)	-0.0333 (0.0462)	-0.0388 (0.0440)	-0.0388 (0.0440)
<i>Change CerAdv* 1400</i>	-0.0139 (0.0280)	-0.0460 (0.0509)	-0.0485 (0.0513)	-0.0511 (0.0518)	-0.0598 (0.0496)	-0.0460 (0.0509)	-0.0460 (0.0509)
<i>Change CerAdv* 1450</i>	-0.0235 (0.0303)	-0.0451 (0.0619)	-0.0479 (0.0624)	-0.0509 (0.0628)	-0.0573 (0.0622)	-0.0451 (0.0619)	-0.0451 (0.0619)
<i>Change CerAdv* 1500</i>	-0.00977 (0.0414)	-0.0452 (0.0688)	-0.0488 (0.0692)	-0.0524 (0.0698)	-0.0629 (0.0686)	-0.0452 (0.0688)	-0.0452 (0.0688)
<i>Change CerAdv* 1550</i>	0.102* (0.0618)	0.221** (0.108)	0.128 (0.111)	0.193* (0.110)	0.225** (0.112)	0.207* (0.110)	0.128 (0.107)
<i>Change CerAdv* 1600</i>	0.130* (0.0690)	0.379*** (0.125)	0.286** (0.124)	0.350*** (0.128)	0.397*** (0.130)	0.364*** (0.127)	0.286** (0.121)
<i>Change CerAdv* 1650</i>	0.177** (0.0756)	0.505*** (0.123)	0.412*** (0.122)	0.477*** (0.124)	0.514*** (0.128)	0.491*** (0.124)	0.412*** (0.121)
<i>Change CerAdv* 1700</i>	0.194** (0.0794)	0.511*** (0.129)	0.418*** (0.126)	0.482*** (0.131)	0.459*** (0.134)	0.497*** (0.131)	0.418*** (0.125)
<i>Change CerAdv* 1750</i>	0.198** (0.0807)	0.578*** (0.109)	0.485*** (0.109)	0.549*** (0.111)	0.532*** (0.114)	0.563*** (0.110)	0.485*** (0.108)
<i>Change CerAdv* 1800</i>	0.199** (0.0813)	0.544*** (0.110)	0.450*** (0.110)	0.516*** (0.112)	0.496*** (0.115)	0.529*** (0.111)	0.450*** (0.109)
<i>Change CerAdv* 1850</i>	0.205** (0.0834)	0.504*** (0.112)	0.409*** (0.111)	0.476*** (0.114)	0.455*** (0.118)	0.490*** (0.114)	0.411*** (0.110)
Controls (x Year FE):							
<i>Change LandProd</i>	NO	YES	YES	YES	YES	YES	YES
Precipitation	NO	NO	YES	NO	NO	NO	NO
Temperature	NO	NO	NO	YES	NO	NO	NO
Elevation	NO	NO	NO	NO	YES	NO	NO
Ruggedness	NO	NO	NO	NO	NO	YES	NO
Abs Latitude	NO	NO	NO	NO	NO	NO	YES
COUNTRY FE	YES	YES	YES	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES	YES	YES	YES
r2	0.717	0.733	0.750	0.732	0.730	0.736	0.745
N	2718	2718	2700	2664	2610	2718	2718

The table reports panel OLS estimates and the unit of observation is the territory delimited by modern-country borders every 50 years. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.8 Archaeological evidence of complex civilizations and wild relatives: robustness checks

Table F.35: Wild Relatives of Domesticated Crops and the Location of Ancient Cities. Robustness Checks: Excluding Continents one-by-one

Dep. variable: presence of cities/large settlements founded by AD 400					
	(1)	(2)	(3)	(4)	(5)
<i>WR Cer</i>	0.239*** (0.0495)	0.220*** (0.0503)	0.102*** (0.0292)	0.224*** (0.0392)	0.199*** (0.0385)
<i>WR RT</i>	0.00373 (0.0138)	0.0160 (0.0173)	-0.0224 (0.0142)	0.0100 (0.0197)	0.00537 (0.0157)
<i>WR Cer and RT</i>	-0.00320 (0.0178)	0.0386 (0.0256)	-0.0117 (0.0141)	0.0392* (0.0236)	0.0267 (0.0217)
CONTINENT FE	YES	YES	YES	YES	YES
Sample excludes	Africa	Asia	Europe	America	Oceania
r ²	0.176	0.182	0.0953	0.144	0.144
N	14487	13902	13201	11598	16265

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.36: Wild Relatives of Domesticated Crops and the Location of Ancient Cities. Robustness Checks: Logit regressions

Dep. variable: presence of cities/large settlements founded by AD 400				
	(1)	(2)	(3)	(4)
<i>WR Cer</i>	2.963*** (0.391)	2.852*** (0.438)	2.877*** (0.616)	1.698*** (0.324)
<i>WR RT</i>		-0.771 (0.650)	-0.0138 (0.686)	-1.331*** (0.429)
<i>WR Cer and RT</i>		-0.912 (0.676)	-0.191 (0.868)	-0.759 (0.823)
COUNTRY FE	NO	NO	YES	NO
CONTINENT FE	NO	NO	NO	YES
N	17076	17076	17076	17076

The table reports cross-sectional Logistic estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.37: Wild Relatives of Domesticated Crops and the Location of Ancient Cities. Robustness Checks: Controlling for Geography.

Dependent variable is the presence of cities/large settlements founded by AD 400							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>WR Cer</i>	0.210*** (0.0388)	0.189*** (0.0397)	0.195*** (0.0380)	0.196*** (0.0376)	0.198*** (0.0371)	0.207*** (0.0429)	0.194*** (0.0352)
<i>WR RT</i>	0.0423*** (0.0161)	0.00105 (0.0141)	0.00451 (0.0149)	0.00559 (0.0151)	0.0102 (0.0114)	0.0286 (0.0213)	0.0181 (0.0174)
<i>WR Cer and RT</i>	0.0638*** (0.0204)	0.0208 (0.0192)	0.0229 (0.0204)	0.0250 (0.0204)	0.0301* (0.0162)	0.0585** (0.0282)	0.0565** (0.0230)
Precipitation	-0.0244*** (0.00850)						
Temperature		0.00996 (0.0127)					
Elevation			-0.00696 (0.00535)				
Ruggedness				0.00173 (0.00691)			
Abs Latitude					0.000277 (0.000713)		
Irrigation Potential						0.0277*** (0.0103)	
Plow Advantage							0.0383*** (0.00824)
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
r2	0.153	0.148	0.145	0.145	0.145	0.171	0.165
N	16850	16267	17076	17076	17076	9086	17076

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.9 Archaeological evidence of complex civilizations and distance from areas of independent domestication of cereal grains

The estimates in Table 5 show that distance from the nearest area of independent domestication of a cereal grain, $DistanceCer_i$, is negatively correlated with urbanization. Moreover, when controlling for $DistanceCer_i$, the distance from the nearest area of independent adoption of agriculture is not correlated with urbanization, thus suggesting that farming crops different than cereals might be irrelevant to the development of complex hierarchies. In this subsection we present a series of additional robustness checks.

In table F.38, we show that results are robust to controlling sequentially for precipitation, temperature, elevation, and ruggedness. In Table F.39, we extend the analysis of Table 5 by controlling for: absolute latitude, irrigation potential and, productivity advantage from the plow,. In the last two columns of the table, we limit the analysis to Asia and Africa and exclude deserts. The results are robust with one exception. When controlling for absolute latitude the coefficient on distance to the areas of cereal domestication drops by half and is no longer statistically significant (column 1). The great majority of the centers of cereal domestication are concentrated in a very limited latitude band (between 10 and 40 degrees N), in which there are almost no centers of domestication of roots and tubers. Thus, the data limit our ability to disentangle the impact of distance to the closest cereal domestication center and absolute latitude. There are, however, a couple of exceptions to this rule: in Lingnan (South China, lat: approx. 25 N) yams and taro were domesticated but not cereals, and in the Sudanese savannah (lat: approx. 5 N) sorghum was domesticated. When limiting the analysis to Africa and Asia, distance to the closest centers of early domestication of cereals matters, while distance to the closest centers of early domestication of roots and tubers does not, even after controlling for absolute latitude (column 2).

In Table F.40, we replace the left hand side variable of ancient cities with alternative archaeological sites (which predate the 476 CE fall of the Roman Empire in the Old World, or the 1492 discovery of the Americas in the New World), that are reasonable proxies for hierarchy/civilization. Column 1 documents a negative and statistically significant correlation between the presence of ancient archaeological sites and the distance from the nearest area of domestication of cereals. As before, distance to the nearest center of domestication of other crops does not seem to matter when controlling for distance to centers of cereal adoption. This result is confirmed when archaeological sites are pyramids, ancient temples, ancient mines, ancient palaces and ancient sculptured stones, but not when the archaeological sites are ancient standing stones. The results reported in this table are also valid when examining the extensive margin and focusing on the number of archaeological

sites in each cell (Table F.41) and when excluding Europe (Table F.42).

As another robustness check, in Table F.43, we repeat the analysis presented in Table 5, but using GAEZ data. The presence of an ancient city is regressed on land productivity and the productivity advantage of cereals over roots and tubers. Column 2 illustrates that a higher productivity advantage of cereals over roots and tubers is associated with an increase in the probability of finding an ancient city in that area. In addition, it shows that if we control for cereal advantage, land productivity is uncorrelated with the location of ancient cities. Results are robust to adding continent fixed effects (column 3), for country fixed effects (column 4) and when using the Reitsma and Seto (2016) data on the location of large settlements founded before either 450 AD or 500 BC (columns 5-8).

Table F.38: The Origin of the Neolithic Transition and the Location of Ancient Cities. Robustness checks: Controlling for Geography.

Dependent variable is the presence of cities/large settlements founded by AD 400					
	(1)	(2)	(3)	(4)	(5)
<i>DistanceCer</i>	-0.00141** (0.000601)	-0.000710 (0.000922)	-0.00148** (0.000679)	-0.00132** (0.000630)	-0.000710 (0.00112)
<i>DistanceAgr</i>	0.000250 (0.000576)	0.000340 (0.000653)	0.000323 (0.000609)	0.000101 (0.000617)	0.000339 (0.000662)
Precipitation	0.00166 (0.00772)				
Temperature		0.0413** (0.0186)			
Elevation			-0.00706 (0.00848)		
Ruggedness				-0.00597 (0.00662)	
Abs Latitude					-0.00159 (0.00110)
CONTINENT FE	YES	YES	YES	YES	YES
r ²	0.0497	0.0593	0.0505	0.0501	0.0548
N	15862	15833	15927	15927	15927

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.39: The Origin of the Neolithic Transition and the Location of Ancient Cities. Robustness checks: controlling for irrigation potential and the plow, excluding Europe and deserts.

Dependent variable is the presence of cities/large settlements founded by AD 400				
	(1)	(2)	(3)	(4)
<i>DistanceCer</i>	-0.00294*** (0.00103)	-0.00136* (0.000710)	-0.00144*** (0.000245)	-0.00264*** (0.000922)
<i>DistanceAgr</i>	0.000528 (0.00125)	0.000271 (0.000664)	0.000564** (0.000276)	0.000369 (0.00103)
Irrigation Potential	0.00527 (0.0104)			
Plow Advantage		0.0358*** (0.0100)		
CONTINENT FE	YES	YES	YES	YES
r2	0.105	0.0689	0.0642	0.0836
N	8214	15927	12052	8942

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. In column 3 the sample excludes Europe, in column 4 the sample excludes deserts. Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.40: The Origin of the Neolithic Transition and Archeological Ruins.

Dependent variable is a dummy that identifies evidence of:							
	ancient archaeolog. sites (1)	pyramids (2)	ancient temples (3)	ancient mines (4)	ancient palaces (5)	ancient sculptured stones (6)	ancient standing stones (7)
<i>DistanceCer</i>	-0.00279*** (0.000824)	-0.000282 (0.000187)	-0.000636** (0.000311)	-0.000210** (0.000106)	-0.000132** (0.0000550)	-0.000232** (0.000108)	-0.0000152 (0.0000706)
<i>DistanceAgr</i>	0.000864 (0.000753)	0.000105 (0.000146)	0.000316 (0.000330)	0.0000109 (0.000144)	0.0000689 (0.0000487)	0.000166 (0.000105)	0.00000243 (0.000119)
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
r2	0.0328	0.00451	0.0105	0.00294	0.00189	0.00930	0.0187
N	15927	15927	15927	15927	15927	15927	15927

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. The dependent variable is a dummy that takes the value of one if there is archaeological evidence of either ancient sites from the Stone Age (column 1), or ancient pyramids or mastaba (column 2), or ancient temples (column 3), or ancient mines or quarries (column 4), or ancient palaces (column 5), or ancient sculptured stones (column 6). or ancient standing stones (column 7). Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.41: The Origin of the Neolithic Transition and Archaeological Ruins.

	Dependent variable is the log (1+ number of archaeological ruin)						
	ancient archaeolog. sites	pyramids	ancient temples	ancient mines	ancient palaces	ancient sculptured stones	ancient standing stones
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>DistanceCer</i>	-0.00359*** (0.00114)	-0.000264 (0.000170)	-0.000556* (0.000306)	-0.000153** (0.0000740)	-0.000109** (0.0000446)	-0.000193** (0.0000932)	-0.0000245 (0.0000652)
<i>DistanceAgr</i>	0.00120 (0.000981)	0.0000880 (0.000133)	0.000167 (0.000332)	0.0000140 (0.000101)	0.0000581 (0.0000382)	0.000152 (0.0000977)	0.0000752 (0.000157)
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
r2	0.0266	0.00256	0.00784	0.00260	0.00159	0.00800	0.0176
N	15927	15927	15927	15927	15927	15927	15927

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. The dependent variable is log of one plus the number of either ancient sites from the Stone Age (column 1), or ancient pyramids or mastaba (column 2), or ancient temples (column 3), or ancient mines or quarries (column 4), or ancient palaces (column 5), or ancient sculptured stones (column 6), or ancient standing stones (column 7). Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.42: The Origin of the Neolithic Transition and Archaeological Ruins. Robustness Checks: Excluding Europe

	Dependent variable is a dummy that identifies evidence of						
	ancient archaeolog. sites	pyramids	ancient temples	ancient mines	ancient palaces	ancient sculptured stones	ancient standing stones
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>DistanceCer</i>	-0.00279*** (0.000826)	-0.000281 (0.000181)	-0.000637** (0.000311)	-0.000208* (0.000122)	-0.000132** (0.0000558)	-0.000231** (0.000104)	-0.0000128 (0.0000637)
<i>DistanceAgr</i>	0.000776 (0.000528)	0.0000616 (0.000109)	0.000362 (0.000328)	-0.0000761 (0.000141)	0.0000597 (0.0000476)	0.000124 (0.0000768)	-0.0000923 (0.0000663)
CONTINENT FE	YES	YES	YES	YES	YES	YES	YES
r2	0.0410	0.00524	0.0135	0.00405	0.00265	0.00520	0.00416
N	12052	12052	12052	12052	12052	12052	12052

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. The sample excludes Europe. The dependent variable is a dummy that takes the value of one if there is archeological evidence of either ancient sites from the Stone Age (column 1), or ancient pyramids or mastaba (column 2), or ancient temples (column 3), or ancient mines or quarries (column 4), or ancient palaces (column 5), or ancient sculptured stones (column 6), or ancient standing stones (column 7). Robust standard errors, clustered at the country-level, in parentheses *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.43: Potential Crop Yields and the Location of Ancient Cities.

	Dependent variable is the presence of cities founded by:							
	classical antiquity				AD 450		500 BC	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>CerAdv</i>	0.0469*** (0.0143)	0.145*** (0.0388)	0.129*** (0.0380)	0.0340** (0.0162)	0.0174*** (0.00621)	0.0161** (0.00635)	0.00882*** (0.00335)	0.00875** (0.00350)
<i>LandProd</i>		-0.0864*** (0.0267)	-0.0744*** (0.0256)	-0.0126 (0.0128)	-0.0108** (0.00427)	-0.0106** (0.00462)	-0.00579** (0.00242)	-0.00640** (0.00278)
CONTINENT FE	NO	NO	YES	NO	NO	YES	NO	YES
COUNTRY FE	NO	NO	NO	YES	NO	NO	NO	NO
r2	0.0498	0.0841	0.0986	0.451	0.00758	0.0159	0.00316	0.0103
N	15927	15927	15927	15927	15927	15927	15927	15927

The table reports cross-sectional OLS estimates and the unit of observation is the 1x1 decimal degree square. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

Table F.44: Potential Crop Yields and the Location of Ancient Cities. Difference-in-difference analysis

	Dependent variable is:			
	archaeol. site (dummy)	Log(1+ # archaeol. site)	ancient settlem. (dummy)	Log(1+ # ancient settlem.)
	(1)	(2)	(3)	(4)
<i>Post X CerAdv</i>	0.0683** (0.0293)	0.117*** (0.0417)	0.0668** (0.0283)	0.101*** (0.0365)
<i>Post X LandProd</i>	-0.0266 (0.0208)	-0.0581** (0.0281)	-0.0226 (0.0203)	-0.0456* (0.0247)
CELL FE	YES	YES	YES	YES
TIME FE	YES	YES	YES	YES
r2	0.0710	0.0800	0.0738	0.0765
N	31854	31854	31854	31854

The table reports difference-in-difference OLS regressions. The unit of observation is the 1x1 decimal degree square either before or after the Neolithic transition. Robust standard errors, clustered at the country-level, in parentheses. *** significant at less than 1 percent; ** significant at 5 percent; * significant at 10 percent.

F.10 Figures

Figure F.1: Density plot of the focal year for societies in Murdock’s Ethnographic Atlas.

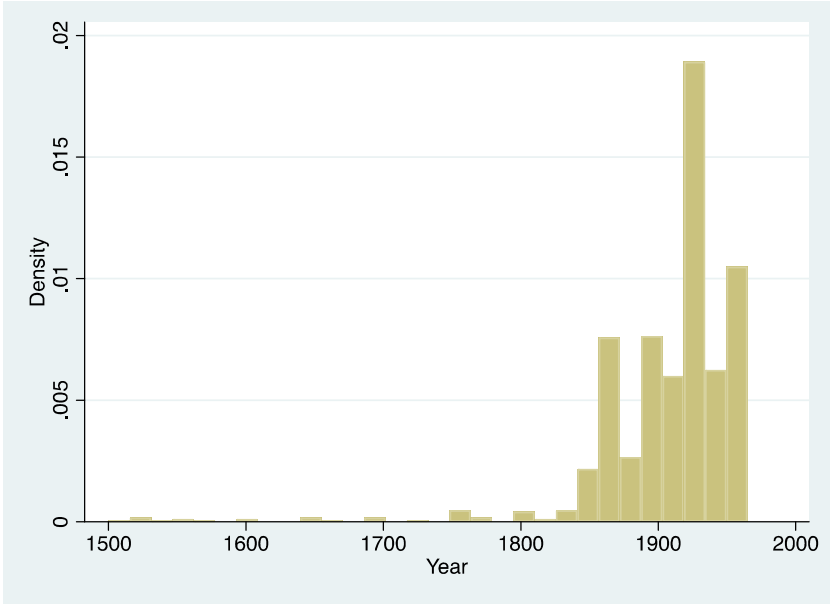


Figure F.2: Farming surplus in pre-industrial societies

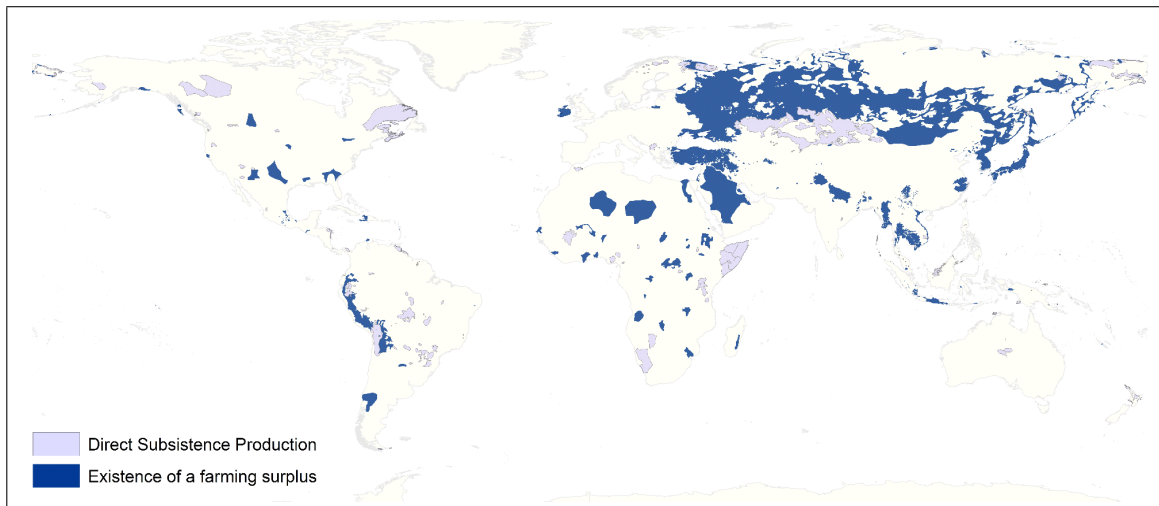


Figure F.3: Years of colonization



Figure F.4: Cities founded before 450 AD

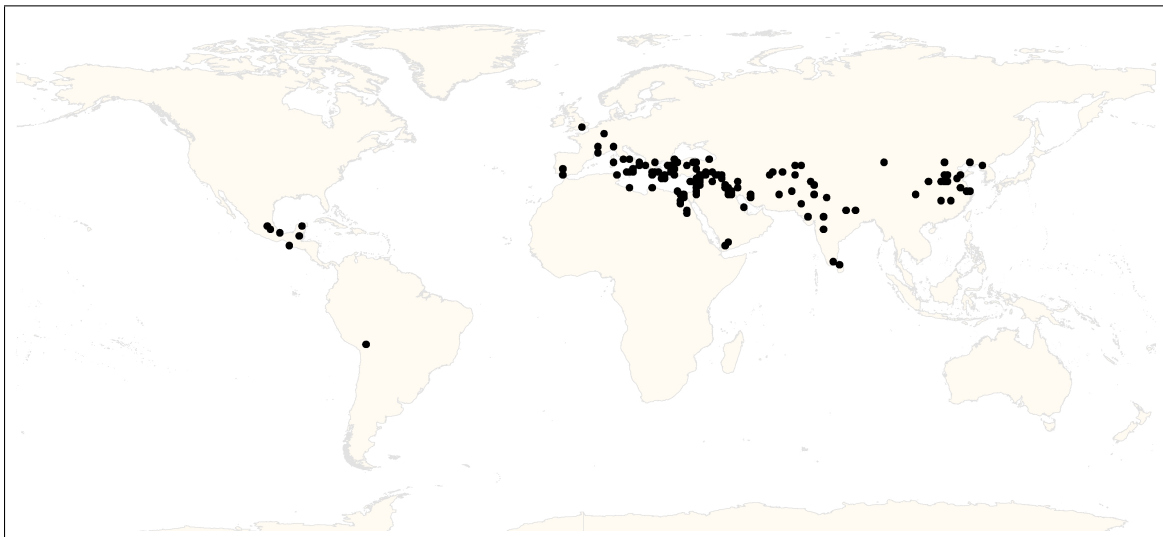


Figure F.5: Cities founded before 500 BC

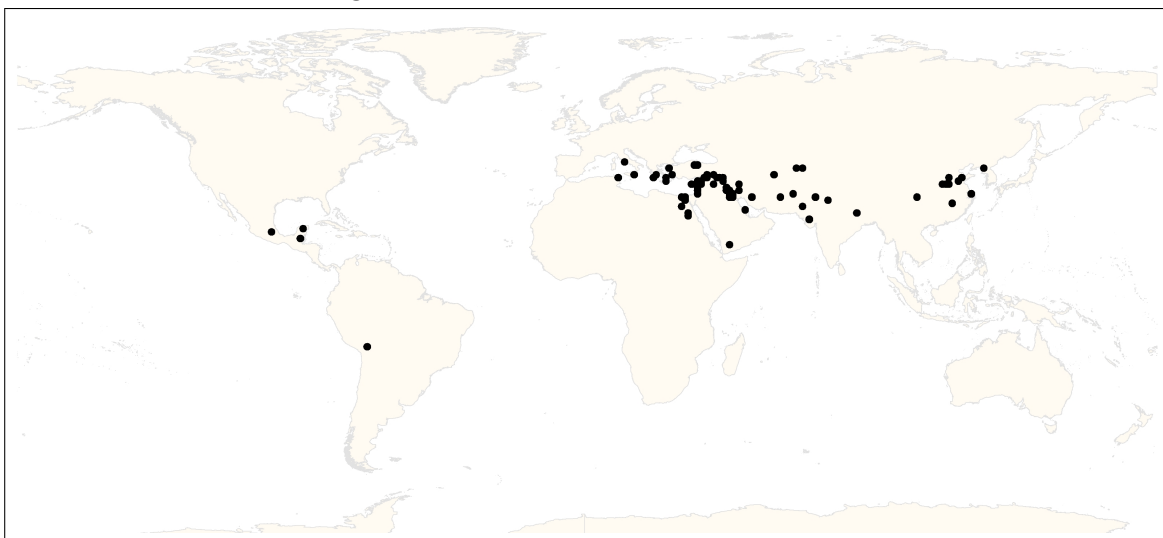


Figure F.6: Potential yields (calories per hectare) from cereal grains.

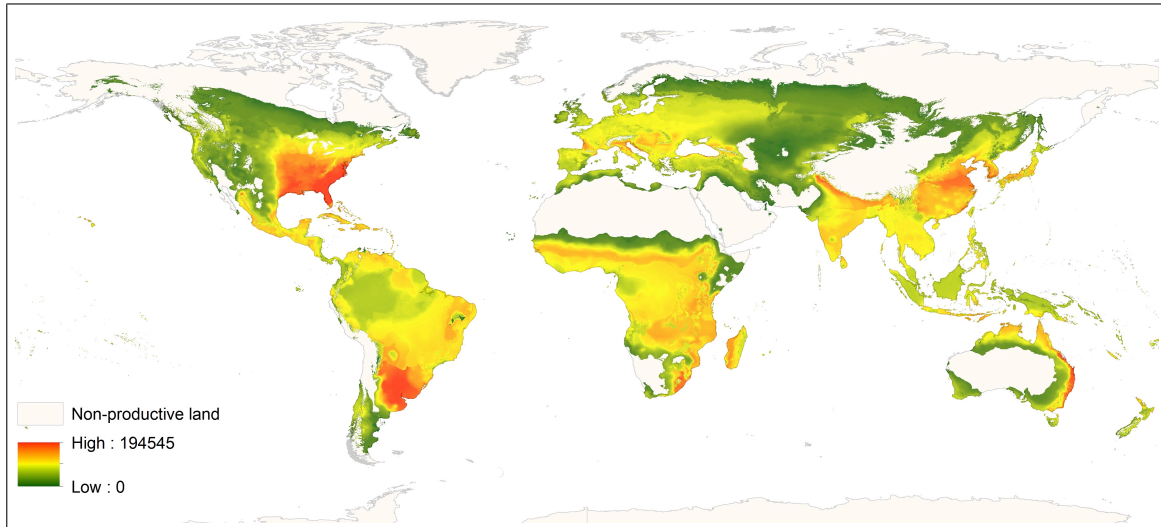


Figure F.7: Potential yields (calories per hectare) from roots and tubers

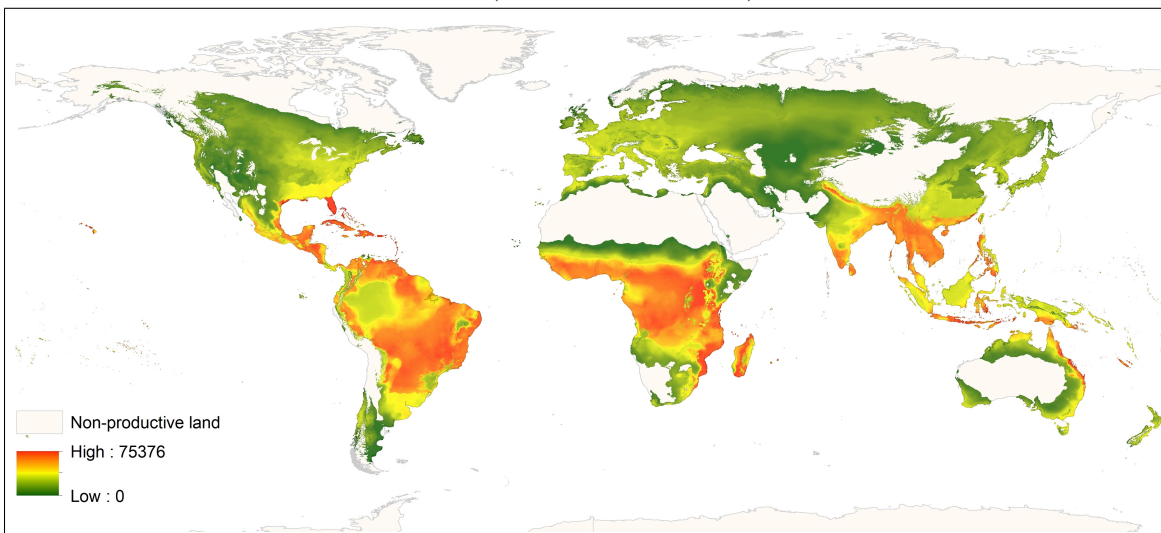


Figure F.8: Optimal crop in terms of caloric yields among cereals, roots and tubers

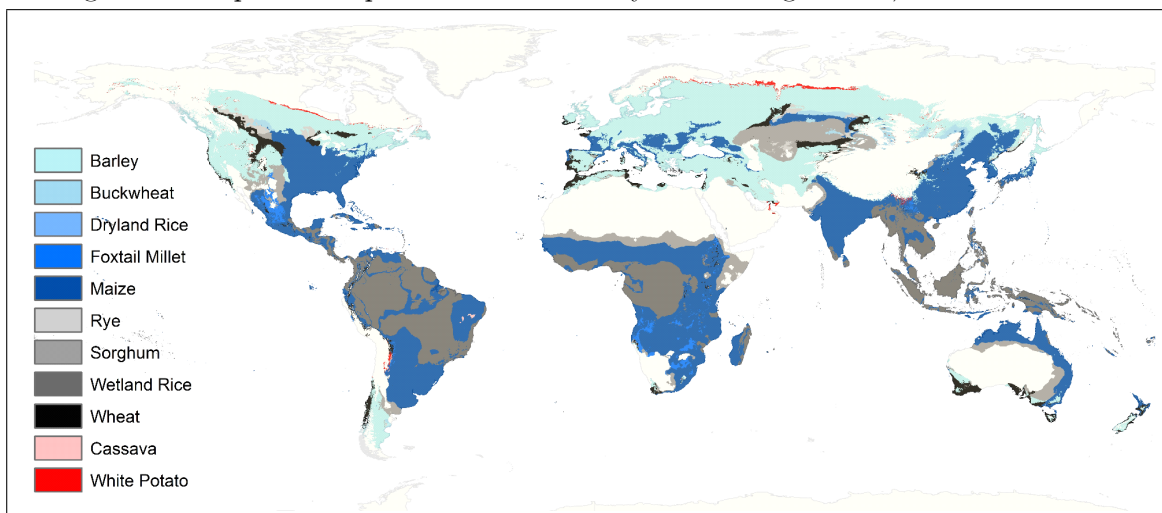


Figure F.9:

Figure F.10: Box plot: productivity advantage of cereals and hierarchy in pre-industrial societies

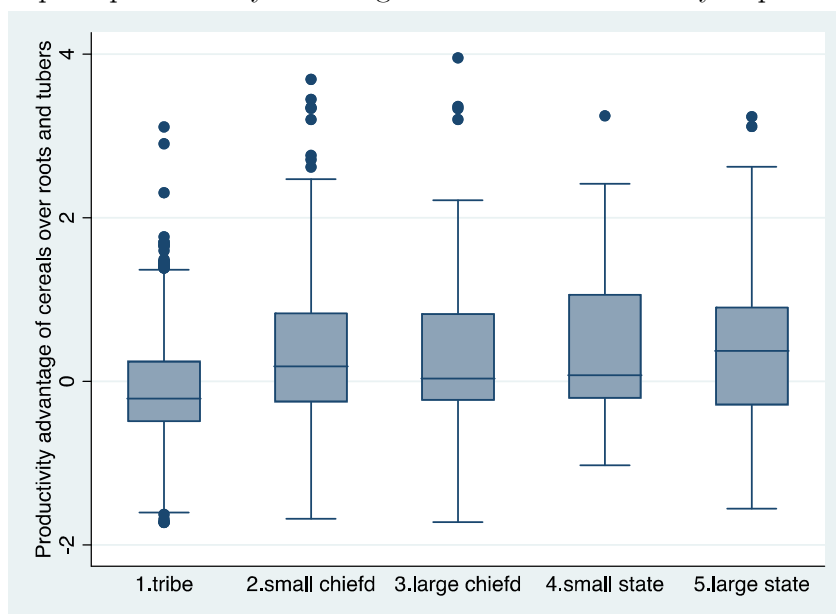


Figure F.11: Box plot: cereals and hierarchy in classical antiquity

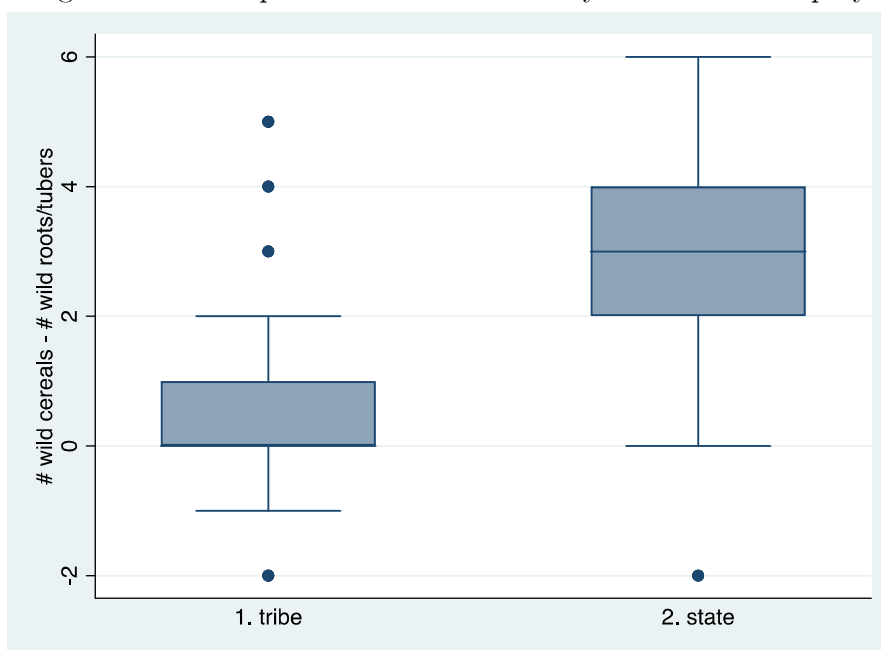


Figure F.12: Box plot: distance from areas of origin of agriculture and large human settlements founded before 400AD

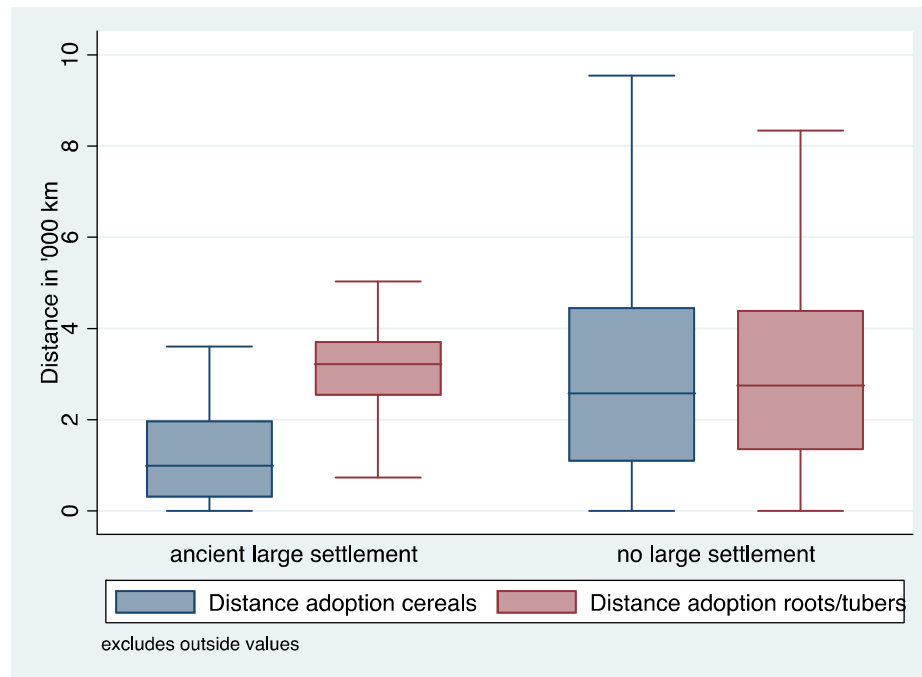


Figure F.13: Cities founded before 500 BC and centers of independent domestication

